

```

In [ ]: # =====
# Multimodal ResNet-50 with Concatenation Fusion
# - Uses 7-Fold CV  $\approx$  70% Train / 15% Val / 15% Test Overall
# - Implements ResNet-50 as the image backbone within the established multimodal pi
# - MODIFIED: Swapped the previous backbone with ResNet-50 and adjusted the model c
# =====

import os, time, json, warnings
from copy import deepcopy
from collections import defaultdict
from itertools import cycle
import numpy as np
import pandas as pd

import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader, Dataset
from torchvision import transforms, models
from PIL import Image

from sklearn.model_selection import StratifiedKFold, StratifiedShuffleSplit
from sklearn.metrics import (
    classification_report, accuracy_score, cohen_kappa_score,
    matthews_corrcoef, roc_auc_score, confusion_matrix, roc_curve, auc
)
from sklearn.preprocessing import label_binarize

from torch.amp import GradScaler, autocast

warnings.filterwarnings('ignore')
torch.backends.cudnn.benchmark = True

# Optional FLOPs (best effort)
try:
    from thop import profile
    HAVE_THOP = True
except Exception:
    HAVE_THOP = False

# Optional plotting libs
try:
    import matplotlib.pyplot as plt
    import seaborn as sns
    HAVE_PLOTTING = True
except ImportError:
    HAVE_PLOTTING = False

# -----
# 1) CONFIG – UPDATE THESE TWO PATHS
# -----
class Config:
    DATA_DIR = r"D:\AIUB\Neural\Synthetic_Skin_Disease_Dataset" # <-- update to

```

```

ORIGINAL_METADATA_CSV = r"D:\AIUB\Neural\complete_metadata_m.csv" # <-- update

# === MODEL SWAP: ResNet-50 ===
MODEL_NAME = "resnet50"
RESULTS_DIR = "metadata_Multimodal_ResNet50_Concat_KFold" # <-- New directory f

IMG_SIZE = 224
BATCH_SIZE = 128
LEARNING_RATE = 1e-4      # A good starting point for ResNet-50
WEIGHT_DECAY = 1e-4
EPOCHS = 200
NUM_CLASSES = 5
DEVICE = "cuda" if torch.cuda.is_available() else "cpu"
USE_AMP = True
NUM_WORKERS = 0
PATIENCE = 20
WARMUP_EPOCHS = 10
LABEL_SMOOTHING = 0.1
PARTIALLY_UNFREEZE = True # Fine-tuning the last block of ResNet is recommended

# K-Fold settings for ~70/15/15 overall
N_SPLITS = 7
VAL_WITHIN_REMAINDER = 0.15 / (1.0 - 1.0/7.0) # ≈ 0.175

# Metadata settings
META_DROPOUT = 0.3
META_HIDDEN_DIM = 128
META_OUT_DIM = 256

# Fairness: minimum subgroup size for stable metrics
MIN_SUPPORT = 15

config = Config()
os.makedirs(config.RESULTS_DIR, exist_ok=True)
print("Device:", config.DEVICE)
print(f"Model: {config.MODEL_NAME.capitalize()} with Concatenation Fusion")
print(f"K-Fold: N_SPLITS={config.N_SPLITS} (~Test {100/config.N_SPLITS:.1f}%), Val_
if not HAVE_PLOTTING:
    print("Warning: matplotlib and/or seaborn not found. Plots will be skipped.")

# -----
# 2) CSV STANDARDIZATION
# -----
def standardize_metadata_columns(df: pd.DataFrame) -> pd.DataFrame:
    """Standardizes metadata column names and creates necessary fields."""
    df = df.copy()
    if 'filepath' in df.columns:
        df['filepath'] = df['filepath'].astype(str).str.replace('\\', '/', regex=False)
    if 'filename' not in df.columns:
        if 'filepath' in df.columns:
            df['filename'] = df['filepath'].apply(lambda p: os.path.basename(str(p)))
        else:
            raise ValueError("CSV must include 'filepath' or 'filename'.")
    if 'label' not in df.columns and 'filepath' in df.columns:
        df['label'] = df['filepath'].str.split('/').str[1]

```

```

df['label'] = df['label'].astype(str)
if 'age_group' not in df.columns:
    if 'age' in df.columns:
        age_num = pd.to_numeric(df['age'], errors='coerce')
        bins = [-1, 17, 29, 44, 64, 200]
        names = ['child', 'young_adult', 'adult', 'middle_aged', 'senior']
        df['age_group'] = pd.cut(age_num, bins=bins, labels=names).astype(str).
    else:
        df['age_group'] = 'unknown'
if 'gender' not in df.columns:
    if 'dominant_gender' in df.columns:
        g = df['dominant_gender'].astype(str).str.strip().str.lower()
        df['gender'] = g.map({'man': 'male', 'male': 'male', 'woman': 'female', 'fema
    else:
        df['gender'] = 'unknown'
if 'skin_tone' not in df.columns:
    if 'dominant_race' in df.columns:
        df['skin_tone'] = df['dominant_race'].astype(str).str.strip().str.lower
    else:
        df['skin_tone'] = 'unknown'
required = ['filename', 'label', 'age_group', 'gender', 'skin_tone']
if any(c not in df.columns for c in required):
    raise ValueError(f"Missing required columns after standardization.")
return df

# -----
# 3) DATASET & VOCAB
# -----
class GPUAugOnlyNorm(nn.Module):
    """Module for GPU-side normalization."""
    def __init__(self):
        super().__init__()
        self.normalize = transforms.Normalize(mean=[0.485, 0.456, 0.406], std=[0.22
@torch.no_grad()
    def forward(self, x): return self.normalize(x)

class MetaVocab:
    """Creates and manages vocabularies for metadata fields."""
    def __init__(self, df_train, cols=('age_group', 'gender', 'skin_tone')):
        self.cols = cols
        self.unk_token = "<UNK>"
        self.vocabs = {}
        for c in cols:
            uniq = sorted([self.unk_token] + sorted(df_train[c].dropna().astype(str)
            self.vocabs[c] = {v:i for i,v in enumerate(uniq)}
        self.sizes = {c: len(self.vocabs[c]) for c in cols}
    def encode_row(self, row):
        out = {}
        for c in self.cols:
            v = str(row[c]) if pd.notnull(row[c]) else self.unk_token
            out[c] = self.vocabs[c].get(v, self.vocabs[c][self.unk_token])
        return out

class MultimodalDermDataset(Dataset):
    """Dataset class for images and metadata, with optional preloading."""

```

```

def __init__(self, config, df, class_to_idx, meta_vocab, transform=None):
    self.config = config
    self.df = df.reset_index(drop=True)
    self.class_to_idx = class_to_idx
    self.transform = transform
    self.meta_vocab = meta_vocab
    self.image_paths, self.labels, self.meta = [], [], []
    use_filepath = 'filepath' in self.df.columns
    for _, row in self.df.iterrows():
        if use_filepath and pd.notnull(row.get('filepath', None)):
            raw = str(row['filepath'])
            p = raw if os.path.isabs(raw) else os.path.join(config.DATA_DIR, raw)
        else:
            p = os.path.join(config.DATA_DIR, str(row['split']), str(row['label']))
        self.image_paths.append(p)
        self.labels.append(self.class_to_idx[str(row['label'])])
        self.meta.append(self.meta_vocab.encode_row(row))
    self.images = []
    for p in self.image_paths:
        with Image.open(p).convert("RGB") as im:
            self.images.append(im.copy())

def __len__(self): return len(self.labels)
def __getitem__(self, idx):
    image = self.images[idx]
    if self.transform: image = self.transform(image)
    return image, self.labels[idx], self.meta[idx]

# -----
# 4) MODEL (ResNet-50 + Metadata Concatenation)
# -----
class MetadataEncoder(nn.Module):
    """Encodes categorical metadata into a dense feature vector."""
    def __init__(self, meta_vocab, emb_dims=None, hidden=128, out_dim=256, dropout=0.5):
        super().__init__()
        sizes = meta_vocab.sizes
        if emb_dims is None:
            emb_dims = {
                'age_group': min(16, max(4, sizes['age_group'])),
                'gender': min(8, max(3, sizes['gender'])),
                'skin_tone': min(16, max(4, sizes['skin_tone'])),
            }
        self.emb_age = nn.Embedding(sizes['age_group'], emb_dims['age_group'])
        self.emb_gen = nn.Embedding(sizes['gender'], emb_dims['gender'])
        self.emb_tone = nn.Embedding(sizes['skin_tone'], emb_dims['skin_tone'])
        concat_dim = sum(emb_dims.values())
        self.mlp = nn.Sequential(
            nn.Linear(concat_dim, hidden), nn.GELU(), nn.Dropout(dropout),
            nn.Linear(hidden, out_dim), nn.GELU()
        )
    def forward(self, meta_idx):
        a = self.emb_age(meta_idx['age_group'])
        g = self.emb_gen(meta_idx['gender'])
        s = self.emb_tone(meta_idx['skin_tone'])
        return self.mlp(torch.cat([a, g, s], dim=-1))

```

```

class ResNet50MultimodalConcat(nn.Module):
    """
    A multimodal model using a pretrained ResNet-50 backbone.
    Features from the image backbone's convolutional base are concatenated
    with features from the metadata encoder and passed to a final classifier.
    """
    def __init__(self, num_classes, meta_vocab, meta_hidden_dim, meta_out_dim, meta_dropout):
        super().__init__()
        # Image backbone (ResNet-50)
        self.backbone = models.resnet50(weights=models.ResNet50_Weights.IMAGENET1K_V1)

        # Freeze all backbone parameters initially
        for param in self.backbone.parameters():
            param.requires_grad = False

        # Unfreeze the final residual block (layer4) if configured
        if config.PARTIALLY_UNFREEZE:
            print("Unfreezing ResNet-50 layer4 for fine-tuning.")
            for param in self.backbone.layer4.parameters():
                param.requires_grad = True

        # The feature dimension from ResNet-50's convolutional base (before the final fully connected layer)
        image_feature_dim = self.backbone.fc.in_features
        self.backbone.fc = nn.Identity() # Remove the original classifier

        # Metadata encoder
        self.meta_encoder = MetadataEncoder(meta_vocab,
                                            hidden=meta_hidden_dim,
                                            out_dim=meta_out_dim,
                                            dropout=meta_dropout)

        # Fusion and Classifier Head
        self.classifier = nn.Sequential(
            nn.Linear(image_feature_dim + meta_out_dim, 1024),
            nn.BatchNorm1d(1024),
            nn.ReLU(),
            nn.Dropout(0.5),
            nn.Linear(1024, 512),
            nn.BatchNorm1d(512),
            nn.ReLU(),
            nn.Dropout(0.5),
            nn.Linear(512, num_classes)
        )

    def forward(self, pixel_values, meta_idx):
        # Image stream
        img_features = self.backbone(pixel_values)
        img_features = torch.flatten(img_features, 1)

        # Metadata stream
        meta_features = self.meta_encoder(meta_idx)

        # Fusion by simple concatenation
        combined_features = torch.cat((img_features, meta_features), dim=1)

```

```

        # Classification
        output = self.classifier(combined_features)
        return output

# -----
# 5) TRAIN / EVAL HELPERS
# -----
def _to_device_meta(batch_meta, device):
    if isinstance(batch_meta, dict):
        ag = torch.as_tensor(batch_meta['age_group'], dtype=torch.long, device=device)
        ge = torch.as_tensor(batch_meta['gender'], dtype=torch.long, device=device)
        st = torch.as_tensor(batch_meta['skin_tone'], dtype=torch.long, device=device)
        return {'age_group': ag, 'gender': ge, 'skin_tone': st}
    else:
        ag = torch.tensor([m['age_group'] for m in batch_meta], dtype=torch.long, device=device)
        ge = torch.tensor([m['gender'] for m in batch_meta], dtype=torch.long, device=device)
        st = torch.tensor([m['skin_tone'] for m in batch_meta], dtype=torch.long, device=device)
        return {'age_group': ag, 'gender': ge, 'skin_tone': st}

def make_dataloaders_from_df(config, df_std):
    """Creates dataloaders for Train, Val, and Test splits from a DataFrame."""
    classes = sorted(df_std[df_std['split']=='Train']['label'].astype(str).unique())
    class_to_idx = {c:i for i,c in enumerate(classes)}
    config.NUM_CLASSES = len(classes)
    meta_vocab = MetaVocab(df_std[df_std['split']=='Train'])

    train_tfm = transforms.Compose([
        transforms.RandomResizedCrop(config.IMG_SIZE, scale=(0.8, 1.0)),
        transforms.RandomHorizontalFlip(),
        transforms.RandomApply([transforms.ColorJitter(0.4,0.4,0.2,0.1)], p=0.8),
        transforms.RandomRotation(15),
        transforms.ToTensor(),
    ])
    eval_tfm = transforms.Compose([
        transforms.Resize((config.IMG_SIZE, config.IMG_SIZE)),
        transforms.ToTensor(),
    ])
    dsets = {s: MultimodalDermDataset(config, df_std[df_std['split']==s], class_to_idx[s], eval_tfm) for s, t in [('Train', train_tfm), ('Val', eval_tfm), ('Test', eval_tfm)]}
    dls = {s: DataLoader(dsets[s], batch_size=config.BATCH_SIZE, shuffle=(s=='Train'), num_workers=config.NUM_WORKERS, pin_memory=True) for s in dsets}
    return dls, {k: len(v) for k,v in dsets.items()}, classes, meta_vocab

def plot_and_save_history(hist, fold_id, results_dir):
    """Plots training and validation loss and accuracy and saves them to a file."""
    if not HAVE_PLOTTING: return

    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(18, 6))
    epochs = range(1, len(hist['train_loss']) + 1)
    ax1.plot(epochs, hist['train_loss'], 'bo-', label='Training loss')
    ax1.plot(epochs, hist['val_loss'], 'ro-', label='Validation loss')
    ax1.set_title(f'Training and Validation Loss (Fold {fold_id})'); ax1.set_xlabel('Epoch')
    ax1.legend(); ax1.grid(True)
    ax2.plot(epochs, hist['train_acc'], 'bo-', label='Training accuracy')
    ax2.plot(epochs, hist['val_acc'], 'ro-', label='Validation accuracy')

```

```

ax2.set_title(f'Training and Validation Accuracy (Fold {fold_id})'); ax2.set_xlabel('Epoch')
ax2.legend(); ax2.grid(True)
plt.tight_layout()
history_path = os.path.join(results_dir, f'training_history_fold_{fold_id}.png')
plt.savefig(history_path, dpi=300); plt.show(); plt.close()
print(f"Saved training history plot to: {history_path}")

def train_model(model, dls, sizes, num_epochs, fold_id, class_weights):
    start = time.time()
    best = deepcopy(model.state_dict())
    best_val_loss = float('inf')
    patience = 0
    history = defaultdict(list)
    scaler = GradScaler(enabled=config.USE_AMP)
    gpu_norm = GPUAugOnlyNorm().to(config.DEVICE)
    optimizer = optim.AdamW(filter(lambda p: p.requires_grad, model.parameters()),
                                lr=config.LEARNING_RATE, weight_decay=config.WEIGHT_DECAY)

    def lr_lambda(ep):
        if ep < config.WARMUP_EPOCHS: return float(ep+1)/config.WARMUP_EPOCHS
        return max(0.0, 0.5 * (1.0 + np.cos(np.pi * (ep - config.WARMUP_EPOCHS) / (
            num_epochs - config.WARMUP_EPOCHS))))
    scheduler = optim.lr_scheduler.LambdaLR(optimizer, lr_lambda)

    criterion = nn.CrossEntropyLoss(weight=class_weights, label_smoothing=config.LABEL_SMOOTHING)
    print(f"Using weighted CrossEntropyLoss for training.")

    best_model_path = os.path.join(config.RESULTS_DIR, f'best_model_fold_{fold_id}.pth')

    print(f"\nSTART TRAINING ({config.MODEL_NAME.capitalize()} + Concat) for {num_epochs} epochs")
    for epoch in range(num_epochs):
        for phase in ['Train', 'Val']:
            model.train(phase == 'Train')
            running_loss, running_corrects = 0.0, 0
            for images, labels, metas in dls[phase]:
                images = gpu_norm(images.to(config.DEVICE))
                labels = labels.to(config.DEVICE, dtype=torch.long)
                meta_idx = _to_device_meta(metas, config.DEVICE)
                optimizer.zero_grad(set_to_none=True)
                with autocast(device_type=config.DEVICE.split(':')[0], enabled=config.USE_AMP):
                    logits = model(pixel_values=images, meta_idx=meta_idx)
                    loss = criterion(logits, labels)
                if phase == 'Train':
                    scaler.scale(loss).backward(); scaler.step(optimizer); scaler.update()
                running_loss += loss.item() * images.size(0)
                running_corrects += (logits.argmax(1) == labels).sum().item()

            epoch_loss = running_loss / sizes[phase]
            epoch_acc = running_corrects / sizes[phase]
            history[f'{phase.lower()}_loss'].append(epoch_loss)
            history[f'{phase.lower()}_acc'].append(epoch_acc)

            if phase == 'Val':
                if epoch_loss < best_val_loss:
                    best_val_loss = epoch_loss; best = deepcopy(model.state_dict())
                    torch.save(model.state_dict(), best_model_path)
                else:
                    patience += 1

```

```

        scheduler.step()
        print(f"Epoch {epoch+1}/{num_epochs} | Train Loss: {history['train_loss'][-1]:.4f} Val Loss: {history['val_loss'][-1]:.4f} Acc: {history['val_acc'][-1]:.4f}")
        if patience >= config.PATIENCE:
            print("Early stopping triggered."); break

    print(f"Training complete in {(time.time()-start)/60.0:.1f} minutes. Best Val Loss: {best_val_loss:.4f}")
    print(f"Best model for fold saved to: {best_model_path}")
    model.load_state_dict(best)
    plot_and_save_history(history, fold_id, config.RESULTS_DIR)
    return model, history

# -----
# 6) EVALUATION & FAIRNESS METRICS
# -----
def _print_section(title):
    print("\n" + "="*80); print(title.upper()); print("="*80 + "\n")

def _group_fairness_metrics(y_true, y_pred, groups, num_classes, min_support=None):
    uniq_groups = sorted(pd.Series(groups).astype(str).unique())
    per_group_acc, per_group_tprs, counts = {}, {}, {}
    y_true, y_pred = np.asarray(y_true), np.asarray(y_pred)
    for g in uniq_groups:
        mask = (np.asarray(groups) == g); counts[g] = int(mask.sum())
        yt, yp = y_true[mask], y_pred[mask]
        per_group_acc[g] = (yt == yp).mean() if len(yt) > 0 else np.nan
        tprs = [( (yt == c) & (yp == c)).sum() / ((yt == c).sum() or 1) ) for c in range(num_classes)]
        per_group_tprs[g] = np.array(tprs, dtype=float)
    filt_keys = [g for g in uniq_groups if counts[g] >= min_support] if min_support else uniq_groups
    wga = float(np.nanmin([per_group_acc.get(g, np.nan) for g in filt_keys])) if min_support else None
    max_gap, worst_pair = 0.0, (None, None)
    for i in range(len(filt_keys)):
        for j in range(i + 1, len(filt_keys)):
            gi, gj = filt_keys[i], filt_keys[j]
            a, b = per_group_tprs[gi], per_group_tprs[gj]
            mask = ~(np.isnan(a) | np.isnan(b))
            if mask.sum() > 0:
                gap = float(np.nanmean(np.abs(a[mask] - b[mask])))
                if gap > max_gap: max_gap, worst_pair = gap, (gi, gj)
    return {'wga': wga, 'eod': max_gap, 'worst_pair': worst_pair, 'keys_used': filt_keys}

def plot_and_save_roc_curves(y_true_bin, y_prob, class_names, fold_id, results_dir):
    if not HAVE_PLOTTING: return
    n_classes = len(class_names)
    fpr, tpr, roc_auc = dict(), dict(), dict()
    for i in range(n_classes):
        fpr[i], tpr[i], _ = roc_curve(y_true_bin[:, i], y_prob[:, i])
        roc_auc[i] = auc(fpr[i], tpr[i])
    all_fpr = np.unique(np.concatenate([fpr[i] for i in range(n_classes)]))
    mean_tpr = np.zeros_like(all_fpr)
    for i in range(n_classes): mean_tpr += np.interp(all_fpr, fpr[i], tpr[i])
    mean_tpr /= n_classes
    fpr["macro"], tpr["macro"], roc_auc["macro"] = all_fpr, mean_tpr, auc(all_fpr, mean_tpr)
    plt.figure(figsize=(12, 10))
    colors = cycle(['aqua', 'darkorange', 'cornflowerblue', 'green', 'red', 'purple'])
    for i, color in zip(range(n_classes), colors):

```

```

        plt.plot(fpr[i], tpr[i], color=color, lw=2, label=f'ROC of {class_names[i]}')
    plt.plot(fpr["macro"], tpr["macro"], label=f'Macro-average ROC (AUC = {roc_auc["macro"]})')
    plt.plot([0, 1], [0, 1], 'k--', lw=2); plt.xlim([0.0, 1.0]); plt.ylim([0.0, 1.0])
    plt.xlabel('False Positive Rate'); plt.ylabel('True Positive Rate'); plt.title('ROC Curve')
    plt.legend(loc="lower right"); plt.grid(True)
    roc_path = os.path.join(results_dir, f'roc_curves_fold_{fold_id}.png')
    plt.savefig(roc_path, dpi=300); plt.show(); plt.close()
    print(f"Saved ROC curves plot to: {roc_path}")

def plot_and_save_scalar_metrics(metrics_dict, fold_id, results_dir):
    if not HAVE_PLOTTING: return
    names, values = list(metrics_dict.keys()), list(metrics_dict.values())
    plt.figure(figsize=(10, 6))
    bars = sns.barplot(x=names, y=values, palette="viridis")
    plt.title(f'Overall Scalar Metrics Summary - Fold {fold_id}'); plt.ylabel('Score')
    for bar in bars.patches:
        plt.text(bar.get_x() + bar.get_width() / 2, bar.get_height() + 0.01, f'{bar.get_y():.2f}')
    scalar_path = os.path.join(results_dir, f'scalar_metrics_fold_{fold_id}.png')
    plt.savefig(scalar_path, dpi=300); plt.show(); plt.close()
    print(f"Saved scalar metrics plot to: {scalar_path}")

def evaluate_and_visualize(model, dl_test, class_names, df_fold, fold_id):
    model.eval()
    gpu_norm = GPUAugOnlyNorm().to(config.DEVICE)
    y_true, y_pred, y_prob = [], [], []
    with torch.no_grad():
        for images, labels, metas in dl_test:
            images = gpu_norm(images.to(config.DEVICE))
            meta_idx = _to_device_meta(metas, config.DEVICE)
            logits = model(pixel_values=images, meta_idx=meta_idx)
            y_true.extend(labels.numpy())
            y_pred.extend(logits.argmax(1).cpu().numpy())
            y_prob.extend(torch.softmax(logits, dim=1).cpu().numpy())
    y_true, y_pred, y_prob = np.array(y_true), np.array(y_pred), np.array(y_prob)

    _print_section(f"OVERALL PERFORMANCE (FOLD {fold_id})")
    print(classification_report(y_true, y_pred, target_names=class_names, zero_division=0))

    acc = accuracy_score(y_true, y_pred)
    kappa = cohen_kappa_score(y_true, y_pred)
    mcc = matthews_corrcoef(y_true, y_pred)
    try:
        y_true_bin = label_binarize(y_true, classes=np.arange(len(class_names)))
        roc_auc_macro = roc_auc_score(y_true_bin, y_prob, average='macro', multi_class='ovo')
    except Exception: y_true_bin, roc_auc_macro = None, float('nan')
    print(f"\n--- Overall Scalar Metrics ---\nAccuracy: {acc:.4f}\nMacro ROC AUC: {roc_auc_macro:.4f}")

    if HAVE_PLOTTING:
        cm = confusion_matrix(y_true, y_pred, labels=np.arange(len(class_names)))
        cm_norm = np.nan_to_num(cm.astype('float') / cm.sum(axis=1)[:, np.newaxis])
        plt.figure(figsize=(12, 10))
        sns.heatmap(cm_norm, annot=True, fmt='.2f', cmap='Blues', xticklabels=class_names)
        plt.title(f'Normalized Confusion Matrix (Fold {fold_id})'); plt.ylabel('True Labels')
        cm_path = os.path.join(config.RESULTS_DIR, f'confusion_matrix_fold_{fold_id}.png')
        plt.savefig(cm_path, dpi=300); plt.show(); plt.close()
        print(f"Saved normalized confusion matrix to: {cm_path}")

```

```

        if y_true_bin is not None: plot_and_save_roc_curves(y_true_bin, y_prob, cla
        plot_and_save_scalar_metrics({'Accuracy': acc, 'Macro AUC': roc_auc_macro,

df_test = df_fold[df_fold['split'] == 'Test'].reset_index(drop=True)
df_test = df_test.iloc[:len(y_true)] # Align rows just in case
age_disp = df_test['age_group'].map({'child': '0-17', 'young_adult': '18-59', '

full_labels = np.arange(len(class_names))

for group_name, series in [('GENDER', df_test['gender']), ('AGE', age_disp), ('
    _print_section(f"PERFORMANCE BY {group_name}")
    for val in sorted(series.unique()):
        if val == 'unknown': continue
        mask = (series == val).values
        if mask.sum() > 0:
            print(f"--- Subgroup: {val} ---\n")
            print(classification_report(y_true[mask], y_pred[mask],
                                        target_names=class_names,
                                        labels=full_labels,
                                        zero_division=0))

groups_composite = (age_disp.astype(str) + "|" + df_test['gender'].astype(str)
fair_all = _group_fairness_metrics(y_true, y_pred, groups_composite, len(class_
fair_filt = _group_fairness_metrics(y_true, y_pred, groups_composite, len(class
print("\n" + "="*80 + f"\nFAIRNESS SUMMARY (ALL GROUPS) | WGA: {fair_all['wga']}
print("\n" + "="*80 + f"\nFAIRNESS SUMMARY (FILTERED, n>={config.MIN_SUPPORT})

def acc_counts(s, yt, yp): return {str(g): ((yt[s==g] == yp[s==g]).sum(), (s==g
pooled_info = {'gender': acc_counts(df_test['gender'], y_true, y_pred), 'age':

return {'accuracy': acc, 'roc_auc_macro': roc_auc_macro, 'kappa': kappa, 'mcc':

# -----
# 7) K-FOLD CROSS-VALIDATION ORCHESTRATION
# -----
def build_kfold_splits(df_std, n_splits, val_within_remainder, seed=42):
    df, y = df_std.copy().reset_index(drop=True), df_std['label'].astype(str).value
    skf = StratifiedKFold(n_splits=n_splits, shuffle=True, random_state=seed)
    folds = []
    for i, (trainval_idx, test_idx) in enumerate(skf.split(np.arange(len(df)), y),
        sss2 = StratifiedShuffleSplit(n_splits=1, test_size=val_within_remainder, r
        tr_rel, val_rel = next(sss2.split(trainval_idx, y[trainval_idx]))
        dfi = df.copy(); dfi['split'] = 'Ignore'
        dfi.loc[trainval_idx[tr_rel], 'split'], dfi.loc[trainval_idx[val_rel], 'spl
        folds.append(dfi[dfi['split'] != 'Ignore'])
    return folds

def run_one_fold(fold_df, fold_id, config):
    print("\n" + "="*80 + f"\n### FOLD {fold_id} ###\n" + "="*88)
    dls, sizes, class_names, meta_vocab = make_data loaders_from_df(config, fold_df)

    # Calculate class weights for the current fold's training data
    train_labels = fold_df[fold_df['split']=='Train']['label']
    class_counts = train_labels.value_counts().sort_index().reindex(class_names, fi
    weights = 1.0 / (class_counts + 1e-6)

```

```

class_weights = torch.tensor(weights.values, dtype=torch.float).to(config.DEVICE)
class_weights = class_weights / class_weights.sum() * len(class_names)

# Initialize the correct model based on the config
if config.MODEL_NAME == 'resnet50':
    model = ResNet50MultimodalConcat(
        num_classes=len(class_names),
        meta_vocab=meta_vocab,
        meta_hidden_dim=config.META_HIDDEN_DIM,
        meta_out_dim=config.META_OUT_DIM,
        meta_dropout=config.META_DROPOUT
    ).to(config.DEVICE)
else:
    raise ValueError(f"Model name '{config.MODEL_NAME}' not recognized.")

trainable_params = sum(p.numel() for p in model.parameters() if p.requires_grad)
total_params = sum(p.numel() for p in model.parameters())
print(f"Total params: {total_params/1e6:.2f}M | Trainable params: {trainable_params/1e6:.2f}M")

model, _ = train_model(model, dls, sizes, num_epochs=config.EPOCHS, fold_id=fold_id)
return evaluate_and_visualize(model, dls['Test'], class_names, fold_df, fold_id)

def kfold_cross_val(df_std, config):
    folds = build_kfold_splits(df_std, config.N_SPLITS, config.VAL_WITHIN_REMAINDER)
    per_fold = [run_one_fold(fold, i + 1, config) for i, fold in enumerate(folds)]
    perf = pd.DataFrame(per_fold)

    def mean_std(col): return (perf[col].mean(), perf[col].std())
    summary = {k: mean_std(k) for k in ['accuracy', 'roc_auc_macro', 'kappa', 'mcc']}

    print("\n" + "="*88 + f"\nCross-validated over {config.N_SPLITS} folds\n" + "=")
    for k, (m, s) in summary.items(): print(f"{k.replace('_', ' ').title():<20}: {m:.4f} ± {s:.4f}")

    def merge_pooled(key):
        agg = defaultdict(lambda: [0, 0])
        for m in per_fold:
            for g, (c, t) in m['pooled_groups'][key].items(): agg[g][0] += c; agg[g][1] += t
        return sorted([(g, c/t, t) for g, (c, t) in agg.items() if t>0], key=lambda x: x[1])

    def print_table(title, rows):
        print(f"\n--- {title} ---"); print(f"{'Group':>35s} {'Acc':>8s} {'Count':>8s}")
        for g, acc, cnt in rows:
            if cnt >= config.MIN_SUPPORT: print(f"{g:>35s} {acc:>8.3f} {cnt:>8d}")

    for key, name in [('gender', 'GENDER'), ('age', 'AGE'), ('skin_tone', 'SKIN TONE')]:
        print_table(f"POOLED OVERALL ACCURACY BY {name} (n>={config.MIN_SUPPORT})",
                    merge_pooled(key))

    perf['fold'] = range(1, len(perf) + 1); perf.set_index('fold', inplace=True)
    perf.to_csv(os.path.join(config.RESULTS_DIR, "kfold_performance.csv"))
    pd.DataFrame({k: {'mean': v[0], 'std': v[1]} for k, v in summary.items()}).T.to_csv(
        os.path.join(config.RESULTS_DIR, "kfold_summary.csv"))
    print(f"\nResults, models, and plots saved to '{config.RESULTS_DIR}'")

# -----
# 8) MAIN EXECUTION
# -----

```

```
if __name__ == '__main__':  
    try:  
        df_orig = pd.read_csv(config.ORIGINAL_METADATA_CSV)  
        df_std = standardize_metadata_columns(df_orig)  
        print(f"Loaded and standardized metadata: {len(df_std)} rows.")  
        kfold_cross_val(df_std, config)  
    except FileNotFoundError:  
        print(f"ERROR: Metadata file not found at '{config.ORIGINAL_METADATA_CSV}'")  
    except Exception as e:  
        print(f"An unexpected error occurred: {e}")  
        import traceback  
        traceback.print_exc()
```

Device: cuda
Model: Resnet50 with Concatenation Fusion
K-Fold: N_SPLITS=7 (~Test 14.3%), Val_in_remainder=0.175
Loaded and standardized metadata: 565 rows.

```
#####  
####  
### FOLD 1 ###  
#####  
####  
Unfreezing ResNet-50 layer4 for fine-tuning.  
Total params: 26.43M | Trainable params: 17.89M (67.68%)  
Using weighted CrossEntropyLoss for training.
```

```
START TRAINING (Resnet50 + Concat) for 200 epochs...  
Epoch 1/200 | Train Loss: 1.9571 Acc: 0.2180 | Val Loss: 1.8291 Acc: 0.2235 | LR: 2.00e-05  
Epoch 2/200 | Train Loss: 1.9430 Acc: 0.2206 | Val Loss: 1.8308 Acc: 0.3412 | LR: 3.00e-05  
Epoch 3/200 | Train Loss: 1.8466 Acc: 0.2231 | Val Loss: 1.8239 Acc: 0.4235 | LR: 4.00e-05  
Epoch 4/200 | Train Loss: 1.8100 Acc: 0.2957 | Val Loss: 1.8070 Acc: 0.4588 | LR: 5.00e-05  
Epoch 5/200 | Train Loss: 1.7236 Acc: 0.3183 | Val Loss: 1.7762 Acc: 0.5412 | LR: 6.00e-05  
Epoch 6/200 | Train Loss: 1.6482 Acc: 0.3509 | Val Loss: 1.7234 Acc: 0.6000 | LR: 7.00e-05  
Epoch 7/200 | Train Loss: 1.5749 Acc: 0.4135 | Val Loss: 1.6583 Acc: 0.6471 | LR: 8.00e-05  
Epoch 8/200 | Train Loss: 1.4758 Acc: 0.4937 | Val Loss: 1.5693 Acc: 0.6588 | LR: 9.00e-05  
Epoch 9/200 | Train Loss: 1.3784 Acc: 0.5414 | Val Loss: 1.4803 Acc: 0.6588 | LR: 1.00e-04  
Epoch 10/200 | Train Loss: 1.3228 Acc: 0.5614 | Val Loss: 1.3988 Acc: 0.6471 | LR: 1.00e-04  
Epoch 11/200 | Train Loss: 1.3098 Acc: 0.6190 | Val Loss: 1.3388 Acc: 0.6706 | LR: 1.00e-04  
Epoch 12/200 | Train Loss: 1.2297 Acc: 0.6291 | Val Loss: 1.2896 Acc: 0.6941 | LR: 1.00e-04  
Epoch 13/200 | Train Loss: 1.1736 Acc: 0.7043 | Val Loss: 1.2551 Acc: 0.6706 | LR: 9.99e-05  
Epoch 14/200 | Train Loss: 1.1522 Acc: 0.6892 | Val Loss: 1.2196 Acc: 0.6824 | LR: 9.99e-05  
Epoch 15/200 | Train Loss: 1.1210 Acc: 0.7043 | Val Loss: 1.1882 Acc: 0.7059 | LR: 9.98e-05  
Epoch 16/200 | Train Loss: 1.0614 Acc: 0.7694 | Val Loss: 1.1530 Acc: 0.7059 | LR: 9.98e-05  
Epoch 17/200 | Train Loss: 1.0797 Acc: 0.7644 | Val Loss: 1.1206 Acc: 0.7176 | LR: 9.97e-05  
Epoch 18/200 | Train Loss: 1.0438 Acc: 0.7669 | Val Loss: 1.0959 Acc: 0.7176 | LR: 9.96e-05  
Epoch 19/200 | Train Loss: 1.0085 Acc: 0.8070 | Val Loss: 1.0758 Acc: 0.7294 | LR: 9.94e-05  
Epoch 20/200 | Train Loss: 1.0020 Acc: 0.8170 | Val Loss: 1.0570 Acc: 0.7176 | LR: 9.93e-05  
Epoch 21/200 | Train Loss: 0.9939 Acc: 0.8371 | Val Loss: 1.0372 Acc: 0.7412 | LR:
```

9.92e-05

Epoch 22/200 | Train Loss: 0.9852 Acc: 0.8195 | Val Loss: 1.0215 Acc: 0.7765 | LR:

9.90e-05

Epoch 23/200 | Train Loss: 0.9644 Acc: 0.8396 | Val Loss: 1.0158 Acc: 0.7647 | LR:

9.88e-05

Epoch 24/200 | Train Loss: 0.9505 Acc: 0.8396 | Val Loss: 1.0095 Acc: 0.7765 | LR:

9.87e-05

Epoch 25/200 | Train Loss: 0.9401 Acc: 0.8521 | Val Loss: 1.0012 Acc: 0.7765 | LR:

9.85e-05

Epoch 26/200 | Train Loss: 0.9300 Acc: 0.8596 | Val Loss: 1.0000 Acc: 0.7647 | LR:

9.83e-05

Epoch 27/200 | Train Loss: 0.9078 Acc: 0.8847 | Val Loss: 0.9943 Acc: 0.7647 | LR:

9.80e-05

Epoch 28/200 | Train Loss: 0.8855 Acc: 0.9023 | Val Loss: 0.9920 Acc: 0.7765 | LR:

9.78e-05

Epoch 29/200 | Train Loss: 0.9022 Acc: 0.8897 | Val Loss: 0.9878 Acc: 0.7882 | LR:

9.76e-05

Epoch 30/200 | Train Loss: 0.8992 Acc: 0.8847 | Val Loss: 0.9765 Acc: 0.7882 | LR:

9.73e-05

Epoch 31/200 | Train Loss: 0.8788 Acc: 0.9148 | Val Loss: 0.9673 Acc: 0.8000 | LR:

9.70e-05

Epoch 32/200 | Train Loss: 0.8847 Acc: 0.9148 | Val Loss: 0.9609 Acc: 0.8118 | LR:

9.67e-05

Epoch 33/200 | Train Loss: 0.8908 Acc: 0.9198 | Val Loss: 0.9556 Acc: 0.8118 | LR:

9.64e-05

Epoch 34/200 | Train Loss: 0.8555 Acc: 0.9198 | Val Loss: 0.9461 Acc: 0.8118 | LR:

9.61e-05

Epoch 35/200 | Train Loss: 0.8356 Acc: 0.9298 | Val Loss: 0.9391 Acc: 0.8118 | LR:

9.58e-05

Epoch 36/200 | Train Loss: 0.8529 Acc: 0.9148 | Val Loss: 0.9340 Acc: 0.8235 | LR:

9.55e-05

Epoch 37/200 | Train Loss: 0.8550 Acc: 0.9373 | Val Loss: 0.9285 Acc: 0.8235 | LR:

9.51e-05

Epoch 38/200 | Train Loss: 0.8457 Acc: 0.9323 | Val Loss: 0.9252 Acc: 0.8118 | LR:

9.47e-05

Epoch 39/200 | Train Loss: 0.8307 Acc: 0.9474 | Val Loss: 0.9154 Acc: 0.8353 | LR:

9.44e-05

Epoch 40/200 | Train Loss: 0.8304 Acc: 0.9248 | Val Loss: 0.9128 Acc: 0.8353 | LR:

9.40e-05

Epoch 41/200 | Train Loss: 0.8252 Acc: 0.9348 | Val Loss: 0.9101 Acc: 0.8471 | LR:

9.36e-05

Epoch 42/200 | Train Loss: 0.8245 Acc: 0.9624 | Val Loss: 0.9066 Acc: 0.8353 | LR:

9.32e-05

Epoch 43/200 | Train Loss: 0.8147 Acc: 0.9599 | Val Loss: 0.9049 Acc: 0.8353 | LR:

9.27e-05

Epoch 44/200 | Train Loss: 0.8461 Acc: 0.9474 | Val Loss: 0.9067 Acc: 0.8235 | LR:

9.23e-05

Epoch 45/200 | Train Loss: 0.8110 Acc: 0.9549 | Val Loss: 0.9070 Acc: 0.8235 | LR:

9.19e-05

Epoch 46/200 | Train Loss: 0.8088 Acc: 0.9524 | Val Loss: 0.9062 Acc: 0.8353 | LR:

9.14e-05

Epoch 47/200 | Train Loss: 0.8162 Acc: 0.9649 | Val Loss: 0.9022 Acc: 0.8706 | LR:

9.09e-05

Epoch 48/200 | Train Loss: 0.8122 Acc: 0.9574 | Val Loss: 0.9050 Acc: 0.8471 | LR:

9.05e-05

Epoch 49/200 | Train Loss: 0.8103 Acc: 0.9599 | Val Loss: 0.9004 Acc: 0.8588 | LR:

9.00e-05

Epoch 50/200 | Train Loss: 0.8416 Acc: 0.9398 | Val Loss: 0.9002 Acc: 0.8588 | LR:

8.95e-05

Epoch 51/200 | Train Loss: 0.7953 Acc: 0.9699 | Val Loss: 0.9013 Acc: 0.8471 | LR:

8.89e-05

Epoch 52/200 | Train Loss: 0.7954 Acc: 0.9749 | Val Loss: 0.8988 Acc: 0.8471 | LR:

8.84e-05

Epoch 53/200 | Train Loss: 0.7950 Acc: 0.9574 | Val Loss: 0.8991 Acc: 0.8471 | LR:

8.79e-05

Epoch 54/200 | Train Loss: 0.7959 Acc: 0.9749 | Val Loss: 0.8979 Acc: 0.8353 | LR:

8.73e-05

Epoch 55/200 | Train Loss: 0.7840 Acc: 0.9749 | Val Loss: 0.8970 Acc: 0.8471 | LR:

8.68e-05

Epoch 56/200 | Train Loss: 0.8037 Acc: 0.9749 | Val Loss: 0.8975 Acc: 0.8353 | LR:

8.62e-05

Epoch 57/200 | Train Loss: 0.7904 Acc: 0.9799 | Val Loss: 0.8988 Acc: 0.8353 | LR:

8.56e-05

Epoch 58/200 | Train Loss: 0.8068 Acc: 0.9649 | Val Loss: 0.9045 Acc: 0.8471 | LR:

8.51e-05

Epoch 59/200 | Train Loss: 0.7927 Acc: 0.9774 | Val Loss: 0.9072 Acc: 0.8471 | LR:

8.45e-05

Epoch 60/200 | Train Loss: 0.7995 Acc: 0.9699 | Val Loss: 0.9007 Acc: 0.8471 | LR:

8.39e-05

Epoch 61/200 | Train Loss: 0.7901 Acc: 0.9674 | Val Loss: 0.8962 Acc: 0.8471 | LR:

8.33e-05

Epoch 62/200 | Train Loss: 0.7938 Acc: 0.9774 | Val Loss: 0.8919 Acc: 0.8588 | LR:

8.26e-05

Epoch 63/200 | Train Loss: 0.7882 Acc: 0.9749 | Val Loss: 0.8904 Acc: 0.8588 | LR:

8.20e-05

Epoch 64/200 | Train Loss: 0.7783 Acc: 0.9699 | Val Loss: 0.8870 Acc: 0.8471 | LR:

8.14e-05

Epoch 65/200 | Train Loss: 0.7812 Acc: 0.9699 | Val Loss: 0.8858 Acc: 0.8588 | LR:

8.07e-05

Epoch 66/200 | Train Loss: 0.7854 Acc: 0.9699 | Val Loss: 0.8873 Acc: 0.8353 | LR:

8.01e-05

Epoch 67/200 | Train Loss: 0.7854 Acc: 0.9699 | Val Loss: 0.8867 Acc: 0.8588 | LR:

7.94e-05

Epoch 68/200 | Train Loss: 0.7869 Acc: 0.9774 | Val Loss: 0.8848 Acc: 0.8706 | LR:

7.87e-05

Epoch 69/200 | Train Loss: 0.7704 Acc: 0.9850 | Val Loss: 0.8776 Acc: 0.8588 | LR:

7.80e-05

Epoch 70/200 | Train Loss: 0.7732 Acc: 0.9799 | Val Loss: 0.8762 Acc: 0.8471 | LR:

7.73e-05

Epoch 71/200 | Train Loss: 0.7804 Acc: 0.9774 | Val Loss: 0.8798 Acc: 0.8353 | LR:

7.67e-05

Epoch 72/200 | Train Loss: 0.7732 Acc: 0.9875 | Val Loss: 0.8734 Acc: 0.8471 | LR:

7.59e-05

Epoch 73/200 | Train Loss: 0.7760 Acc: 0.9749 | Val Loss: 0.8694 Acc: 0.8588 | LR:

7.52e-05

Epoch 74/200 | Train Loss: 0.7641 Acc: 0.9875 | Val Loss: 0.8701 Acc: 0.8471 | LR:

7.45e-05

Epoch 75/200 | Train Loss: 0.7883 Acc: 0.9950 | Val Loss: 0.8689 Acc: 0.8471 | LR:

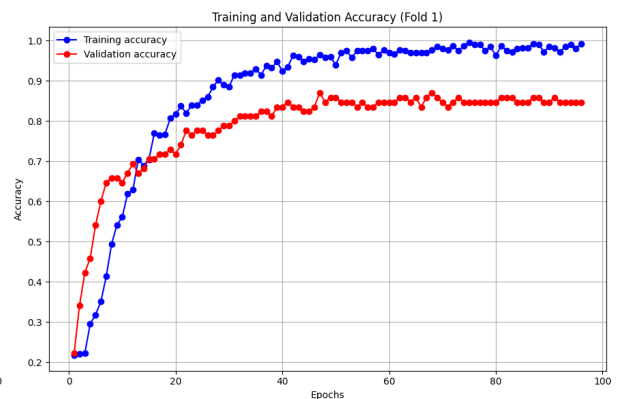
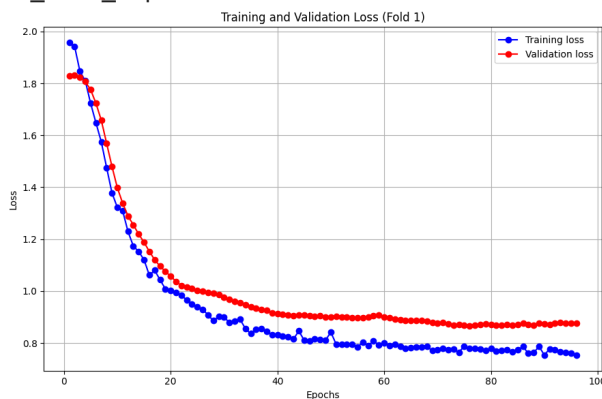
7.38e-05

Epoch 76/200 | Train Loss: 0.7798 Acc: 0.9900 | Val Loss: 0.8669 Acc: 0.8471 | LR:

7.31e-05

Epoch 77/200 | Train Loss: 0.7794 Acc: 0.9900 | Val Loss: 0.8693 Acc: 0.8471 | LR:

7.23e-05
Epoch 78/200 | Train Loss: 0.7764 Acc: 0.9749 | Val Loss: 0.8715 Acc: 0.8471 | LR: 7.16e-05
Epoch 79/200 | Train Loss: 0.7718 Acc: 0.9850 | Val Loss: 0.8724 Acc: 0.8471 | LR: 7.08e-05
Epoch 80/200 | Train Loss: 0.7790 Acc: 0.9624 | Val Loss: 0.8708 Acc: 0.8471 | LR: 7.01e-05
Epoch 81/200 | Train Loss: 0.7674 Acc: 0.9875 | Val Loss: 0.8672 Acc: 0.8588 | LR: 6.93e-05
Epoch 82/200 | Train Loss: 0.7723 Acc: 0.9749 | Val Loss: 0.8674 Acc: 0.8588 | LR: 6.86e-05
Epoch 83/200 | Train Loss: 0.7727 Acc: 0.9724 | Val Loss: 0.8701 Acc: 0.8588 | LR: 6.78e-05
Epoch 84/200 | Train Loss: 0.7673 Acc: 0.9799 | Val Loss: 0.8691 Acc: 0.8471 | LR: 6.70e-05
Epoch 85/200 | Train Loss: 0.7738 Acc: 0.9825 | Val Loss: 0.8716 Acc: 0.8471 | LR: 6.62e-05
Epoch 86/200 | Train Loss: 0.7862 Acc: 0.9825 | Val Loss: 0.8758 Acc: 0.8471 | LR: 6.55e-05
Epoch 87/200 | Train Loss: 0.7599 Acc: 0.9925 | Val Loss: 0.8712 Acc: 0.8588 | LR: 6.47e-05
Epoch 88/200 | Train Loss: 0.7621 Acc: 0.9900 | Val Loss: 0.8684 Acc: 0.8588 | LR: 6.39e-05
Epoch 89/200 | Train Loss: 0.7869 Acc: 0.9724 | Val Loss: 0.8763 Acc: 0.8471 | LR: 6.31e-05
Epoch 90/200 | Train Loss: 0.7535 Acc: 0.9850 | Val Loss: 0.8747 Acc: 0.8471 | LR: 6.23e-05
Epoch 91/200 | Train Loss: 0.7776 Acc: 0.9825 | Val Loss: 0.8711 Acc: 0.8588 | LR: 6.15e-05
Epoch 92/200 | Train Loss: 0.7751 Acc: 0.9724 | Val Loss: 0.8772 Acc: 0.8471 | LR: 6.07e-05
Epoch 93/200 | Train Loss: 0.7653 Acc: 0.9850 | Val Loss: 0.8779 Acc: 0.8471 | LR: 5.99e-05
Epoch 94/200 | Train Loss: 0.7645 Acc: 0.9900 | Val Loss: 0.8762 Acc: 0.8471 | LR: 5.90e-05
Epoch 95/200 | Train Loss: 0.7595 Acc: 0.9799 | Val Loss: 0.8771 Acc: 0.8471 | LR: 5.82e-05
Epoch 96/200 | Train Loss: 0.7534 Acc: 0.9925 | Val Loss: 0.8766 Acc: 0.8471 | LR: 5.74e-05
Early stopping triggered.
Training complete in 6.8 minutes. Best Val Loss: 0.8669
Best model for fold saved to: new_results_Multimodal_ResNet50_Concat_KFold\best_model_fold_1.pth



Saved training history plot to: new_results_Multimodal_ResNet50_Concat_KFold\training_history_fold_1.png

=====

OVERALL PERFORMANCE (FOLD 1)

=====

	precision	recall	f1-score	support
Herpes	1.00	0.71	0.83	7
Melanoma	0.75	1.00	0.86	3
Monkeypox	1.00	0.89	0.94	18
Sarampion	0.89	0.86	0.88	29
Varicela	0.82	0.96	0.88	24
accuracy			0.89	81
macro avg	0.89	0.88	0.88	81
weighted avg	0.90	0.89	0.89	81

--- Overall Scalar Metrics ---

Accuracy: 0.8889

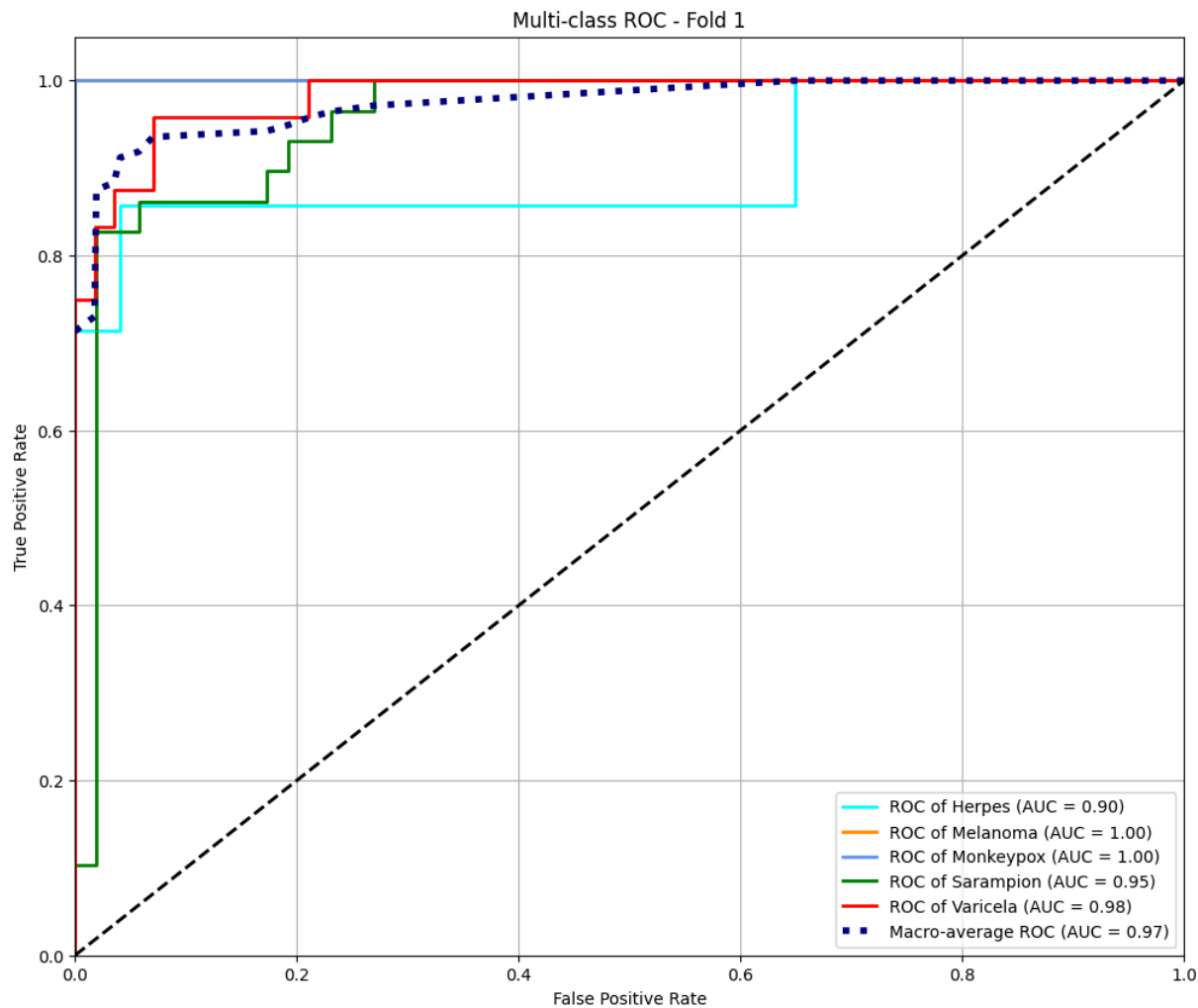
Macro ROC AUC: 0.9676

Cohen's Kappa: 0.8463

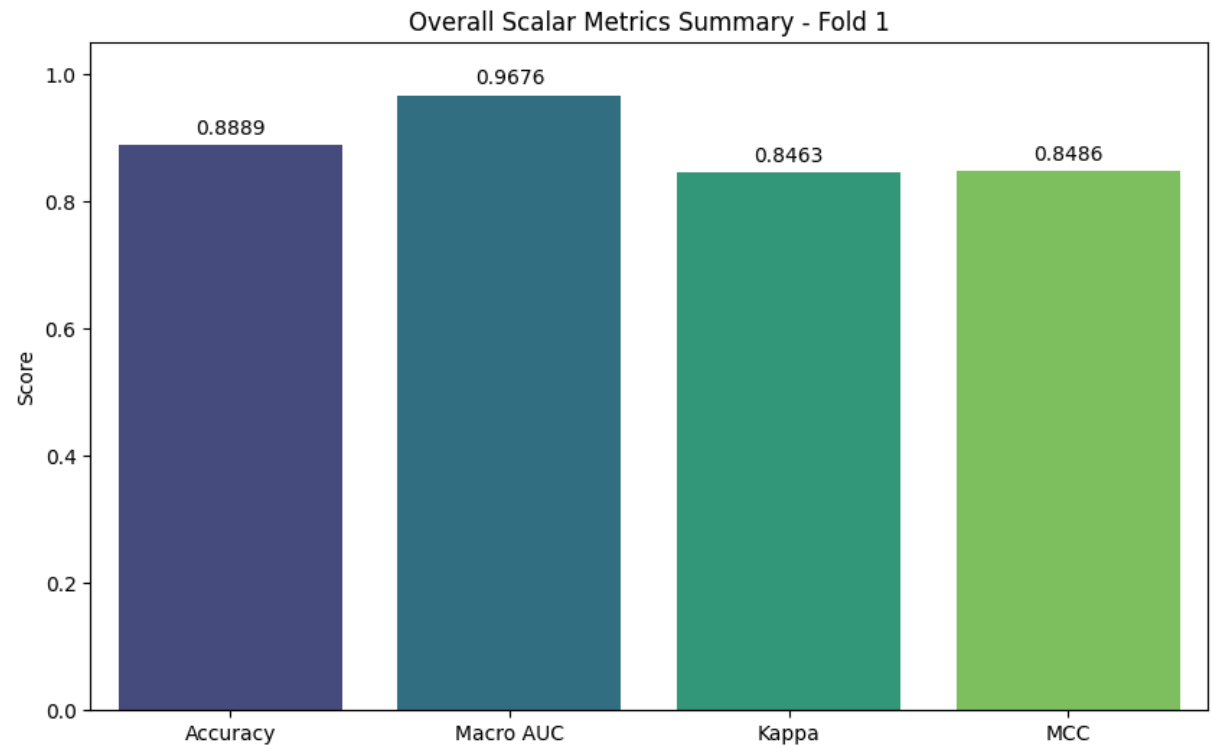
Matthews CorrCoef: 0.8486



Saved normalized confusion matrix to: new_results_Multimodal_ResNet50_Concat_KFold\confusion_matrix_fold_1.png



Saved ROC curves plot to: new_results_Multimodal_ResNet50_Concat_KFold\roc_curves_fold_1.png



Saved scalar metrics plot to: new_results_Multimodal_ResNet50_Concat_KFold\scalar_metrics_fold_1.png

=====

PERFORMANCE BY GENDER

=====

--- Subgroup: female ---

	precision	recall	f1-score	support
Herpes	1.00	0.67	0.80	3
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	2
Sarampion	0.92	0.92	0.92	12
Varicela	0.91	1.00	0.95	10
accuracy			0.93	27
macro avg	0.77	0.72	0.73	27
weighted avg	0.93	0.93	0.92	27

--- Subgroup: male ---

	precision	recall	f1-score	support
Herpes	1.00	0.75	0.86	4
Melanoma	0.75	1.00	0.86	3
Monkeypox	1.00	0.88	0.93	16
Sarampion	0.88	0.82	0.85	17
Varicela	0.76	0.93	0.84	14
accuracy			0.87	54
macro avg	0.88	0.88	0.87	54
weighted avg	0.89	0.87	0.87	54

=====

PERFORMANCE BY AGE

=====

--- Subgroup: 0-17 ---

	precision	recall	f1-score	support
Herpes	1.00	1.00	1.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	2
Sarampion	1.00	0.75	0.86	4
Varicela	0.80	1.00	0.89	4
accuracy			0.91	11
macro avg	0.76	0.75	0.75	11
weighted avg	0.93	0.91	0.91	11

--- Subgroup: 18-59 ---

	precision	recall	f1-score	support
Herpes	1.00	0.67	0.80	6
Melanoma	0.75	1.00	0.86	3
Monkeypox	1.00	0.88	0.93	16
Sarampion	0.88	0.88	0.88	25
Varicela	0.83	0.95	0.88	20
accuracy			0.89	70
macro avg	0.89	0.87	0.87	70
weighted avg	0.90	0.89	0.89	70

=====

PERFORMANCE BY SKIN TONE

=====

--- Subgroup: asian ---

	precision	recall	f1-score	support
Herpes	1.00	1.00	1.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	3
Sarampion	1.00	1.00	1.00	1
Varicela	1.00	1.00	1.00	4
accuracy			1.00	9
macro avg	0.80	0.80	0.80	9
weighted avg	1.00	1.00	1.00	9

--- Subgroup: black ---

	precision	recall	f1-score	support
Herpes	1.00	1.00	1.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	0.86	0.92	7
Sarampion	0.50	1.00	0.67	1
Varicela	0.00	0.00	0.00	0
accuracy			0.89	9
macro avg	0.50	0.57	0.52	9
weighted avg	0.94	0.89	0.90	9

--- Subgroup: indian ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	3
Sarampion	0.80	1.00	0.89	4
Varicela	0.00	0.00	0.00	0
accuracy			0.88	8

macro avg	0.36	0.40	0.38	8
weighted avg	0.78	0.88	0.82	8

--- Subgroup: latino hispanic ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	1
Sarampion	1.00	1.00	1.00	4
Varicela	0.00	0.00	0.00	0
accuracy			1.00	5
macro avg	0.40	0.40	0.40	5
weighted avg	1.00	1.00	1.00	5

--- Subgroup: middle eastern ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	1
Sarampion	0.00	0.00	0.00	0
Varicela	0.00	0.00	0.00	0
accuracy			1.00	1
macro avg	0.20	0.20	0.20	1
weighted avg	1.00	1.00	1.00	1

--- Subgroup: white ---

	precision	recall	f1-score	support
Herpes	1.00	0.75	0.86	4
Melanoma	0.75	1.00	0.86	3
Monkeypox	1.00	0.67	0.80	3
Sarampion	0.94	0.79	0.86	19
Varicela	0.79	0.95	0.86	20
accuracy			0.86	49
macro avg	0.90	0.83	0.85	49
weighted avg	0.88	0.86	0.86	49

```
=====
FAIRNESS SUMMARY (ALL GROUPS) | WGA: 0.5000 EOD: 0.8000
Worst pair: 0-17|female|asian vs 18-59|male|asian
=====
```

```
=====
FAIRNESS SUMMARY (FILTERED, n>=15) | WGA: 0.7826 EOD: 0.5018
Worst pair: 18-59|female|white vs 18-59|male|white
Groups: ['18-59|female|white', '18-59|male|white']
=====
```

```
#####  
####  
### FOLD 2 ###  
#####  
####
```

Unfreezing ResNet-50 layer4 for fine-tuning.

Total params: 26.43M | Trainable params: 17.89M (67.68%)

Using weighted CrossEntropyLoss for training.

START TRAINING (Resnet50 + Concat) for 200 epochs...

Epoch 1/200	Train Loss: 2.0138	Acc: 0.1830	Val Loss: 1.8259	Acc: 0.3882	LR: 2.00e-05
Epoch 2/200	Train Loss: 1.9134	Acc: 0.2381	Val Loss: 1.8249	Acc: 0.3765	LR: 3.00e-05
Epoch 3/200	Train Loss: 1.8663	Acc: 0.2907	Val Loss: 1.8234	Acc: 0.3765	LR: 4.00e-05
Epoch 4/200	Train Loss: 1.8084	Acc: 0.2957	Val Loss: 1.8167	Acc: 0.4000	LR: 5.00e-05
Epoch 5/200	Train Loss: 1.7647	Acc: 0.3584	Val Loss: 1.7985	Acc: 0.4471	LR: 6.00e-05
Epoch 6/200	Train Loss: 1.6472	Acc: 0.4311	Val Loss: 1.7664	Acc: 0.5765	LR: 7.00e-05
Epoch 7/200	Train Loss: 1.5073	Acc: 0.4185	Val Loss: 1.7185	Acc: 0.6353	LR: 8.00e-05
Epoch 8/200	Train Loss: 1.5175	Acc: 0.5038	Val Loss: 1.6598	Acc: 0.6941	LR: 9.00e-05
Epoch 9/200	Train Loss: 1.3580	Acc: 0.5714	Val Loss: 1.5946	Acc: 0.7059	LR: 1.00e-04
Epoch 10/200	Train Loss: 1.2994	Acc: 0.6090	Val Loss: 1.5337	Acc: 0.7294	LR: 1.00e-04
Epoch 11/200	Train Loss: 1.2648	Acc: 0.5965	Val Loss: 1.4721	Acc: 0.7176	LR: 1.00e-04
Epoch 12/200	Train Loss: 1.2273	Acc: 0.6541	Val Loss: 1.4202	Acc: 0.6941	LR: 1.00e-04
Epoch 13/200	Train Loss: 1.1598	Acc: 0.7018	Val Loss: 1.3874	Acc: 0.6706	LR: 9.99e-05
Epoch 14/200	Train Loss: 1.1135	Acc: 0.7068	Val Loss: 1.3564	Acc: 0.6706	LR: 9.99e-05
Epoch 15/200	Train Loss: 1.0672	Acc: 0.7494	Val Loss: 1.3387	Acc: 0.6941	LR: 9.98e-05
Epoch 16/200	Train Loss: 1.0706	Acc: 0.7318	Val Loss: 1.3274	Acc: 0.7176	LR: 9.98e-05
Epoch 17/200	Train Loss: 1.0553	Acc: 0.7719	Val Loss: 1.3168	Acc: 0.7294	LR: 9.97e-05
Epoch 18/200	Train Loss: 1.0135	Acc: 0.7870	Val Loss: 1.2947	Acc: 0.7294	LR: 9.96e-05
Epoch 19/200	Train Loss: 1.0036	Acc: 0.7895	Val Loss: 1.2640	Acc: 0.7412	LR: 9.94e-05
Epoch 20/200	Train Loss: 0.9792	Acc: 0.7970	Val Loss: 1.2666	Acc: 0.7412	LR: 9.93e-05
Epoch 21/200	Train Loss: 0.9628	Acc: 0.8221	Val Loss: 1.2610	Acc: 0.7412	LR: 9.92e-05
Epoch 22/200	Train Loss: 0.9943	Acc: 0.7845	Val Loss: 1.2568	Acc: 0.7529	LR: 9.90e-05
Epoch 23/200	Train Loss: 0.9474	Acc: 0.8145	Val Loss: 1.2401	Acc: 0.7529	LR:

9.88e-05
Epoch 24/200 | Train Loss: 0.9538 Acc: 0.8221 | Val Loss: 1.2346 Acc: 0.7647 | LR: 9.87e-05
Epoch 25/200 | Train Loss: 0.9263 Acc: 0.8446 | Val Loss: 1.2323 Acc: 0.7882 | LR: 9.85e-05
Epoch 26/200 | Train Loss: 0.9191 Acc: 0.8797 | Val Loss: 1.2187 Acc: 0.8000 | LR: 9.83e-05
Epoch 27/200 | Train Loss: 0.9043 Acc: 0.8722 | Val Loss: 1.2165 Acc: 0.7882 | LR: 9.80e-05
Epoch 28/200 | Train Loss: 0.8863 Acc: 0.8922 | Val Loss: 1.2064 Acc: 0.7882 | LR: 9.78e-05
Epoch 29/200 | Train Loss: 0.8932 Acc: 0.8722 | Val Loss: 1.2043 Acc: 0.8000 | LR: 9.76e-05
Epoch 30/200 | Train Loss: 0.8908 Acc: 0.8897 | Val Loss: 1.2161 Acc: 0.8118 | LR: 9.73e-05
Epoch 31/200 | Train Loss: 0.9039 Acc: 0.8972 | Val Loss: 1.1875 Acc: 0.8235 | LR: 9.70e-05
Epoch 32/200 | Train Loss: 0.8733 Acc: 0.8972 | Val Loss: 1.1668 Acc: 0.8353 | LR: 9.67e-05
Epoch 33/200 | Train Loss: 0.8878 Acc: 0.9373 | Val Loss: 1.1508 Acc: 0.8235 | LR: 9.64e-05
Epoch 34/200 | Train Loss: 0.8620 Acc: 0.8872 | Val Loss: 1.1436 Acc: 0.8353 | LR: 9.61e-05
Epoch 35/200 | Train Loss: 0.8524 Acc: 0.9298 | Val Loss: 1.1353 Acc: 0.8353 | LR: 9.58e-05
Epoch 36/200 | Train Loss: 0.8339 Acc: 0.9398 | Val Loss: 1.1406 Acc: 0.8235 | LR: 9.55e-05
Epoch 37/200 | Train Loss: 0.8521 Acc: 0.9173 | Val Loss: 1.1505 Acc: 0.8353 | LR: 9.51e-05
Epoch 38/200 | Train Loss: 0.8375 Acc: 0.9398 | Val Loss: 1.1437 Acc: 0.8471 | LR: 9.47e-05
Epoch 39/200 | Train Loss: 0.8380 Acc: 0.9398 | Val Loss: 1.1305 Acc: 0.8471 | LR: 9.44e-05
Epoch 40/200 | Train Loss: 0.8451 Acc: 0.9398 | Val Loss: 1.1321 Acc: 0.8471 | LR: 9.40e-05
Epoch 41/200 | Train Loss: 0.8274 Acc: 0.9449 | Val Loss: 1.1230 Acc: 0.8471 | LR: 9.36e-05
Epoch 42/200 | Train Loss: 0.8494 Acc: 0.9474 | Val Loss: 1.1168 Acc: 0.8353 | LR: 9.32e-05
Epoch 43/200 | Train Loss: 0.8247 Acc: 0.9474 | Val Loss: 1.1068 Acc: 0.8353 | LR: 9.27e-05
Epoch 44/200 | Train Loss: 0.8220 Acc: 0.9474 | Val Loss: 1.0765 Acc: 0.8353 | LR: 9.23e-05
Epoch 45/200 | Train Loss: 0.8250 Acc: 0.9449 | Val Loss: 1.0588 Acc: 0.8353 | LR: 9.19e-05
Epoch 46/200 | Train Loss: 0.8157 Acc: 0.9424 | Val Loss: 1.0561 Acc: 0.8588 | LR: 9.14e-05
Epoch 47/200 | Train Loss: 0.8283 Acc: 0.9624 | Val Loss: 1.0486 Acc: 0.8588 | LR: 9.09e-05
Epoch 48/200 | Train Loss: 0.7929 Acc: 0.9699 | Val Loss: 1.0496 Acc: 0.8706 | LR: 9.05e-05
Epoch 49/200 | Train Loss: 0.8191 Acc: 0.9549 | Val Loss: 1.0540 Acc: 0.8706 | LR: 9.00e-05
Epoch 50/200 | Train Loss: 0.8139 Acc: 0.9699 | Val Loss: 1.0442 Acc: 0.8706 | LR: 8.95e-05
Epoch 51/200 | Train Loss: 0.8115 Acc: 0.9699 | Val Loss: 1.0434 Acc: 0.8588 | LR:

8.89e-05
Epoch 52/200 | Train Loss: 0.8117 Acc: 0.9499 | Val Loss: 1.0414 Acc: 0.8824 | LR: 8.84e-05
Epoch 53/200 | Train Loss: 0.8115 Acc: 0.9599 | Val Loss: 1.0427 Acc: 0.8706 | LR: 8.79e-05
Epoch 54/200 | Train Loss: 0.7778 Acc: 0.9799 | Val Loss: 1.0506 Acc: 0.8706 | LR: 8.73e-05
Epoch 55/200 | Train Loss: 0.8035 Acc: 0.9699 | Val Loss: 1.0527 Acc: 0.8706 | LR: 8.68e-05
Epoch 56/200 | Train Loss: 0.7909 Acc: 0.9749 | Val Loss: 1.0596 Acc: 0.8588 | LR: 8.62e-05
Epoch 57/200 | Train Loss: 0.7932 Acc: 0.9699 | Val Loss: 1.0602 Acc: 0.8706 | LR: 8.56e-05
Epoch 58/200 | Train Loss: 0.8025 Acc: 0.9699 | Val Loss: 1.0626 Acc: 0.8588 | LR: 8.51e-05
Epoch 59/200 | Train Loss: 0.7998 Acc: 0.9674 | Val Loss: 1.0621 Acc: 0.8706 | LR: 8.45e-05
Epoch 60/200 | Train Loss: 0.7814 Acc: 0.9774 | Val Loss: 1.0522 Acc: 0.8706 | LR: 8.39e-05
Epoch 61/200 | Train Loss: 0.7993 Acc: 0.9774 | Val Loss: 1.0427 Acc: 0.8706 | LR: 8.33e-05
Epoch 62/200 | Train Loss: 0.7726 Acc: 0.9799 | Val Loss: 1.0465 Acc: 0.8941 | LR: 8.26e-05
Epoch 63/200 | Train Loss: 0.7712 Acc: 0.9774 | Val Loss: 1.0335 Acc: 0.8706 | LR: 8.20e-05
Epoch 64/200 | Train Loss: 0.7809 Acc: 0.9850 | Val Loss: 1.0317 Acc: 0.8706 | LR: 8.14e-05
Epoch 65/200 | Train Loss: 0.7772 Acc: 0.9850 | Val Loss: 1.0222 Acc: 0.8824 | LR: 8.07e-05
Epoch 66/200 | Train Loss: 0.7708 Acc: 0.9724 | Val Loss: 1.0175 Acc: 0.8824 | LR: 8.01e-05
Epoch 67/200 | Train Loss: 0.7813 Acc: 0.9699 | Val Loss: 1.0157 Acc: 0.8941 | LR: 7.94e-05
Epoch 68/200 | Train Loss: 0.7757 Acc: 0.9699 | Val Loss: 1.0332 Acc: 0.8824 | LR: 7.87e-05
Epoch 69/200 | Train Loss: 0.7675 Acc: 0.9925 | Val Loss: 1.0349 Acc: 0.8824 | LR: 7.80e-05
Epoch 70/200 | Train Loss: 0.7881 Acc: 0.9774 | Val Loss: 1.0393 Acc: 0.8706 | LR: 7.73e-05
Epoch 71/200 | Train Loss: 0.7669 Acc: 0.9850 | Val Loss: 1.0414 Acc: 0.8706 | LR: 7.67e-05
Epoch 72/200 | Train Loss: 0.7674 Acc: 0.9850 | Val Loss: 1.0360 Acc: 0.8824 | LR: 7.59e-05
Epoch 73/200 | Train Loss: 0.7677 Acc: 0.9799 | Val Loss: 1.0301 Acc: 0.8824 | LR: 7.52e-05
Epoch 74/200 | Train Loss: 0.7845 Acc: 0.9674 | Val Loss: 1.0225 Acc: 0.8824 | LR: 7.45e-05
Epoch 75/200 | Train Loss: 0.7572 Acc: 0.9900 | Val Loss: 1.0273 Acc: 0.8824 | LR: 7.38e-05
Epoch 76/200 | Train Loss: 0.7999 Acc: 0.9749 | Val Loss: 1.0327 Acc: 0.8824 | LR: 7.31e-05
Epoch 77/200 | Train Loss: 0.7793 Acc: 0.9925 | Val Loss: 1.0347 Acc: 0.8824 | LR: 7.23e-05
Epoch 78/200 | Train Loss: 0.7716 Acc: 0.9799 | Val Loss: 1.0319 Acc: 0.8824 | LR: 7.16e-05
Epoch 79/200 | Train Loss: 0.7758 Acc: 0.9825 | Val Loss: 1.0211 Acc: 0.8824 | LR:

7.08e-05
Epoch 80/200 | Train Loss: 0.7772 Acc: 0.9850 | Val Loss: 1.0216 Acc: 0.8706 | LR: 7.01e-05
Epoch 81/200 | Train Loss: 0.7641 Acc: 0.9900 | Val Loss: 1.0135 Acc: 0.8706 | LR: 6.93e-05
Epoch 82/200 | Train Loss: 0.7694 Acc: 0.9825 | Val Loss: 1.0292 Acc: 0.8824 | LR: 6.86e-05
Epoch 83/200 | Train Loss: 0.7555 Acc: 0.9925 | Val Loss: 1.0206 Acc: 0.8706 | LR: 6.78e-05
Epoch 84/200 | Train Loss: 0.7621 Acc: 0.9850 | Val Loss: 1.0176 Acc: 0.8941 | LR: 6.70e-05
Epoch 85/200 | Train Loss: 0.7669 Acc: 0.9875 | Val Loss: 1.0157 Acc: 0.8824 | LR: 6.62e-05
Epoch 86/200 | Train Loss: 0.7925 Acc: 0.9850 | Val Loss: 1.0086 Acc: 0.8941 | LR: 6.55e-05
Epoch 87/200 | Train Loss: 0.7670 Acc: 0.9825 | Val Loss: 1.0110 Acc: 0.8941 | LR: 6.47e-05
Epoch 88/200 | Train Loss: 0.7716 Acc: 0.9900 | Val Loss: 1.0112 Acc: 0.8941 | LR: 6.39e-05
Epoch 89/200 | Train Loss: 0.7839 Acc: 0.9825 | Val Loss: 1.0247 Acc: 0.8941 | LR: 6.31e-05
Epoch 90/200 | Train Loss: 0.7722 Acc: 0.9749 | Val Loss: 1.0255 Acc: 0.8941 | LR: 6.23e-05
Epoch 91/200 | Train Loss: 0.7747 Acc: 0.9900 | Val Loss: 1.0066 Acc: 0.8941 | LR: 6.15e-05
Epoch 92/200 | Train Loss: 0.7614 Acc: 0.9900 | Val Loss: 1.0013 Acc: 0.8941 | LR: 6.07e-05
Epoch 93/200 | Train Loss: 0.7713 Acc: 0.9850 | Val Loss: 1.0120 Acc: 0.8941 | LR: 5.99e-05
Epoch 94/200 | Train Loss: 0.7646 Acc: 0.9875 | Val Loss: 1.0074 Acc: 0.8941 | LR: 5.90e-05
Epoch 95/200 | Train Loss: 0.7658 Acc: 0.9875 | Val Loss: 1.0060 Acc: 0.8941 | LR: 5.82e-05
Epoch 96/200 | Train Loss: 0.7528 Acc: 0.9850 | Val Loss: 0.9959 Acc: 0.9059 | LR: 5.74e-05
Epoch 97/200 | Train Loss: 0.7557 Acc: 0.9799 | Val Loss: 0.9958 Acc: 0.8941 | LR: 5.66e-05
Epoch 98/200 | Train Loss: 0.7532 Acc: 0.9950 | Val Loss: 0.9942 Acc: 0.8941 | LR: 5.58e-05
Epoch 99/200 | Train Loss: 0.7592 Acc: 0.9900 | Val Loss: 0.9932 Acc: 0.8941 | LR: 5.50e-05
Epoch 100/200 | Train Loss: 0.7601 Acc: 0.9875 | Val Loss: 1.0024 Acc: 0.8941 | LR: 5.41e-05
Epoch 101/200 | Train Loss: 0.7726 Acc: 0.9850 | Val Loss: 0.9950 Acc: 0.8941 | LR: 5.33e-05
Epoch 102/200 | Train Loss: 0.7605 Acc: 0.9850 | Val Loss: 0.9928 Acc: 0.8941 | LR: 5.25e-05
Epoch 103/200 | Train Loss: 0.7657 Acc: 0.9825 | Val Loss: 0.9943 Acc: 0.8941 | LR: 5.17e-05
Epoch 104/200 | Train Loss: 0.7599 Acc: 0.9900 | Val Loss: 1.0096 Acc: 0.8941 | LR: 5.08e-05
Epoch 105/200 | Train Loss: 0.7587 Acc: 0.9850 | Val Loss: 1.0054 Acc: 0.8941 | LR: 5.00e-05
Epoch 106/200 | Train Loss: 0.7602 Acc: 0.9875 | Val Loss: 1.0049 Acc: 0.8941 | LR: 4.92e-05
Epoch 107/200 | Train Loss: 0.7481 Acc: 0.9925 | Val Loss: 1.0139 Acc: 0.8941 | LR:

4.83e-05

Epoch 108/200 | Train Loss: 0.7517 Acc: 0.9875 | Val Loss: 1.0160 Acc: 0.8941 | LR:

4.75e-05

Epoch 109/200 | Train Loss: 0.7672 Acc: 0.9825 | Val Loss: 1.0295 Acc: 0.8706 | LR:

4.67e-05

Epoch 110/200 | Train Loss: 0.7634 Acc: 0.9950 | Val Loss: 1.0274 Acc: 0.8706 | LR:

4.59e-05

Epoch 111/200 | Train Loss: 0.7412 Acc: 0.9925 | Val Loss: 1.0271 Acc: 0.8706 | LR:

4.50e-05

Epoch 112/200 | Train Loss: 0.7617 Acc: 0.9850 | Val Loss: 1.0218 Acc: 0.8706 | LR:

4.42e-05

Epoch 113/200 | Train Loss: 0.7513 Acc: 0.9850 | Val Loss: 1.0266 Acc: 0.8824 | LR:

4.34e-05

Epoch 114/200 | Train Loss: 0.7539 Acc: 0.9950 | Val Loss: 1.0241 Acc: 0.8588 | LR:

4.26e-05

Epoch 115/200 | Train Loss: 0.7420 Acc: 0.9975 | Val Loss: 1.0157 Acc: 0.8706 | LR:

4.18e-05

Epoch 116/200 | Train Loss: 0.7579 Acc: 0.9875 | Val Loss: 1.0169 Acc: 0.8588 | LR:

4.10e-05

Epoch 117/200 | Train Loss: 0.7491 Acc: 0.9900 | Val Loss: 1.0229 Acc: 0.8471 | LR:

4.01e-05

Epoch 118/200 | Train Loss: 0.7541 Acc: 0.9925 | Val Loss: 1.0163 Acc: 0.8588 | LR:

3.93e-05

Epoch 119/200 | Train Loss: 0.7631 Acc: 0.9925 | Val Loss: 1.0071 Acc: 0.8588 | LR:

3.85e-05

Epoch 120/200 | Train Loss: 0.7551 Acc: 0.9900 | Val Loss: 1.0029 Acc: 0.8706 | LR:

3.77e-05

Epoch 121/200 | Train Loss: 0.7432 Acc: 0.9900 | Val Loss: 1.0091 Acc: 0.8706 | LR:

3.69e-05

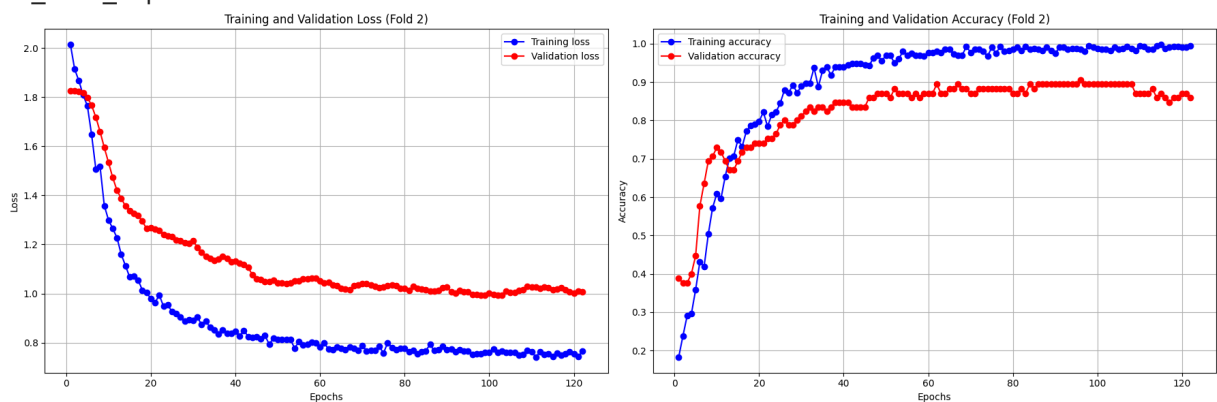
Epoch 122/200 | Train Loss: 0.7644 Acc: 0.9950 | Val Loss: 1.0067 Acc: 0.8588 | LR:

3.61e-05

Early stopping triggered.

Training complete in 8.9 minutes. Best Val Loss: 0.9928

Best model for fold saved to: new_results_Multimodal_ResNet50_Concat_KFold\best_model_fold_2.pth



Saved training history plot to: new_results_Multimodal_ResNet50_Concat_KFold\training_history_fold_2.png

=====

OVERALL PERFORMANCE (FOLD 2)

=====

	precision	recall	f1-score	support
Herpes	0.78	1.00	0.88	7
Melanoma	1.00	1.00	1.00	3
Monkeypox	0.94	0.83	0.88	18
Sarampion	0.92	0.83	0.87	29
Varicela	0.85	0.96	0.90	24
accuracy			0.89	81
macro avg	0.90	0.92	0.91	81
weighted avg	0.90	0.89	0.89	81

--- Overall Scalar Metrics ---

Accuracy: 0.8889

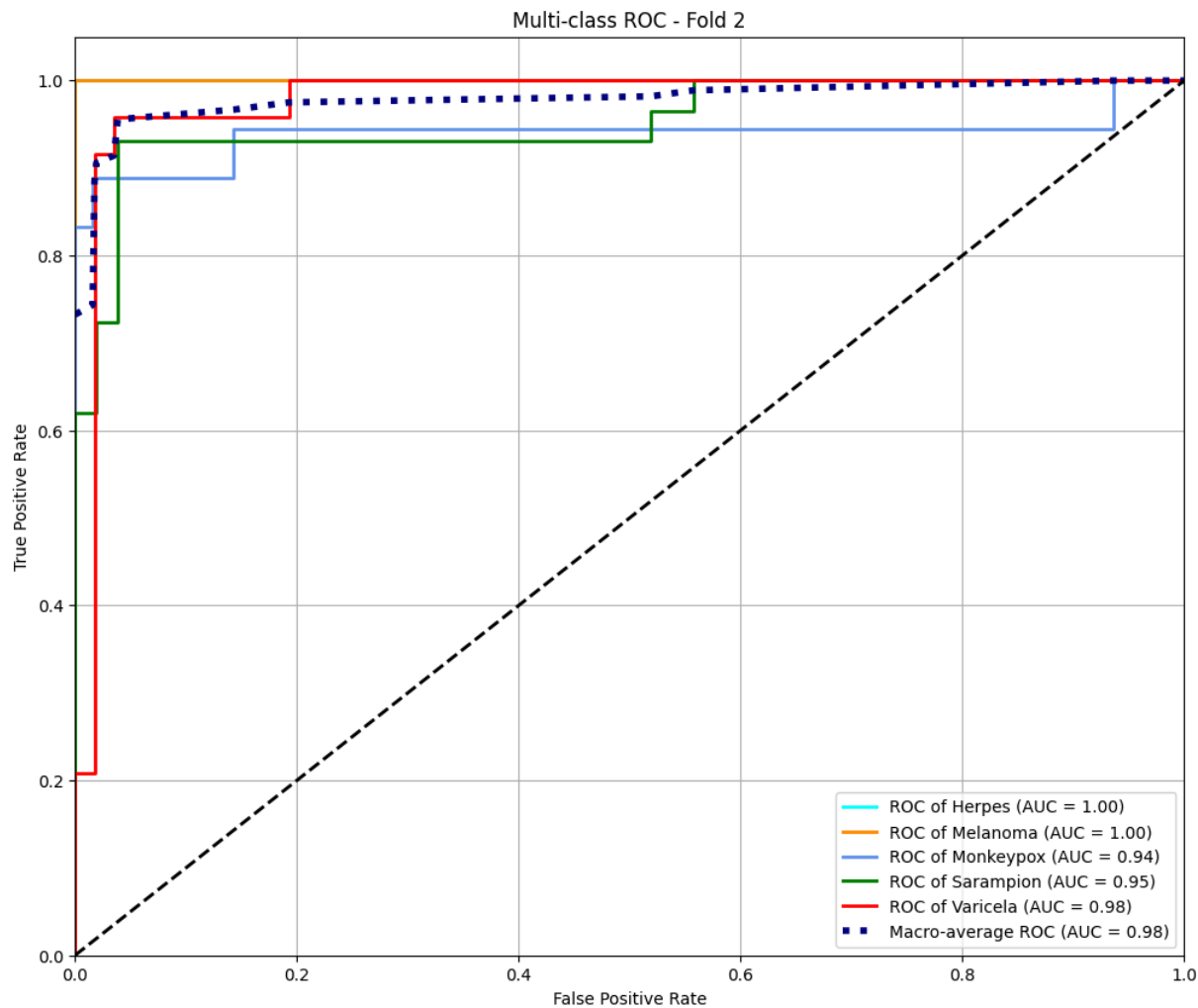
Macro ROC AUC: 0.9740

Cohen's Kappa: 0.8481

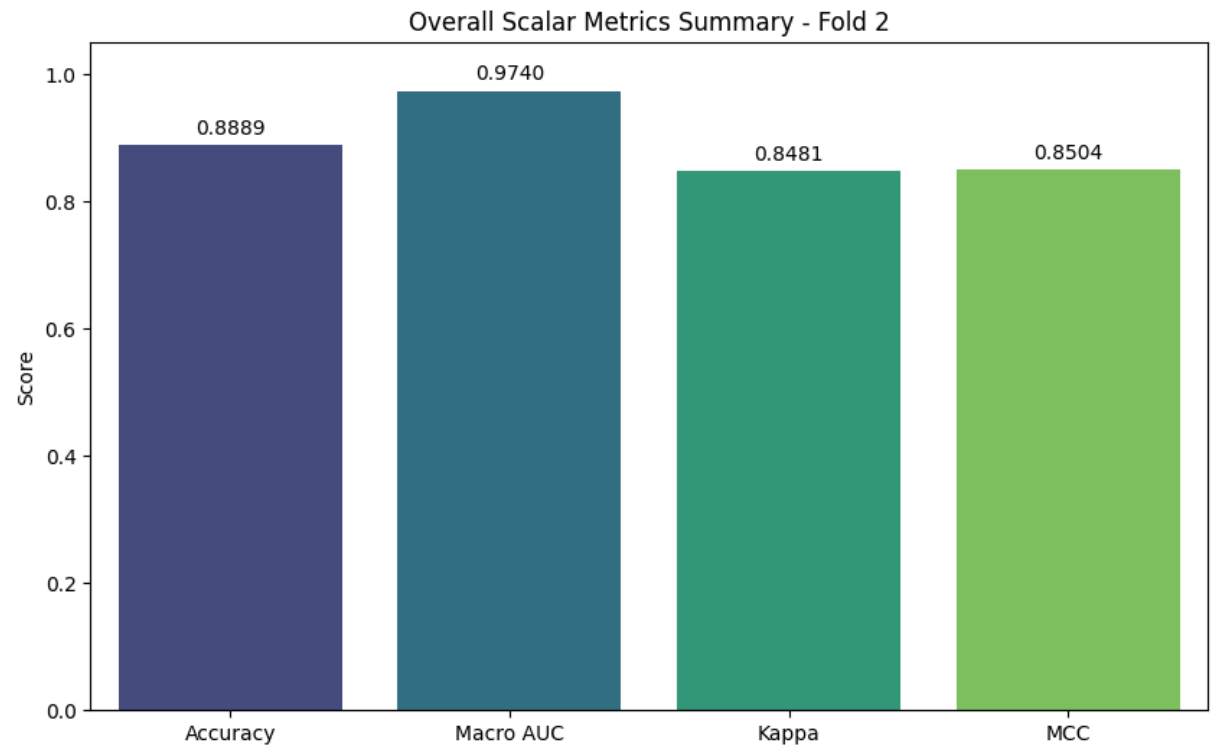
Matthews CorrCoef: 0.8504



Saved normalized confusion matrix to: new_results_Multimodal_ResNet50_Concat_KFold\confusion_matrix_fold_2.png



Saved ROC curves plot to: new_results_Multimodal_ResNet50_Concat_KFold\roc_curves_fold_2.png



Saved scalar metrics plot to: new_results_Multimodal_ResNet50_Concat_KFold\scalar_metrics_fold_2.png

=====

PERFORMANCE BY GENDER

=====

--- Subgroup: female ---

	precision	recall	f1-score	support
Herpes	1.00	1.00	1.00	2
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	0.50	0.67	4
Sarampion	0.93	0.93	0.93	14
Varicela	0.87	1.00	0.93	13
accuracy			0.91	33
macro avg	0.76	0.69	0.70	33
weighted avg	0.92	0.91	0.90	33

--- Subgroup: male ---

	precision	recall	f1-score	support
Herpes	0.71	1.00	0.83	5
Melanoma	1.00	1.00	1.00	3
Monkeypox	0.93	0.93	0.93	14
Sarampion	0.92	0.73	0.81	15
Varicela	0.83	0.91	0.87	11
accuracy			0.88	48
macro avg	0.88	0.91	0.89	48
weighted avg	0.89	0.88	0.87	48

=====

PERFORMANCE BY AGE

=====

--- Subgroup: 0-17 ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.00	0.00	0.00	1
Sarampion	1.00	1.00	1.00	3
Varicela	0.80	1.00	0.89	4
accuracy			0.88	8
macro avg	0.36	0.40	0.38	8
weighted avg	0.78	0.88	0.82	8

--- Subgroup: 18-59 ---

	precision	recall	f1-score	support
Herpes	0.75	1.00	0.86	6
Melanoma	1.00	1.00	1.00	3
Monkeypox	0.94	0.88	0.91	17
Sarampion	0.91	0.81	0.86	26
Varicela	0.86	0.95	0.90	20
accuracy			0.89	72
macro avg	0.89	0.93	0.91	72
weighted avg	0.90	0.89	0.89	72

--- Subgroup: 60+ ---

	precision	recall	f1-score	support
Herpes	1.00	1.00	1.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.00	0.00	0.00	0
Sarampion	0.00	0.00	0.00	0
Varicela	0.00	0.00	0.00	0
accuracy			1.00	1
macro avg	0.20	0.20	0.20	1
weighted avg	1.00	1.00	1.00	1

=====

PERFORMANCE BY SKIN TONE

=====

--- Subgroup: asian ---

	precision	recall	f1-score	support
Herpes	1.00	1.00	1.00	3
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	4
Sarampion	1.00	0.50	0.67	2
Varicela	0.67	1.00	0.80	2
accuracy			0.91	11
macro avg	0.73	0.70	0.69	11
weighted avg	0.94	0.91	0.90	11

--- Subgroup: black ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	2
Sarampion	0.00	0.00	0.00	0
Varicela	0.00	0.00	0.00	0
accuracy			1.00	2

macro avg	0.20	0.20	0.20	2
weighted avg	1.00	1.00	1.00	2

--- Subgroup: indian ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	2
Sarampion	1.00	1.00	1.00	2
Varicela	0.00	0.00	0.00	0
accuracy			1.00	4
macro avg	0.40	0.40	0.40	4
weighted avg	1.00	1.00	1.00	4

--- Subgroup: latino hispanic ---

	precision	recall	f1-score	support
Herpes	1.00	1.00	1.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	0.67	0.80	3
Sarampion	0.88	1.00	0.93	7
Varicela	0.00	0.00	0.00	0
accuracy			0.91	11
macro avg	0.57	0.53	0.55	11
weighted avg	0.92	0.91	0.90	11

--- Subgroup: middle eastern ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	3
Sarampion	0.00	0.00	0.00	1
Varicela	0.00	0.00	0.00	1
accuracy			0.60	5
macro avg	0.20	0.20	0.20	5
weighted avg	0.60	0.60	0.60	5

--- Subgroup: white ---

	precision	recall	f1-score	support
Herpes	0.60	1.00	0.75	3
Melanoma	1.00	1.00	1.00	3
Monkeypox	0.67	0.50	0.57	4
Sarampion	1.00	0.82	0.90	17
Varicela	0.91	1.00	0.95	21
accuracy			0.90	48

macro avg	0.84	0.86	0.84	48
weighted avg	0.91	0.90	0.89	48

=====

FAIRNESS SUMMARY (ALL GROUPS) | WGA: 0.6000 EOD: 0.8095

Worst pair: 18-59|female|indian vs 18-59|male|white

=====

=====

FAIRNESS SUMMARY (FILTERED, n>=15) | WGA: 0.8750 EOD: 0.5683

Worst pair: 18-59|female|white vs 18-59|male|white

Groups: ['18-59|female|white', '18-59|male|white']

=====

#####

####

FOLD 3

#####

####

Unfreezing ResNet-50 layer4 for fine-tuning.

Total params: 26.43M | Trainable params: 17.89M (67.68%)

Using weighted CrossEntropyLoss for training.

START TRAINING (Resnet50 + Concat) for 200 epochs...

Epoch 1/200 | Train Loss: 1.9137 Acc: 0.2080 | Val Loss: 1.8198 Acc: 0.1765 | LR: 2.00e-05

Epoch 2/200 | Train Loss: 1.8628 Acc: 0.1805 | Val Loss: 1.8176 Acc: 0.2824 | LR: 3.00e-05

Epoch 3/200 | Train Loss: 1.8970 Acc: 0.2381 | Val Loss: 1.8087 Acc: 0.2588 | LR: 4.00e-05

Epoch 4/200 | Train Loss: 1.8066 Acc: 0.2381 | Val Loss: 1.7931 Acc: 0.4000 | LR: 5.00e-05

Epoch 5/200 | Train Loss: 1.6879 Acc: 0.3208 | Val Loss: 1.7601 Acc: 0.5647 | LR: 6.00e-05

Epoch 6/200 | Train Loss: 1.6741 Acc: 0.2982 | Val Loss: 1.7072 Acc: 0.7176 | LR: 7.00e-05

Epoch 7/200 | Train Loss: 1.5797 Acc: 0.3709 | Val Loss: 1.6300 Acc: 0.7294 | LR: 8.00e-05

Epoch 8/200 | Train Loss: 1.5076 Acc: 0.4586 | Val Loss: 1.5410 Acc: 0.7412 | LR: 9.00e-05

Epoch 9/200 | Train Loss: 1.4133 Acc: 0.5138 | Val Loss: 1.4395 Acc: 0.7176 | LR: 1.00e-04

Epoch 10/200 | Train Loss: 1.3314 Acc: 0.5714 | Val Loss: 1.3432 Acc: 0.7529 | LR: 1.00e-04

Epoch 11/200 | Train Loss: 1.2915 Acc: 0.5965 | Val Loss: 1.2621 Acc: 0.7412 | LR: 1.00e-04

Epoch 12/200 | Train Loss: 1.2380 Acc: 0.6165 | Val Loss: 1.2016 Acc: 0.7412 | LR: 1.00e-04

Epoch 13/200 | Train Loss: 1.2201 Acc: 0.6642 | Val Loss: 1.1583 Acc: 0.7647 | LR: 9.99e-05

Epoch 14/200 | Train Loss: 1.1627 Acc: 0.6516 | Val Loss: 1.1288 Acc: 0.7765 | LR: 9.99e-05

Epoch 15/200 | Train Loss: 1.1282 Acc: 0.6717 | Val Loss: 1.1105 Acc: 0.7765 | LR: 9.98e-05

Epoch 16/200 | Train Loss: 1.1467 Acc: 0.6817 | Val Loss: 1.0933 Acc: 0.7882 | LR:

9.98e-05
Epoch 17/200 | Train Loss: 1.1106 Acc: 0.7118 | Val Loss: 1.0760 Acc: 0.8000 | LR: 9.97e-05
Epoch 18/200 | Train Loss: 1.0529 Acc: 0.7293 | Val Loss: 1.0571 Acc: 0.8235 | LR: 9.96e-05
Epoch 19/200 | Train Loss: 1.0508 Acc: 0.7218 | Val Loss: 1.0456 Acc: 0.8000 | LR: 9.94e-05
Epoch 20/200 | Train Loss: 0.9854 Acc: 0.7920 | Val Loss: 1.0315 Acc: 0.8235 | LR: 9.93e-05
Epoch 21/200 | Train Loss: 1.0157 Acc: 0.8020 | Val Loss: 1.0222 Acc: 0.8471 | LR: 9.92e-05
Epoch 22/200 | Train Loss: 0.9817 Acc: 0.7945 | Val Loss: 1.0108 Acc: 0.8471 | LR: 9.90e-05
Epoch 23/200 | Train Loss: 0.9811 Acc: 0.8221 | Val Loss: 0.9984 Acc: 0.8706 | LR: 9.88e-05
Epoch 24/200 | Train Loss: 0.9541 Acc: 0.8321 | Val Loss: 0.9905 Acc: 0.8588 | LR: 9.87e-05
Epoch 25/200 | Train Loss: 0.9639 Acc: 0.8070 | Val Loss: 0.9872 Acc: 0.8471 | LR: 9.85e-05
Epoch 26/200 | Train Loss: 0.9217 Acc: 0.8647 | Val Loss: 0.9788 Acc: 0.8471 | LR: 9.83e-05
Epoch 27/200 | Train Loss: 0.9500 Acc: 0.8471 | Val Loss: 0.9708 Acc: 0.8588 | LR: 9.80e-05
Epoch 28/200 | Train Loss: 0.9101 Acc: 0.8847 | Val Loss: 0.9648 Acc: 0.8471 | LR: 9.78e-05
Epoch 29/200 | Train Loss: 0.9256 Acc: 0.8797 | Val Loss: 0.9604 Acc: 0.8588 | LR: 9.76e-05
Epoch 30/200 | Train Loss: 0.9016 Acc: 0.8922 | Val Loss: 0.9534 Acc: 0.8471 | LR: 9.73e-05
Epoch 31/200 | Train Loss: 0.9096 Acc: 0.8872 | Val Loss: 0.9594 Acc: 0.8471 | LR: 9.70e-05
Epoch 32/200 | Train Loss: 0.8705 Acc: 0.9073 | Val Loss: 0.9570 Acc: 0.8471 | LR: 9.67e-05
Epoch 33/200 | Train Loss: 0.8787 Acc: 0.8972 | Val Loss: 0.9508 Acc: 0.8588 | LR: 9.64e-05
Epoch 34/200 | Train Loss: 0.8685 Acc: 0.9248 | Val Loss: 0.9455 Acc: 0.8588 | LR: 9.61e-05
Epoch 35/200 | Train Loss: 0.8567 Acc: 0.9023 | Val Loss: 0.9418 Acc: 0.8588 | LR: 9.58e-05
Epoch 36/200 | Train Loss: 0.8540 Acc: 0.9248 | Val Loss: 0.9326 Acc: 0.8706 | LR: 9.55e-05
Epoch 37/200 | Train Loss: 0.8704 Acc: 0.9173 | Val Loss: 0.9248 Acc: 0.8706 | LR: 9.51e-05
Epoch 38/200 | Train Loss: 0.8423 Acc: 0.9348 | Val Loss: 0.9243 Acc: 0.8706 | LR: 9.47e-05
Epoch 39/200 | Train Loss: 0.8487 Acc: 0.9398 | Val Loss: 0.9251 Acc: 0.8706 | LR: 9.44e-05
Epoch 40/200 | Train Loss: 0.8457 Acc: 0.9298 | Val Loss: 0.9191 Acc: 0.9059 | LR: 9.40e-05
Epoch 41/200 | Train Loss: 0.8608 Acc: 0.9248 | Val Loss: 0.9154 Acc: 0.8941 | LR: 9.36e-05
Epoch 42/200 | Train Loss: 0.8378 Acc: 0.9424 | Val Loss: 0.9185 Acc: 0.8824 | LR: 9.32e-05
Epoch 43/200 | Train Loss: 0.8321 Acc: 0.9348 | Val Loss: 0.9269 Acc: 0.8941 | LR: 9.27e-05
Epoch 44/200 | Train Loss: 0.8312 Acc: 0.9348 | Val Loss: 0.9286 Acc: 0.8941 | LR:

9.23e-05

Epoch 45/200 | Train Loss: 0.8296 Acc: 0.9424 | Val Loss: 0.9297 Acc: 0.8941 | LR: 9.19e-05

Epoch 46/200 | Train Loss: 0.8145 Acc: 0.9424 | Val Loss: 0.9295 Acc: 0.9059 | LR: 9.14e-05

Epoch 47/200 | Train Loss: 0.8084 Acc: 0.9424 | Val Loss: 0.9281 Acc: 0.8941 | LR: 9.09e-05

Epoch 48/200 | Train Loss: 0.8130 Acc: 0.9599 | Val Loss: 0.9219 Acc: 0.8941 | LR: 9.05e-05

Epoch 49/200 | Train Loss: 0.8206 Acc: 0.9424 | Val Loss: 0.9214 Acc: 0.9059 | LR: 9.00e-05

Epoch 50/200 | Train Loss: 0.8450 Acc: 0.9148 | Val Loss: 0.9224 Acc: 0.9059 | LR: 8.95e-05

Epoch 51/200 | Train Loss: 0.8386 Acc: 0.9398 | Val Loss: 0.9248 Acc: 0.9059 | LR: 8.89e-05

Epoch 52/200 | Train Loss: 0.8049 Acc: 0.9474 | Val Loss: 0.9243 Acc: 0.9176 | LR: 8.84e-05

Epoch 53/200 | Train Loss: 0.8043 Acc: 0.9624 | Val Loss: 0.9280 Acc: 0.9176 | LR: 8.79e-05

Epoch 54/200 | Train Loss: 0.8039 Acc: 0.9574 | Val Loss: 0.9288 Acc: 0.9059 | LR: 8.73e-05

Epoch 55/200 | Train Loss: 0.7892 Acc: 0.9599 | Val Loss: 0.9250 Acc: 0.8941 | LR: 8.68e-05

Epoch 56/200 | Train Loss: 0.8231 Acc: 0.9574 | Val Loss: 0.9228 Acc: 0.9059 | LR: 8.62e-05

Epoch 57/200 | Train Loss: 0.8063 Acc: 0.9674 | Val Loss: 0.9212 Acc: 0.8941 | LR: 8.56e-05

Epoch 58/200 | Train Loss: 0.8031 Acc: 0.9624 | Val Loss: 0.9190 Acc: 0.8941 | LR: 8.51e-05

Epoch 59/200 | Train Loss: 0.7846 Acc: 0.9724 | Val Loss: 0.9202 Acc: 0.8941 | LR: 8.45e-05

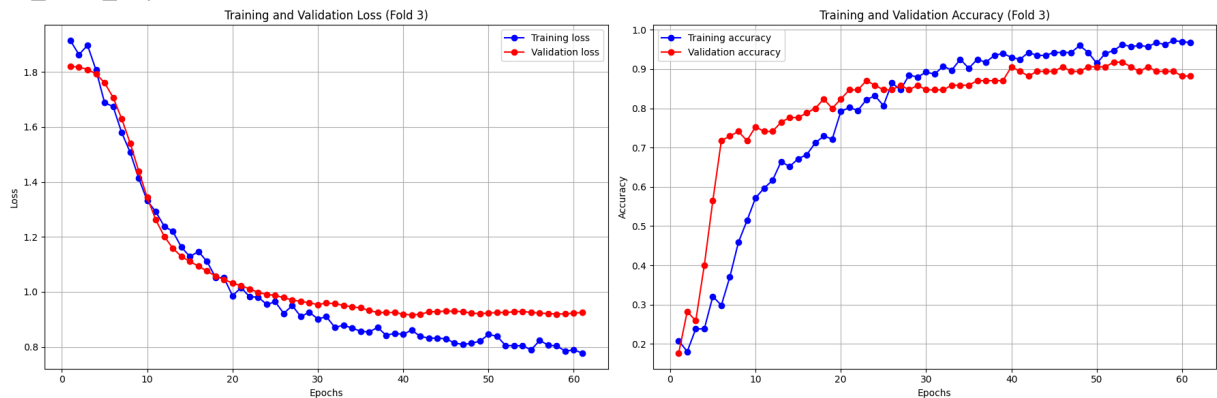
Epoch 60/200 | Train Loss: 0.7887 Acc: 0.9699 | Val Loss: 0.9229 Acc: 0.8824 | LR: 8.39e-05

Epoch 61/200 | Train Loss: 0.7773 Acc: 0.9674 | Val Loss: 0.9249 Acc: 0.8824 | LR: 8.33e-05

Early stopping triggered.

Training complete in 4.6 minutes. Best Val Loss: 0.9154

Best model for fold saved to: new_results_Multimodal_ResNet50_Concat_KFold\best_model_fold_3.pth



Saved training history plot to: new_results_Multimodal_ResNet50_Concat_KFold\training_history_fold_3.png

=====

OVERALL PERFORMANCE (FOLD 3)

=====

	precision	recall	f1-score	support
Herpes	0.60	0.86	0.71	7
Melanoma	1.00	0.67	0.80	3
Monkeypox	0.94	0.89	0.91	18
Sarampion	0.93	0.93	0.93	29
Varicela	0.96	0.92	0.94	24
accuracy			0.90	81
macro avg	0.89	0.85	0.86	81
weighted avg	0.91	0.90	0.90	81

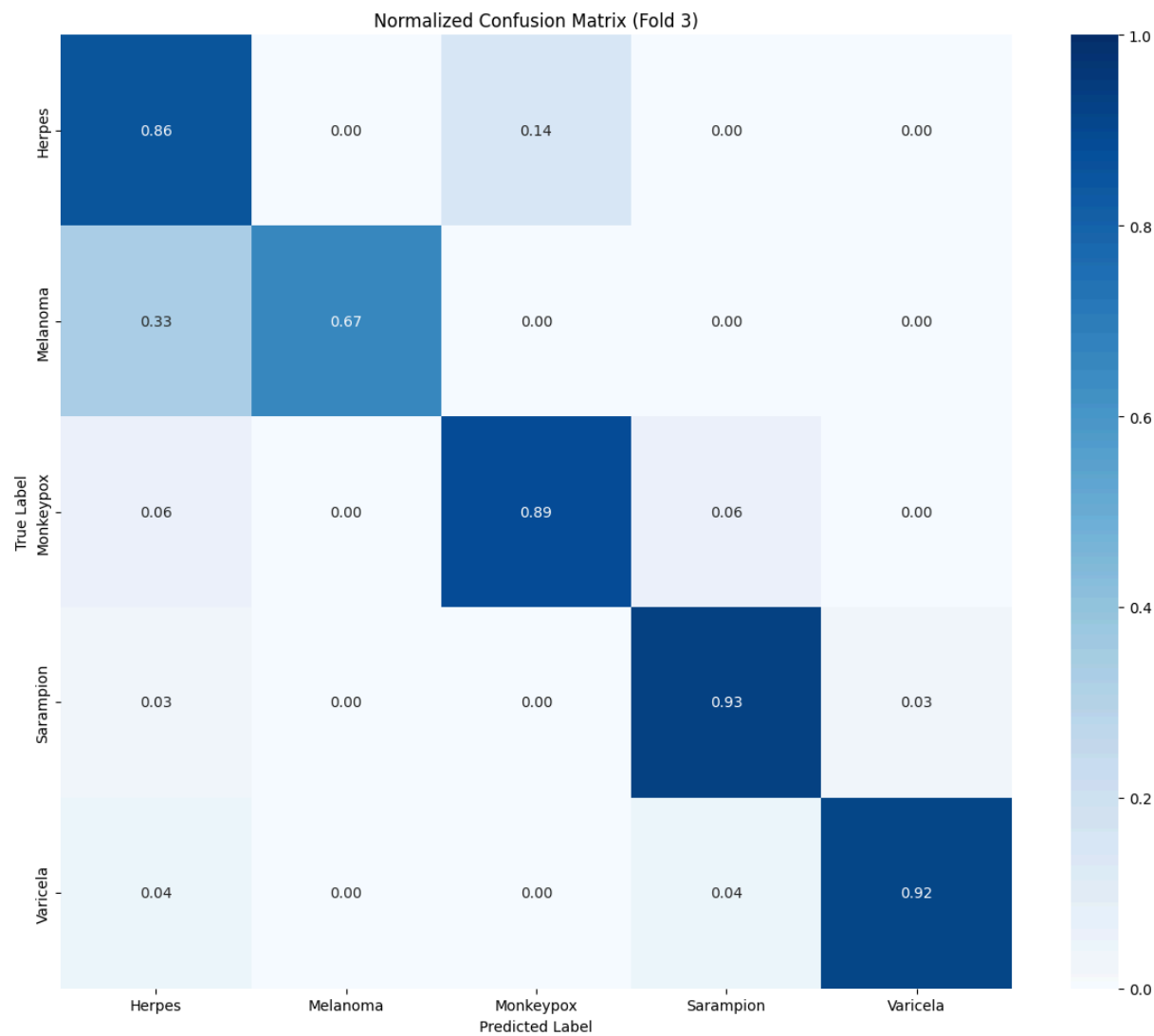
--- Overall Scalar Metrics ---

Accuracy: 0.9012

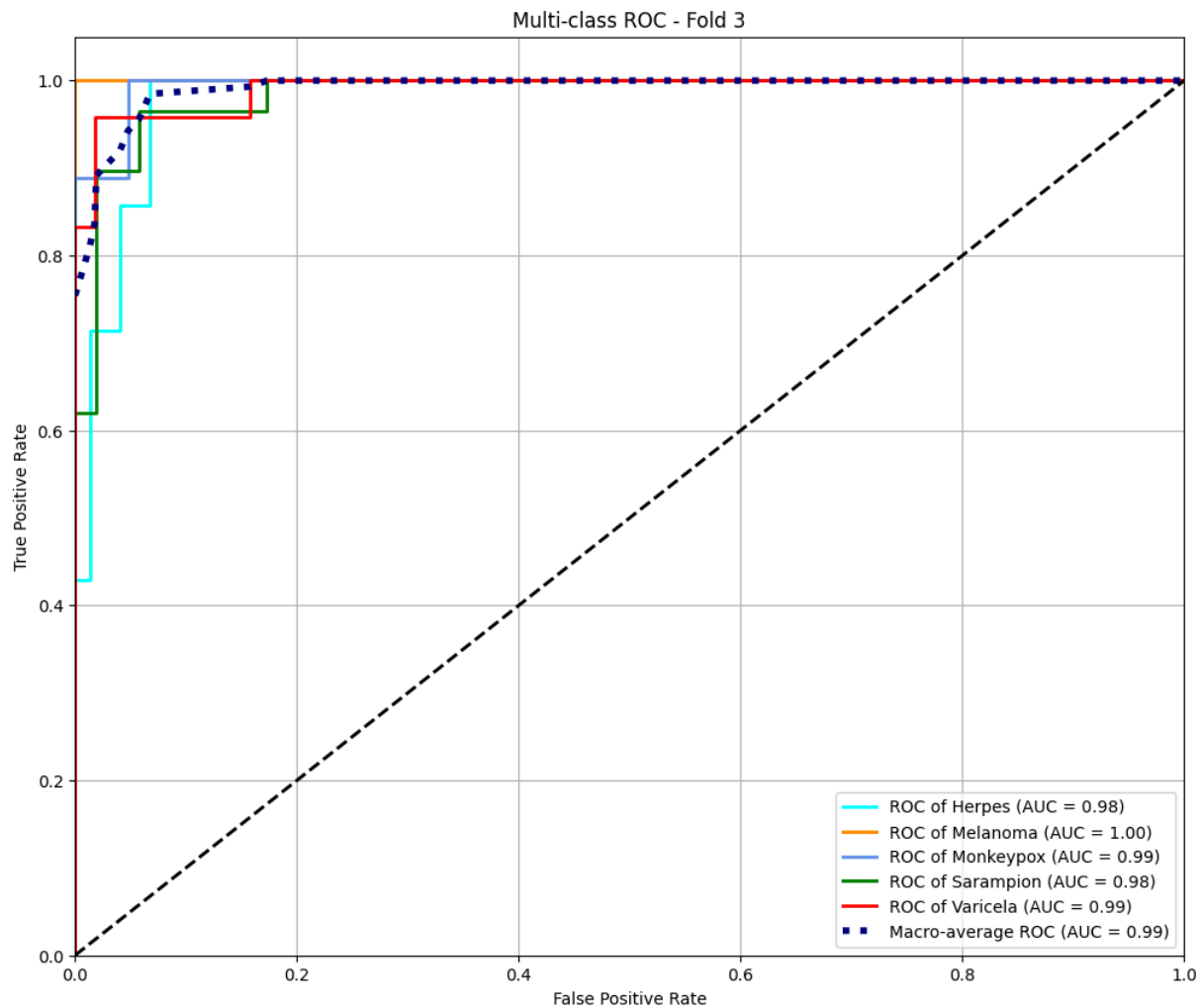
Macro ROC AUC: 0.9903

Cohen's Kappa: 0.8646

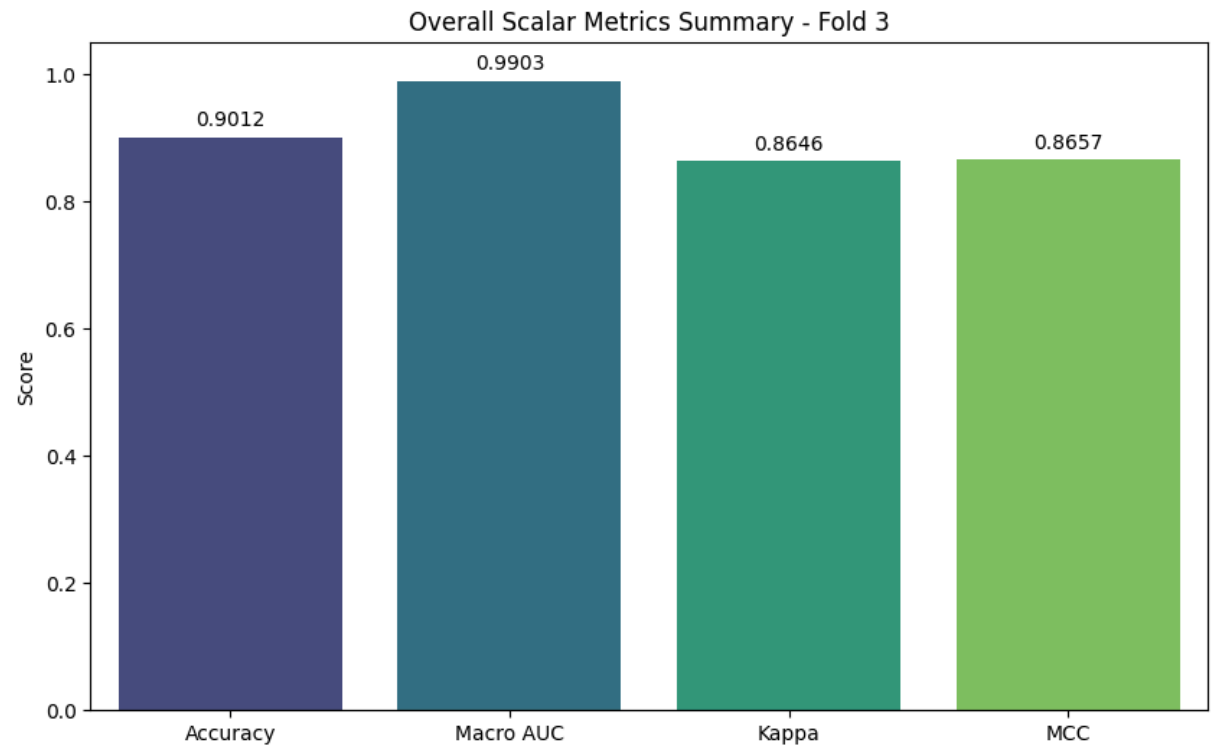
Matthews CorrCoef: 0.8657



Saved normalized confusion matrix to: new_results_Multimodal_ResNet50_Concat_KFold\confusion_matrix_fold_3.png



Saved ROC curves plot to: new_results_Multimodal_ResNet50_Concat_KFold\roc_curves_fold_3.png



Saved scalar metrics plot to: new_results_Multimodal_ResNet50_Concat_KFold\scalar_metrics_fold_3.png

=====

PERFORMANCE BY GENDER

=====

--- Subgroup: female ---

	precision	recall	f1-score	support
Herpes	0.33	1.00	0.50	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	3
Sarampion	0.95	0.91	0.93	22
Varicela	0.89	0.80	0.84	10
accuracy			0.89	36
macro avg	0.63	0.74	0.65	36
weighted avg	0.92	0.89	0.90	36

--- Subgroup: male ---

	precision	recall	f1-score	support
Herpes	0.71	0.83	0.77	6
Melanoma	1.00	0.67	0.80	3
Monkeypox	0.93	0.87	0.90	15
Sarampion	0.88	1.00	0.93	7
Varicela	1.00	1.00	1.00	14
accuracy			0.91	45
macro avg	0.90	0.87	0.88	45
weighted avg	0.92	0.91	0.91	45

=====

PERFORMANCE BY AGE

=====

--- Subgroup: 0-17 ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.00	0.00	0.00	0
Sarampion	1.00	0.67	0.80	3
Varicela	1.00	1.00	1.00	5
accuracy			0.88	8
macro avg	0.40	0.33	0.36	8
weighted avg	1.00	0.88	0.93	8

--- Subgroup: 18-59 ---

	precision	recall	f1-score	support
Herpes	0.67	0.86	0.75	7
Melanoma	1.00	0.67	0.80	3
Monkeypox	0.94	0.89	0.91	18
Sarampion	0.93	0.96	0.94	26
Varicela	0.94	0.89	0.92	19
accuracy			0.90	73
macro avg	0.90	0.85	0.87	73
weighted avg	0.91	0.90	0.91	73

=====

PERFORMANCE BY SKIN TONE

=====

--- Subgroup: asian ---

	precision	recall	f1-score	support
Herpes	1.00	0.50	0.67	2
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.75	1.00	0.86	3
Sarampion	1.00	1.00	1.00	4
Varicela	1.00	1.00	1.00	2
accuracy			0.91	11
macro avg	0.75	0.70	0.70	11
weighted avg	0.93	0.91	0.90	11

--- Subgroup: black ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	3
Sarampion	0.00	0.00	0.00	0
Varicela	0.00	0.00	0.00	0
accuracy			1.00	3
macro avg	0.20	0.20	0.20	3
weighted avg	1.00	1.00	1.00	3

--- Subgroup: indian ---

	precision	recall	f1-score	support
Herpes	0.50	1.00	0.67	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	4
Sarampion	0.00	0.00	0.00	1
Varicela	0.00	0.00	0.00	0
accuracy			0.83	6

macro avg	0.30	0.40	0.33	6
weighted avg	0.75	0.83	0.78	6

--- Subgroup: latino hispanic ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	1
Monkeypox	1.00	1.00	1.00	2
Sarampion	1.00	0.86	0.92	7
Varicela	0.00	0.00	0.00	0
accuracy			0.80	10
macro avg	0.40	0.37	0.38	10
weighted avg	0.90	0.80	0.85	10

--- Subgroup: middle eastern ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	1.00	1.00	1.00	1
Monkeypox	1.00	0.50	0.67	2
Sarampion	0.50	1.00	0.67	1
Varicela	0.00	0.00	0.00	0
accuracy			0.75	4
macro avg	0.50	0.50	0.47	4
weighted avg	0.88	0.75	0.75	4

--- Subgroup: white ---

	precision	recall	f1-score	support
Herpes	0.67	1.00	0.80	4
Melanoma	1.00	1.00	1.00	1
Monkeypox	1.00	0.75	0.86	4
Sarampion	0.94	1.00	0.97	16
Varicela	1.00	0.91	0.95	22
accuracy			0.94	47
macro avg	0.92	0.93	0.92	47
weighted avg	0.95	0.94	0.94	47

```
=====
FAIRNESS SUMMARY (ALL GROUPS) | WGA: 0.0000 EOD: 0.9500
Worst pair: 0-17|female|indian vs 18-59|male|white
=====
```

```
=====
FAIRNESS SUMMARY (FILTERED, n>=15) | WGA: 0.9000 EOD: 0.4167
Worst pair: 18-59|female|white vs 18-59|male|white
Groups: ['18-59|female|white', '18-59|male|white']
=====
```

```
#####  
####  
### FOLD 4 ###  
#####  
####
```

Unfreezing ResNet-50 layer4 for fine-tuning.

Total params: 26.43M | Trainable params: 17.89M (67.68%)

Using weighted CrossEntropyLoss for training.

START TRAINING (Resnet50 + Concat) for 200 epochs...

Epoch 1/200 | Train Loss: 1.8814 Acc: 0.1930 | Val Loss: 1.8095 Acc: 0.0588 | LR: 2.00e-05

Epoch 2/200 | Train Loss: 1.8644 Acc: 0.1754 | Val Loss: 1.8048 Acc: 0.1882 | LR: 3.00e-05

Epoch 3/200 | Train Loss: 1.8130 Acc: 0.2206 | Val Loss: 1.7949 Acc: 0.3294 | LR: 4.00e-05

Epoch 4/200 | Train Loss: 1.7670 Acc: 0.2456 | Val Loss: 1.7721 Acc: 0.3412 | LR: 5.00e-05

Epoch 5/200 | Train Loss: 1.6846 Acc: 0.2732 | Val Loss: 1.7366 Acc: 0.4353 | LR: 6.00e-05

Epoch 6/200 | Train Loss: 1.6081 Acc: 0.3459 | Val Loss: 1.6840 Acc: 0.5412 | LR: 7.00e-05

Epoch 7/200 | Train Loss: 1.4595 Acc: 0.4311 | Val Loss: 1.6203 Acc: 0.6471 | LR: 8.00e-05

Epoch 8/200 | Train Loss: 1.4142 Acc: 0.4411 | Val Loss: 1.5401 Acc: 0.6706 | LR: 9.00e-05

Epoch 9/200 | Train Loss: 1.2955 Acc: 0.5238 | Val Loss: 1.4582 Acc: 0.7294 | LR: 1.00e-04

Epoch 10/200 | Train Loss: 1.2486 Acc: 0.5865 | Val Loss: 1.3735 Acc: 0.7059 | LR: 1.00e-04

Epoch 11/200 | Train Loss: 1.1857 Acc: 0.6291 | Val Loss: 1.3052 Acc: 0.7176 | LR: 1.00e-04

Epoch 12/200 | Train Loss: 1.1484 Acc: 0.6491 | Val Loss: 1.2477 Acc: 0.7412 | LR: 1.00e-04

Epoch 13/200 | Train Loss: 1.1053 Acc: 0.6892 | Val Loss: 1.2032 Acc: 0.7294 | LR: 9.99e-05

Epoch 14/200 | Train Loss: 1.1084 Acc: 0.6892 | Val Loss: 1.1637 Acc: 0.7412 | LR: 9.99e-05

Epoch 15/200 | Train Loss: 1.0582 Acc: 0.7093 | Val Loss: 1.1338 Acc: 0.7529 | LR: 9.98e-05

Epoch 16/200 | Train Loss: 1.0357 Acc: 0.7444 | Val Loss: 1.1122 Acc: 0.7294 | LR: 9.98e-05

Epoch 17/200 | Train Loss: 1.0160 Acc: 0.7669 | Val Loss: 1.0940 Acc: 0.7294 | LR: 9.97e-05

Epoch 18/200 | Train Loss: 1.0091 Acc: 0.7769 | Val Loss: 1.0769 Acc: 0.7412 | LR: 9.96e-05

Epoch 19/200 | Train Loss: 0.9812 Acc: 0.8020 | Val Loss: 1.0668 Acc: 0.7529 | LR: 9.94e-05

Epoch 20/200 | Train Loss: 0.9429 Acc: 0.8095 | Val Loss: 1.0579 Acc: 0.7647 | LR: 9.93e-05

Epoch 21/200 | Train Loss: 0.9526 Acc: 0.8120 | Val Loss: 1.0465 Acc: 0.7647 | LR: 9.92e-05

Epoch 22/200 | Train Loss: 0.9292 Acc: 0.8496 | Val Loss: 1.0409 Acc: 0.7765 | LR: 9.90e-05

Epoch 23/200 | Train Loss: 0.9060 Acc: 0.8446 | Val Loss: 1.0390 Acc: 0.7647 | LR:

9.88e-05
Epoch 24/200 | Train Loss: 0.9011 Acc: 0.8596 | Val Loss: 1.0273 Acc: 0.7882 | LR: 9.87e-05
Epoch 25/200 | Train Loss: 0.9152 Acc: 0.8797 | Val Loss: 1.0163 Acc: 0.8000 | LR: 9.85e-05
Epoch 26/200 | Train Loss: 0.8711 Acc: 0.9023 | Val Loss: 1.0057 Acc: 0.8353 | LR: 9.83e-05
Epoch 27/200 | Train Loss: 0.8821 Acc: 0.8797 | Val Loss: 0.9908 Acc: 0.8471 | LR: 9.80e-05
Epoch 28/200 | Train Loss: 0.8638 Acc: 0.9048 | Val Loss: 0.9877 Acc: 0.8471 | LR: 9.78e-05
Epoch 29/200 | Train Loss: 0.8627 Acc: 0.9123 | Val Loss: 0.9826 Acc: 0.8471 | LR: 9.76e-05
Epoch 30/200 | Train Loss: 0.8482 Acc: 0.9148 | Val Loss: 0.9799 Acc: 0.8471 | LR: 9.73e-05
Epoch 31/200 | Train Loss: 0.8390 Acc: 0.9223 | Val Loss: 0.9741 Acc: 0.8353 | LR: 9.70e-05
Epoch 32/200 | Train Loss: 0.8434 Acc: 0.9198 | Val Loss: 0.9660 Acc: 0.8471 | LR: 9.67e-05
Epoch 33/200 | Train Loss: 0.8627 Acc: 0.9123 | Val Loss: 0.9583 Acc: 0.8588 | LR: 9.64e-05
Epoch 34/200 | Train Loss: 0.8193 Acc: 0.9373 | Val Loss: 0.9471 Acc: 0.8706 | LR: 9.61e-05
Epoch 35/200 | Train Loss: 0.8128 Acc: 0.9398 | Val Loss: 0.9433 Acc: 0.8824 | LR: 9.58e-05
Epoch 36/200 | Train Loss: 0.8234 Acc: 0.9524 | Val Loss: 0.9431 Acc: 0.8588 | LR: 9.55e-05
Epoch 37/200 | Train Loss: 0.8161 Acc: 0.9373 | Val Loss: 0.9423 Acc: 0.8941 | LR: 9.51e-05
Epoch 38/200 | Train Loss: 0.8156 Acc: 0.9449 | Val Loss: 0.9433 Acc: 0.9059 | LR: 9.47e-05
Epoch 39/200 | Train Loss: 0.8025 Acc: 0.9674 | Val Loss: 0.9426 Acc: 0.9059 | LR: 9.44e-05
Epoch 40/200 | Train Loss: 0.8190 Acc: 0.9474 | Val Loss: 0.9378 Acc: 0.9059 | LR: 9.40e-05
Epoch 41/200 | Train Loss: 0.7918 Acc: 0.9549 | Val Loss: 0.9273 Acc: 0.8941 | LR: 9.36e-05
Epoch 42/200 | Train Loss: 0.7895 Acc: 0.9524 | Val Loss: 0.9248 Acc: 0.8941 | LR: 9.32e-05
Epoch 43/200 | Train Loss: 0.7950 Acc: 0.9449 | Val Loss: 0.9224 Acc: 0.8941 | LR: 9.27e-05
Epoch 44/200 | Train Loss: 0.7847 Acc: 0.9549 | Val Loss: 0.9216 Acc: 0.8941 | LR: 9.23e-05
Epoch 45/200 | Train Loss: 0.7876 Acc: 0.9649 | Val Loss: 0.9211 Acc: 0.8824 | LR: 9.19e-05
Epoch 46/200 | Train Loss: 0.7874 Acc: 0.9749 | Val Loss: 0.9156 Acc: 0.8941 | LR: 9.14e-05
Epoch 47/200 | Train Loss: 0.7814 Acc: 0.9749 | Val Loss: 0.9128 Acc: 0.9059 | LR: 9.09e-05
Epoch 48/200 | Train Loss: 0.7778 Acc: 0.9624 | Val Loss: 0.9160 Acc: 0.8941 | LR: 9.05e-05
Epoch 49/200 | Train Loss: 0.7686 Acc: 0.9699 | Val Loss: 0.9154 Acc: 0.9059 | LR: 9.00e-05
Epoch 50/200 | Train Loss: 0.7758 Acc: 0.9749 | Val Loss: 0.9182 Acc: 0.9176 | LR: 8.95e-05
Epoch 51/200 | Train Loss: 0.7771 Acc: 0.9574 | Val Loss: 0.9095 Acc: 0.9176 | LR:

8.89e-05
Epoch 52/200 | Train Loss: 0.7760 Acc: 0.9774 | Val Loss: 0.9087 Acc: 0.9294 | LR: 8.84e-05
Epoch 53/200 | Train Loss: 0.7616 Acc: 0.9699 | Val Loss: 0.9077 Acc: 0.8941 | LR: 8.79e-05
Epoch 54/200 | Train Loss: 0.7703 Acc: 0.9875 | Val Loss: 0.9067 Acc: 0.9059 | LR: 8.73e-05
Epoch 55/200 | Train Loss: 0.7700 Acc: 0.9774 | Val Loss: 0.9096 Acc: 0.9059 | LR: 8.68e-05
Epoch 56/200 | Train Loss: 0.7533 Acc: 0.9850 | Val Loss: 0.9096 Acc: 0.9059 | LR: 8.62e-05
Epoch 57/200 | Train Loss: 0.7782 Acc: 0.9749 | Val Loss: 0.9080 Acc: 0.9059 | LR: 8.56e-05
Epoch 58/200 | Train Loss: 0.7616 Acc: 0.9649 | Val Loss: 0.9059 Acc: 0.8941 | LR: 8.51e-05
Epoch 59/200 | Train Loss: 0.7640 Acc: 0.9674 | Val Loss: 0.9020 Acc: 0.8824 | LR: 8.45e-05
Epoch 60/200 | Train Loss: 0.7614 Acc: 0.9699 | Val Loss: 0.8950 Acc: 0.8941 | LR: 8.39e-05
Epoch 61/200 | Train Loss: 0.7685 Acc: 0.9749 | Val Loss: 0.8920 Acc: 0.8824 | LR: 8.33e-05
Epoch 62/200 | Train Loss: 0.7580 Acc: 0.9674 | Val Loss: 0.8929 Acc: 0.8941 | LR: 8.26e-05
Epoch 63/200 | Train Loss: 0.7777 Acc: 0.9699 | Val Loss: 0.8944 Acc: 0.8941 | LR: 8.20e-05
Epoch 64/200 | Train Loss: 0.7552 Acc: 0.9825 | Val Loss: 0.8953 Acc: 0.9059 | LR: 8.14e-05
Epoch 65/200 | Train Loss: 0.7541 Acc: 0.9850 | Val Loss: 0.8911 Acc: 0.8941 | LR: 8.07e-05
Epoch 66/200 | Train Loss: 0.7664 Acc: 0.9724 | Val Loss: 0.8927 Acc: 0.8941 | LR: 8.01e-05
Epoch 67/200 | Train Loss: 0.7589 Acc: 0.9825 | Val Loss: 0.8897 Acc: 0.8941 | LR: 7.94e-05
Epoch 68/200 | Train Loss: 0.7427 Acc: 0.9850 | Val Loss: 0.8915 Acc: 0.9059 | LR: 7.87e-05
Epoch 69/200 | Train Loss: 0.7589 Acc: 0.9825 | Val Loss: 0.8917 Acc: 0.9059 | LR: 7.80e-05
Epoch 70/200 | Train Loss: 0.7470 Acc: 0.9850 | Val Loss: 0.8867 Acc: 0.9059 | LR: 7.73e-05
Epoch 71/200 | Train Loss: 0.7572 Acc: 0.9799 | Val Loss: 0.8899 Acc: 0.9059 | LR: 7.67e-05
Epoch 72/200 | Train Loss: 0.7509 Acc: 0.9900 | Val Loss: 0.8900 Acc: 0.8941 | LR: 7.59e-05
Epoch 73/200 | Train Loss: 0.7468 Acc: 0.9900 | Val Loss: 0.8932 Acc: 0.8941 | LR: 7.52e-05
Epoch 74/200 | Train Loss: 0.7470 Acc: 0.9875 | Val Loss: 0.8974 Acc: 0.8941 | LR: 7.45e-05
Epoch 75/200 | Train Loss: 0.7468 Acc: 0.9799 | Val Loss: 0.9018 Acc: 0.9059 | LR: 7.38e-05
Epoch 76/200 | Train Loss: 0.7441 Acc: 0.9875 | Val Loss: 0.9119 Acc: 0.9059 | LR: 7.31e-05
Epoch 77/200 | Train Loss: 0.7371 Acc: 0.9799 | Val Loss: 0.9179 Acc: 0.8941 | LR: 7.23e-05
Epoch 78/200 | Train Loss: 0.7546 Acc: 0.9774 | Val Loss: 0.9207 Acc: 0.8941 | LR: 7.16e-05
Epoch 79/200 | Train Loss: 0.7712 Acc: 0.9875 | Val Loss: 0.9226 Acc: 0.8824 | LR:

7.08e-05

Epoch 80/200 | Train Loss: 0.7408 Acc: 0.9875 | Val Loss: 0.9185 Acc: 0.8941 | LR:

7.01e-05

Epoch 81/200 | Train Loss: 0.7459 Acc: 0.9925 | Val Loss: 0.9176 Acc: 0.9059 | LR:

6.93e-05

Epoch 82/200 | Train Loss: 0.7425 Acc: 0.9925 | Val Loss: 0.9110 Acc: 0.9059 | LR:

6.86e-05

Epoch 83/200 | Train Loss: 0.7452 Acc: 0.9900 | Val Loss: 0.9105 Acc: 0.9059 | LR:

6.78e-05

Epoch 84/200 | Train Loss: 0.7374 Acc: 0.9900 | Val Loss: 0.9108 Acc: 0.9059 | LR:

6.70e-05

Epoch 85/200 | Train Loss: 0.7330 Acc: 0.9950 | Val Loss: 0.9099 Acc: 0.8941 | LR:

6.62e-05

Epoch 86/200 | Train Loss: 0.7372 Acc: 0.9900 | Val Loss: 0.9134 Acc: 0.8941 | LR:

6.55e-05

Epoch 87/200 | Train Loss: 0.7458 Acc: 0.9850 | Val Loss: 0.9068 Acc: 0.9059 | LR:

6.47e-05

Epoch 88/200 | Train Loss: 0.7340 Acc: 0.9875 | Val Loss: 0.9095 Acc: 0.9059 | LR:

6.39e-05

Epoch 89/200 | Train Loss: 0.7376 Acc: 0.9875 | Val Loss: 0.9085 Acc: 0.9059 | LR:

6.31e-05

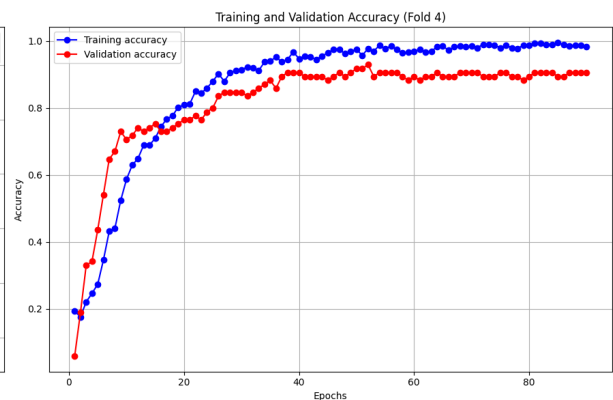
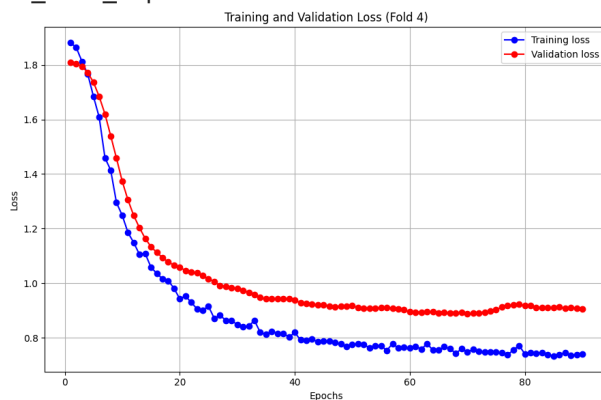
Epoch 90/200 | Train Loss: 0.7397 Acc: 0.9825 | Val Loss: 0.9044 Acc: 0.9059 | LR:

6.23e-05

Early stopping triggered.

Training complete in 6.7 minutes. Best Val Loss: 0.8867

Best model for fold saved to: new_results_Multimodal_ResNet50_Concat_KFold\best_model_fold_4.pth



Saved training history plot to: new_results_Multimodal_ResNet50_Concat_KFold\training_history_fold_4.png

=====

OVERALL PERFORMANCE (FOLD 4)

=====

	precision	recall	f1-score	support
Herpes	0.75	1.00	0.86	6
Melanoma	0.67	1.00	0.80	2
Monkeypox	0.94	0.89	0.92	19
Sarampion	0.84	0.90	0.87	30
Varicela	0.80	0.67	0.73	24
accuracy			0.84	81
macro avg	0.80	0.89	0.83	81
weighted avg	0.84	0.84	0.84	81

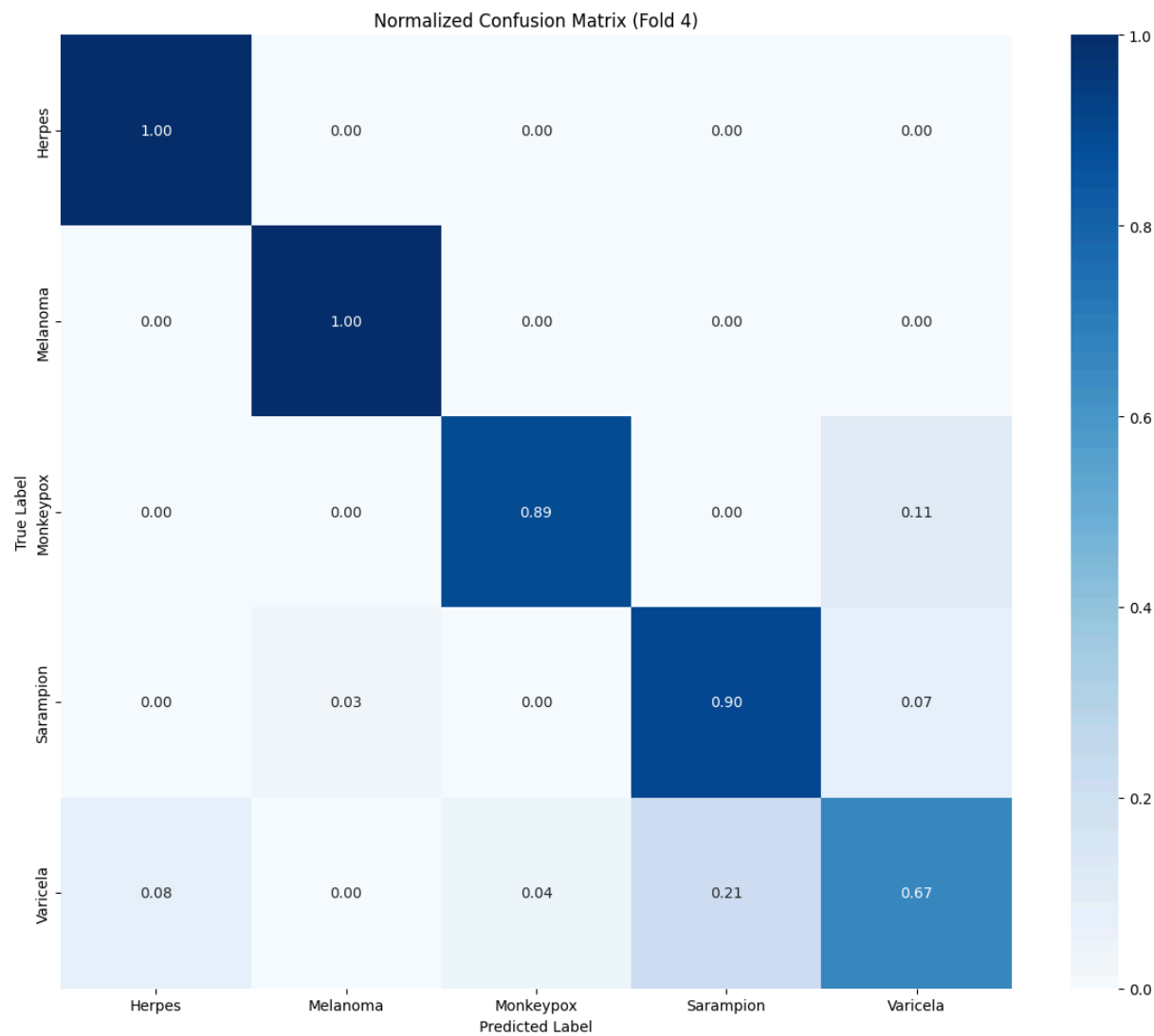
--- Overall Scalar Metrics ---

Accuracy: 0.8395

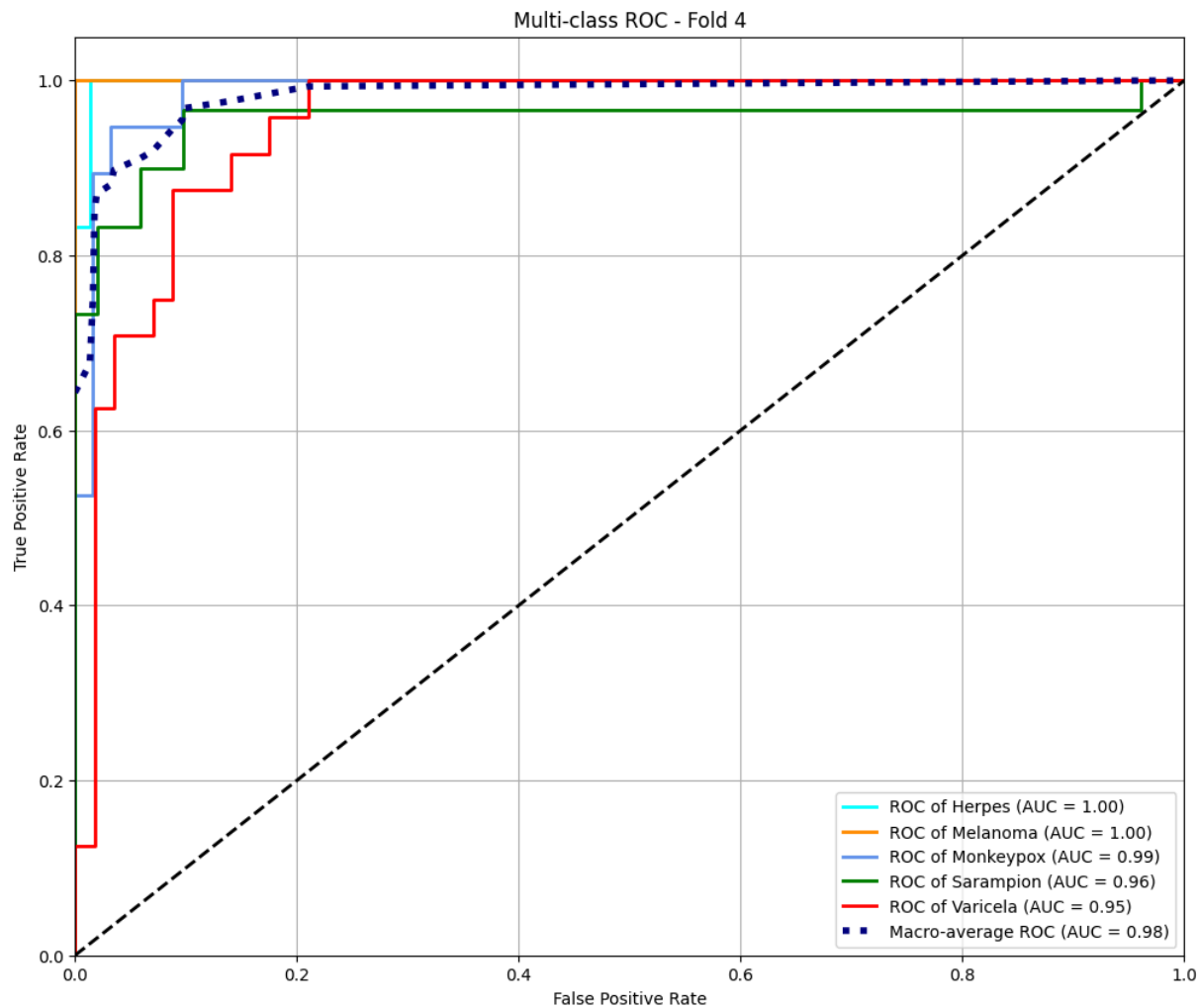
Macro ROC AUC: 0.9786

Cohen's Kappa: 0.7771

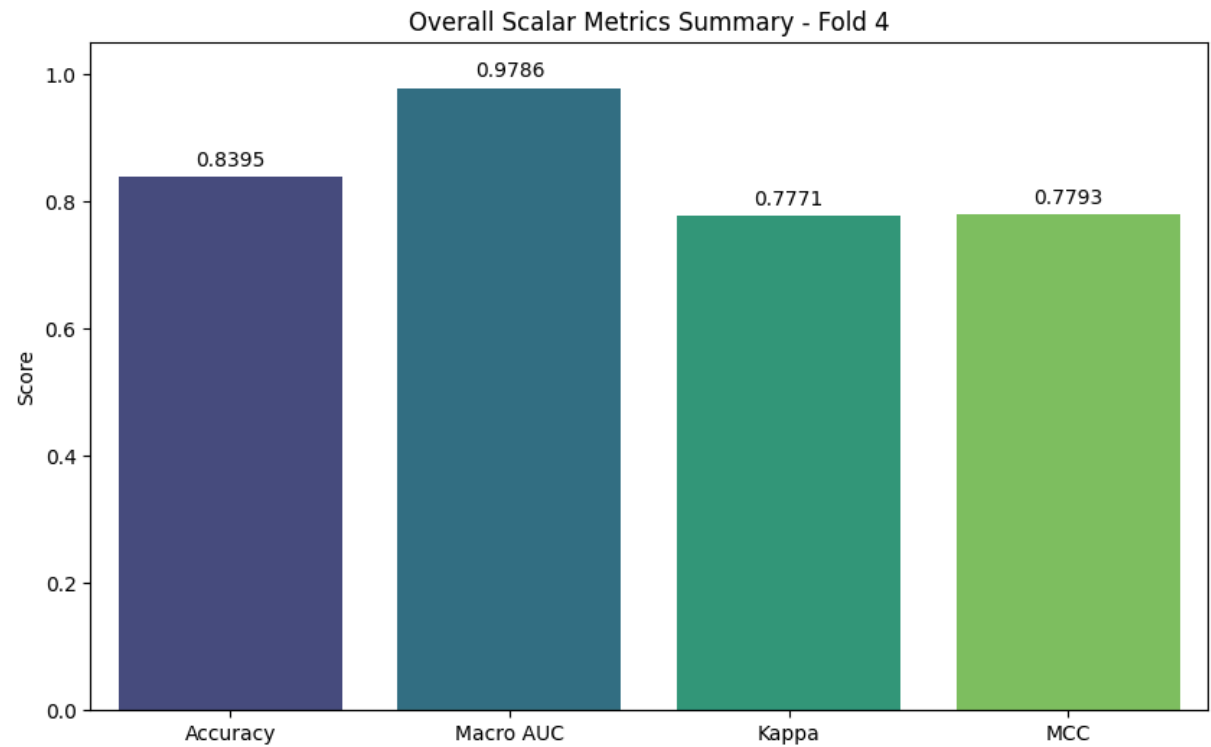
Matthews CorrCoef: 0.7793



Saved normalized confusion matrix to: new_results_Multimodal_ResNet50_Concat_KFold\confusion_matrix_fold_4.png



Saved ROC curves plot to: new_results_Multimodal_ResNet50_Concat_KFold\roc_curves_fold_4.png



Saved scalar metrics plot to: new_results_Multimodal_ResNet50_Concat_KFold\scalar_metrics_fold_4.png

=====

PERFORMANCE BY GENDER

=====

--- Subgroup: female ---

	precision	recall	f1-score	support
Herpes	0.50	1.00	0.67	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	3
Sarampion	0.78	1.00	0.88	14
Varicela	1.00	0.55	0.71	11
accuracy			0.83	29
macro avg	0.66	0.71	0.65	29
weighted avg	0.88	0.83	0.82	29

--- Subgroup: male ---

	precision	recall	f1-score	support
Herpes	0.83	1.00	0.91	5
Melanoma	0.67	1.00	0.80	2
Monkeypox	0.93	0.88	0.90	16
Sarampion	0.93	0.81	0.87	16
Varicela	0.71	0.77	0.74	13
accuracy			0.85	52
macro avg	0.82	0.89	0.84	52
weighted avg	0.86	0.85	0.85	52

=====

PERFORMANCE BY AGE

=====

--- Subgroup: 0-17 ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.00	0.00	0.00	0
Sarampion	1.00	1.00	1.00	2
Varicela	1.00	0.67	0.80	3
accuracy			0.80	5
macro avg	0.40	0.33	0.36	5
weighted avg	1.00	0.80	0.88	5

--- Subgroup: 18-59 ---

	precision	recall	f1-score	support
Herpes	0.86	1.00	0.92	6
Melanoma	0.67	1.00	0.80	2
Monkeypox	0.94	0.89	0.92	19
Sarampion	0.83	0.89	0.86	28
Varicela	0.78	0.67	0.72	21
accuracy			0.84	76
macro avg	0.82	0.89	0.84	76
weighted avg	0.84	0.84	0.84	76

=====

PERFORMANCE BY SKIN TONE

=====

--- Subgroup: asian ---

	precision	recall	f1-score	support
Herpes	0.50	1.00	0.67	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	1
Sarampion	0.75	1.00	0.86	3
Varicela	1.00	0.60	0.75	5
accuracy			0.80	10
macro avg	0.65	0.72	0.65	10
weighted avg	0.88	0.80	0.80	10

--- Subgroup: black ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	3
Sarampion	1.00	1.00	1.00	2
Varicela	0.00	0.00	0.00	0
accuracy			1.00	5
macro avg	0.40	0.40	0.40	5
weighted avg	1.00	1.00	1.00	5

--- Subgroup: indian ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	2
Sarampion	0.80	1.00	0.89	4
Varicela	0.00	0.00	0.00	1
accuracy			0.86	7

macro avg	0.36	0.40	0.38	7
weighted avg	0.74	0.86	0.79	7

--- Subgroup: latino hispanic ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	3
Sarampion	1.00	1.00	1.00	4
Varicela	0.00	0.00	0.00	0
accuracy			1.00	7
macro avg	0.40	0.40	0.40	7
weighted avg	1.00	1.00	1.00	7

--- Subgroup: middle eastern ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	1.00	1.00	1.00	1
Monkeypox	1.00	1.00	1.00	2
Sarampion	1.00	1.00	1.00	1
Varicela	0.00	0.00	0.00	0
accuracy			1.00	4
macro avg	0.60	0.60	0.60	4
weighted avg	1.00	1.00	1.00	4

--- Subgroup: white ---

	precision	recall	f1-score	support
Herpes	0.83	1.00	0.91	5
Melanoma	0.50	1.00	0.67	1
Monkeypox	0.86	0.75	0.80	8
Sarampion	0.81	0.81	0.81	16
Varicela	0.76	0.72	0.74	18
accuracy			0.79	48
macro avg	0.75	0.86	0.79	48
weighted avg	0.80	0.79	0.79	48

```
=====
FAIRNESS SUMMARY (ALL GROUPS) | WGA: 0.0000 EOD: 0.8556
Worst pair: 0-17|male|asian vs 18-59|male|white
=====
```

```
=====
FAIRNESS SUMMARY (FILTERED, n>=15) | WGA: 0.7308 EOD: 0.5222
Worst pair: 18-59|female|white vs 18-59|male|white
Groups: ['18-59|female|white', '18-59|male|white']
=====
```

```
#####  
####  
### FOLD 5 ###  
#####  
####
```

Unfreezing ResNet-50 layer4 for fine-tuning.
Total params: 26.43M | Trainable params: 17.89M (67.68%)
Using weighted CrossEntropyLoss for training.

START TRAINING (Resnet50 + Concat) for 200 epochs...

Epoch 1/200	Train Loss: 1.9133	Acc: 0.2581	Val Loss: 1.8050	Acc: 0.3176	LR: 2.00e-05
Epoch 2/200	Train Loss: 1.8383	Acc: 0.2607	Val Loss: 1.7998	Acc: 0.3412	LR: 3.00e-05
Epoch 3/200	Train Loss: 1.9065	Acc: 0.2481	Val Loss: 1.7868	Acc: 0.3647	LR: 4.00e-05
Epoch 4/200	Train Loss: 1.8187	Acc: 0.2782	Val Loss: 1.7708	Acc: 0.4235	LR: 5.00e-05
Epoch 5/200	Train Loss: 1.8040	Acc: 0.2982	Val Loss: 1.7396	Acc: 0.4824	LR: 6.00e-05
Epoch 6/200	Train Loss: 1.6610	Acc: 0.3634	Val Loss: 1.6956	Acc: 0.5647	LR: 7.00e-05
Epoch 7/200	Train Loss: 1.5689	Acc: 0.4261	Val Loss: 1.6365	Acc: 0.6118	LR: 8.00e-05
Epoch 8/200	Train Loss: 1.4663	Acc: 0.4586	Val Loss: 1.5637	Acc: 0.6235	LR: 9.00e-05
Epoch 9/200	Train Loss: 1.3801	Acc: 0.4987	Val Loss: 1.4788	Acc: 0.6588	LR: 1.00e-04
Epoch 10/200	Train Loss: 1.3355	Acc: 0.5664	Val Loss: 1.3996	Acc: 0.6706	LR: 1.00e-04
Epoch 11/200	Train Loss: 1.2588	Acc: 0.5789	Val Loss: 1.3308	Acc: 0.6471	LR: 1.00e-04
Epoch 12/200	Train Loss: 1.2115	Acc: 0.6190	Val Loss: 1.2770	Acc: 0.6706	LR: 1.00e-04
Epoch 13/200	Train Loss: 1.1618	Acc: 0.6642	Val Loss: 1.2385	Acc: 0.6706	LR: 9.99e-05
Epoch 14/200	Train Loss: 1.1354	Acc: 0.7093	Val Loss: 1.2134	Acc: 0.6941	LR: 9.99e-05
Epoch 15/200	Train Loss: 1.0964	Acc: 0.7218	Val Loss: 1.1859	Acc: 0.7059	LR: 9.98e-05
Epoch 16/200	Train Loss: 1.0761	Acc: 0.7318	Val Loss: 1.1709	Acc: 0.7059	LR: 9.98e-05
Epoch 17/200	Train Loss: 1.0624	Acc: 0.7368	Val Loss: 1.1514	Acc: 0.7412	LR: 9.97e-05
Epoch 18/200	Train Loss: 1.0458	Acc: 0.7419	Val Loss: 1.1301	Acc: 0.7412	LR: 9.96e-05
Epoch 19/200	Train Loss: 1.0216	Acc: 0.7469	Val Loss: 1.1194	Acc: 0.7412	LR: 9.94e-05
Epoch 20/200	Train Loss: 1.0149	Acc: 0.7444	Val Loss: 1.1066	Acc: 0.7647	LR: 9.93e-05
Epoch 21/200	Train Loss: 0.9671	Acc: 0.8120	Val Loss: 1.0925	Acc: 0.7647	LR: 9.92e-05
Epoch 22/200	Train Loss: 0.9926	Acc: 0.8045	Val Loss: 1.0791	Acc: 0.7529	LR: 9.90e-05
Epoch 23/200	Train Loss: 0.9664	Acc: 0.8120	Val Loss: 1.0628	Acc: 0.7529	LR: 9.90e-05

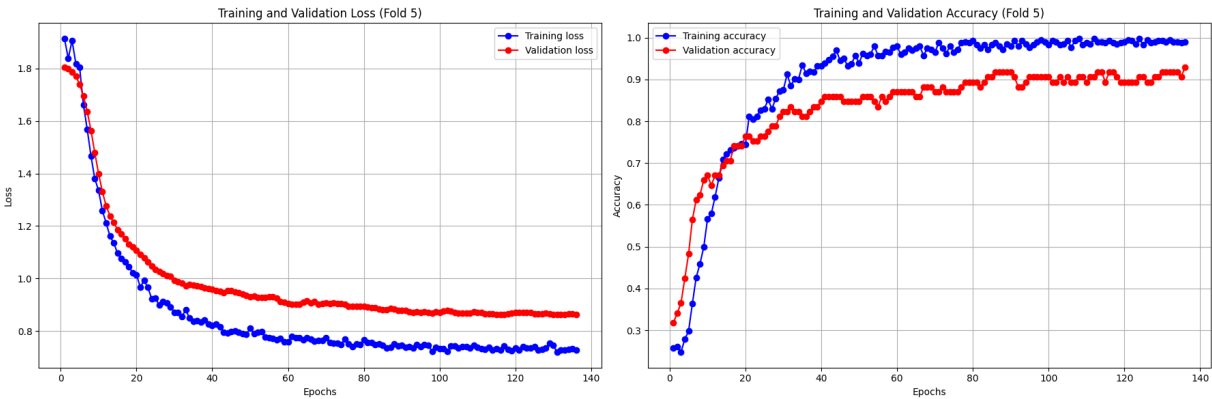
9.88e-05
Epoch 24/200 | Train Loss: 0.9239 Acc: 0.8271 | Val Loss: 1.0464 Acc: 0.7647 | LR: 9.87e-05
Epoch 25/200 | Train Loss: 0.9242 Acc: 0.8296 | Val Loss: 1.0359 Acc: 0.7647 | LR: 9.85e-05
Epoch 26/200 | Train Loss: 0.8998 Acc: 0.8521 | Val Loss: 1.0281 Acc: 0.7765 | LR: 9.83e-05
Epoch 27/200 | Train Loss: 0.9119 Acc: 0.8296 | Val Loss: 1.0180 Acc: 0.7882 | LR: 9.80e-05
Epoch 28/200 | Train Loss: 0.9081 Acc: 0.8546 | Val Loss: 1.0118 Acc: 0.7882 | LR: 9.78e-05
Epoch 29/200 | Train Loss: 0.8909 Acc: 0.8722 | Val Loss: 1.0079 Acc: 0.8118 | LR: 9.76e-05
Epoch 30/200 | Train Loss: 0.8718 Acc: 0.8747 | Val Loss: 0.9926 Acc: 0.8235 | LR: 9.73e-05
Epoch 31/200 | Train Loss: 0.8697 Acc: 0.9123 | Val Loss: 0.9889 Acc: 0.8235 | LR: 9.70e-05
Epoch 32/200 | Train Loss: 0.8558 Acc: 0.8847 | Val Loss: 0.9814 Acc: 0.8353 | LR: 9.67e-05
Epoch 33/200 | Train Loss: 0.8801 Acc: 0.9023 | Val Loss: 0.9735 Acc: 0.8235 | LR: 9.64e-05
Epoch 34/200 | Train Loss: 0.8498 Acc: 0.8997 | Val Loss: 0.9766 Acc: 0.8235 | LR: 9.61e-05
Epoch 35/200 | Train Loss: 0.8359 Acc: 0.9348 | Val Loss: 0.9745 Acc: 0.8118 | LR: 9.58e-05
Epoch 36/200 | Train Loss: 0.8384 Acc: 0.9148 | Val Loss: 0.9720 Acc: 0.8118 | LR: 9.55e-05
Epoch 37/200 | Train Loss: 0.8349 Acc: 0.9198 | Val Loss: 0.9701 Acc: 0.8235 | LR: 9.51e-05
Epoch 38/200 | Train Loss: 0.8414 Acc: 0.9173 | Val Loss: 0.9638 Acc: 0.8353 | LR: 9.47e-05
Epoch 39/200 | Train Loss: 0.8258 Acc: 0.9323 | Val Loss: 0.9610 Acc: 0.8353 | LR: 9.44e-05
Epoch 40/200 | Train Loss: 0.8211 Acc: 0.9323 | Val Loss: 0.9582 Acc: 0.8471 | LR: 9.40e-05
Epoch 41/200 | Train Loss: 0.8274 Acc: 0.9398 | Val Loss: 0.9546 Acc: 0.8588 | LR: 9.36e-05
Epoch 42/200 | Train Loss: 0.8156 Acc: 0.9474 | Val Loss: 0.9526 Acc: 0.8588 | LR: 9.32e-05
Epoch 43/200 | Train Loss: 0.7952 Acc: 0.9549 | Val Loss: 0.9468 Acc: 0.8588 | LR: 9.27e-05
Epoch 44/200 | Train Loss: 0.7927 Acc: 0.9699 | Val Loss: 0.9529 Acc: 0.8588 | LR: 9.23e-05
Epoch 45/200 | Train Loss: 0.7983 Acc: 0.9449 | Val Loss: 0.9534 Acc: 0.8588 | LR: 9.19e-05
Epoch 46/200 | Train Loss: 0.8005 Acc: 0.9499 | Val Loss: 0.9476 Acc: 0.8471 | LR: 9.14e-05
Epoch 47/200 | Train Loss: 0.7944 Acc: 0.9323 | Val Loss: 0.9453 Acc: 0.8471 | LR: 9.09e-05
Epoch 48/200 | Train Loss: 0.7901 Acc: 0.9373 | Val Loss: 0.9403 Acc: 0.8471 | LR: 9.05e-05
Epoch 49/200 | Train Loss: 0.7866 Acc: 0.9574 | Val Loss: 0.9366 Acc: 0.8471 | LR: 9.00e-05
Epoch 50/200 | Train Loss: 0.8104 Acc: 0.9398 | Val Loss: 0.9310 Acc: 0.8471 | LR: 8.95e-05
Epoch 51/200 | Train Loss: 0.7889 Acc: 0.9624 | Val Loss: 0.9334 Acc: 0.8588 | LR:

8.89e-05
Epoch 52/200 | Train Loss: 0.7966 Acc: 0.9574 | Val Loss: 0.9284 Acc: 0.8588 | LR: 8.84e-05
Epoch 53/200 | Train Loss: 0.7987 Acc: 0.9599 | Val Loss: 0.9278 Acc: 0.8588 | LR: 8.79e-05
Epoch 54/200 | Train Loss: 0.7783 Acc: 0.9799 | Val Loss: 0.9280 Acc: 0.8471 | LR: 8.73e-05
Epoch 55/200 | Train Loss: 0.7734 Acc: 0.9574 | Val Loss: 0.9300 Acc: 0.8353 | LR: 8.68e-05
Epoch 56/200 | Train Loss: 0.7728 Acc: 0.9574 | Val Loss: 0.9310 Acc: 0.8588 | LR: 8.62e-05
Epoch 57/200 | Train Loss: 0.7660 Acc: 0.9674 | Val Loss: 0.9255 Acc: 0.8471 | LR: 8.56e-05
Epoch 58/200 | Train Loss: 0.7727 Acc: 0.9649 | Val Loss: 0.9131 Acc: 0.8588 | LR: 8.51e-05
Epoch 59/200 | Train Loss: 0.7596 Acc: 0.9774 | Val Loss: 0.9097 Acc: 0.8706 | LR: 8.45e-05
Epoch 60/200 | Train Loss: 0.7582 Acc: 0.9799 | Val Loss: 0.9056 Acc: 0.8706 | LR: 8.39e-05
Epoch 61/200 | Train Loss: 0.7790 Acc: 0.9599 | Val Loss: 0.9018 Acc: 0.8706 | LR: 8.33e-05
Epoch 62/200 | Train Loss: 0.7758 Acc: 0.9649 | Val Loss: 0.9024 Acc: 0.8706 | LR: 8.26e-05
Epoch 63/200 | Train Loss: 0.7739 Acc: 0.9749 | Val Loss: 0.9027 Acc: 0.8706 | LR: 8.20e-05
Epoch 64/200 | Train Loss: 0.7660 Acc: 0.9699 | Val Loss: 0.9094 Acc: 0.8706 | LR: 8.14e-05
Epoch 65/200 | Train Loss: 0.7736 Acc: 0.9749 | Val Loss: 0.9143 Acc: 0.8588 | LR: 8.07e-05
Epoch 66/200 | Train Loss: 0.7685 Acc: 0.9799 | Val Loss: 0.9078 Acc: 0.8588 | LR: 8.01e-05
Epoch 67/200 | Train Loss: 0.7610 Acc: 0.9574 | Val Loss: 0.9133 Acc: 0.8824 | LR: 7.94e-05
Epoch 68/200 | Train Loss: 0.7632 Acc: 0.9749 | Val Loss: 0.9025 Acc: 0.8824 | LR: 7.87e-05
Epoch 69/200 | Train Loss: 0.7640 Acc: 0.9724 | Val Loss: 0.9035 Acc: 0.8824 | LR: 7.80e-05
Epoch 70/200 | Train Loss: 0.7737 Acc: 0.9649 | Val Loss: 0.9072 Acc: 0.8706 | LR: 7.73e-05
Epoch 71/200 | Train Loss: 0.7568 Acc: 0.9875 | Val Loss: 0.9051 Acc: 0.8706 | LR: 7.67e-05
Epoch 72/200 | Train Loss: 0.7536 Acc: 0.9749 | Val Loss: 0.9073 Acc: 0.8824 | LR: 7.59e-05
Epoch 73/200 | Train Loss: 0.7548 Acc: 0.9624 | Val Loss: 0.9047 Acc: 0.8706 | LR: 7.52e-05
Epoch 74/200 | Train Loss: 0.7492 Acc: 0.9799 | Val Loss: 0.9051 Acc: 0.8706 | LR: 7.45e-05
Epoch 75/200 | Train Loss: 0.7695 Acc: 0.9649 | Val Loss: 0.9016 Acc: 0.8706 | LR: 7.38e-05
Epoch 76/200 | Train Loss: 0.7506 Acc: 0.9724 | Val Loss: 0.8951 Acc: 0.8706 | LR: 7.31e-05
Epoch 77/200 | Train Loss: 0.7404 Acc: 0.9875 | Val Loss: 0.8951 Acc: 0.8824 | LR: 7.23e-05
Epoch 78/200 | Train Loss: 0.7500 Acc: 0.9900 | Val Loss: 0.8953 Acc: 0.8941 | LR: 7.16e-05
Epoch 79/200 | Train Loss: 0.7489 Acc: 0.9875 | Val Loss: 0.8932 Acc: 0.8941 | LR:

7.08e-05
Epoch 80/200 | Train Loss: 0.7671 Acc: 0.9925 | Val Loss: 0.8932 Acc: 0.8941 | LR: 7.01e-05
Epoch 81/200 | Train Loss: 0.7556 Acc: 0.9825 | Val Loss: 0.8908 Acc: 0.8941 | LR: 6.93e-05
Epoch 82/200 | Train Loss: 0.7553 Acc: 0.9749 | Val Loss: 0.8892 Acc: 0.8824 | LR: 6.86e-05
Epoch 83/200 | Train Loss: 0.7485 Acc: 0.9825 | Val Loss: 0.8882 Acc: 0.8941 | LR: 6.78e-05
Epoch 84/200 | Train Loss: 0.7499 Acc: 0.9724 | Val Loss: 0.8827 Acc: 0.9059 | LR: 6.70e-05
Epoch 85/200 | Train Loss: 0.7459 Acc: 0.9825 | Val Loss: 0.8802 Acc: 0.9059 | LR: 6.62e-05
Epoch 86/200 | Train Loss: 0.7365 Acc: 0.9875 | Val Loss: 0.8819 Acc: 0.9176 | LR: 6.55e-05
Epoch 87/200 | Train Loss: 0.7387 Acc: 0.9799 | Val Loss: 0.8864 Acc: 0.9176 | LR: 6.47e-05
Epoch 88/200 | Train Loss: 0.7501 Acc: 0.9724 | Val Loss: 0.8829 Acc: 0.9176 | LR: 6.39e-05
Epoch 89/200 | Train Loss: 0.7428 Acc: 0.9850 | Val Loss: 0.8795 Acc: 0.9176 | LR: 6.31e-05
Epoch 90/200 | Train Loss: 0.7466 Acc: 0.9799 | Val Loss: 0.8774 Acc: 0.9176 | LR: 6.23e-05
Epoch 91/200 | Train Loss: 0.7379 Acc: 0.9925 | Val Loss: 0.8775 Acc: 0.9059 | LR: 6.15e-05
Epoch 92/200 | Train Loss: 0.7397 Acc: 0.9799 | Val Loss: 0.8740 Acc: 0.8824 | LR: 6.07e-05
Epoch 93/200 | Train Loss: 0.7366 Acc: 0.9925 | Val Loss: 0.8711 Acc: 0.8824 | LR: 5.99e-05
Epoch 94/200 | Train Loss: 0.7485 Acc: 0.9850 | Val Loss: 0.8734 Acc: 0.8941 | LR: 5.90e-05
Epoch 95/200 | Train Loss: 0.7407 Acc: 0.9774 | Val Loss: 0.8720 Acc: 0.9059 | LR: 5.82e-05
Epoch 96/200 | Train Loss: 0.7478 Acc: 0.9825 | Val Loss: 0.8722 Acc: 0.9059 | LR: 5.74e-05
Epoch 97/200 | Train Loss: 0.7462 Acc: 0.9900 | Val Loss: 0.8702 Acc: 0.9059 | LR: 5.66e-05
Epoch 98/200 | Train Loss: 0.7217 Acc: 0.9950 | Val Loss: 0.8677 Acc: 0.9059 | LR: 5.58e-05
Epoch 99/200 | Train Loss: 0.7372 Acc: 0.9900 | Val Loss: 0.8732 Acc: 0.9059 | LR: 5.50e-05
Epoch 100/200 | Train Loss: 0.7342 Acc: 0.9825 | Val Loss: 0.8711 Acc: 0.9059 | LR: 5.41e-05
Epoch 101/200 | Train Loss: 0.7338 Acc: 0.9925 | Val Loss: 0.8747 Acc: 0.8941 | LR: 5.33e-05
Epoch 102/200 | Train Loss: 0.7223 Acc: 0.9900 | Val Loss: 0.8778 Acc: 0.8941 | LR: 5.25e-05
Epoch 103/200 | Train Loss: 0.7425 Acc: 0.9825 | Val Loss: 0.8750 Acc: 0.9059 | LR: 5.17e-05
Epoch 104/200 | Train Loss: 0.7439 Acc: 0.9850 | Val Loss: 0.8699 Acc: 0.8941 | LR: 5.08e-05
Epoch 105/200 | Train Loss: 0.7344 Acc: 0.9925 | Val Loss: 0.8674 Acc: 0.9059 | LR: 5.00e-05
Epoch 106/200 | Train Loss: 0.7413 Acc: 0.9774 | Val Loss: 0.8678 Acc: 0.8941 | LR: 4.92e-05
Epoch 107/200 | Train Loss: 0.7394 Acc: 0.9925 | Val Loss: 0.8680 Acc: 0.8941 | LR:

4.83e-05
Epoch 108/200 | Train Loss: 0.7356 Acc: 0.9975 | Val Loss: 0.8686 Acc: 0.9059 | LR: 4.75e-05
Epoch 109/200 | Train Loss: 0.7459 Acc: 0.9825 | Val Loss: 0.8722 Acc: 0.9059 | LR: 4.67e-05
Epoch 110/200 | Train Loss: 0.7374 Acc: 0.9875 | Val Loss: 0.8705 Acc: 0.8941 | LR: 4.59e-05
Epoch 111/200 | Train Loss: 0.7332 Acc: 0.9850 | Val Loss: 0.8719 Acc: 0.9059 | LR: 4.50e-05
Epoch 112/200 | Train Loss: 0.7310 Acc: 0.9975 | Val Loss: 0.8655 Acc: 0.9059 | LR: 4.42e-05
Epoch 113/200 | Train Loss: 0.7387 Acc: 0.9900 | Val Loss: 0.8667 Acc: 0.9176 | LR: 4.34e-05
Epoch 114/200 | Train Loss: 0.7285 Acc: 0.9900 | Val Loss: 0.8653 Acc: 0.9176 | LR: 4.26e-05
Epoch 115/200 | Train Loss: 0.7321 Acc: 0.9875 | Val Loss: 0.8641 Acc: 0.8941 | LR: 4.18e-05
Epoch 116/200 | Train Loss: 0.7264 Acc: 0.9925 | Val Loss: 0.8626 Acc: 0.9176 | LR: 4.10e-05
Epoch 117/200 | Train Loss: 0.7421 Acc: 0.9875 | Val Loss: 0.8638 Acc: 0.9176 | LR: 4.01e-05
Epoch 118/200 | Train Loss: 0.7291 Acc: 0.9850 | Val Loss: 0.8652 Acc: 0.9059 | LR: 3.93e-05
Epoch 119/200 | Train Loss: 0.7247 Acc: 0.9875 | Val Loss: 0.8678 Acc: 0.8941 | LR: 3.85e-05
Epoch 120/200 | Train Loss: 0.7343 Acc: 0.9900 | Val Loss: 0.8700 Acc: 0.8941 | LR: 3.77e-05
Epoch 121/200 | Train Loss: 0.7268 Acc: 0.9950 | Val Loss: 0.8708 Acc: 0.8941 | LR: 3.69e-05
Epoch 122/200 | Train Loss: 0.7402 Acc: 0.9925 | Val Loss: 0.8698 Acc: 0.8941 | LR: 3.61e-05
Epoch 123/200 | Train Loss: 0.7353 Acc: 0.9850 | Val Loss: 0.8715 Acc: 0.8941 | LR: 3.53e-05
Epoch 124/200 | Train Loss: 0.7350 Acc: 0.9975 | Val Loss: 0.8703 Acc: 0.9059 | LR: 3.45e-05
Epoch 125/200 | Train Loss: 0.7402 Acc: 0.9825 | Val Loss: 0.8665 Acc: 0.9059 | LR: 3.38e-05
Epoch 126/200 | Train Loss: 0.7288 Acc: 0.9950 | Val Loss: 0.8656 Acc: 0.8941 | LR: 3.30e-05
Epoch 127/200 | Train Loss: 0.7297 Acc: 0.9875 | Val Loss: 0.8667 Acc: 0.8941 | LR: 3.22e-05
Epoch 128/200 | Train Loss: 0.7344 Acc: 0.9900 | Val Loss: 0.8671 Acc: 0.9059 | LR: 3.14e-05
Epoch 129/200 | Train Loss: 0.7525 Acc: 0.9925 | Val Loss: 0.8661 Acc: 0.9059 | LR: 3.07e-05
Epoch 130/200 | Train Loss: 0.7446 Acc: 0.9925 | Val Loss: 0.8641 Acc: 0.9176 | LR: 2.99e-05
Epoch 131/200 | Train Loss: 0.7204 Acc: 0.9900 | Val Loss: 0.8631 Acc: 0.9176 | LR: 2.92e-05
Epoch 132/200 | Train Loss: 0.7265 Acc: 0.9950 | Val Loss: 0.8635 Acc: 0.9176 | LR: 2.84e-05
Epoch 133/200 | Train Loss: 0.7273 Acc: 0.9900 | Val Loss: 0.8637 Acc: 0.9176 | LR: 2.77e-05
Epoch 134/200 | Train Loss: 0.7313 Acc: 0.9900 | Val Loss: 0.8666 Acc: 0.9176 | LR: 2.69e-05
Epoch 135/200 | Train Loss: 0.7326 Acc: 0.9875 | Val Loss: 0.8653 Acc: 0.9059 | LR:

2.62e-05
Epoch 136/200 | Train Loss: 0.7264 Acc: 0.9900 | Val Loss: 0.8628 Acc: 0.9294 | LR: 2.55e-05
Early stopping triggered.
Training complete in 10.1 minutes. Best Val Loss: 0.8626
Best model for fold saved to: new_results_Multimodal_ResNet50_Concat_KFold\best_model_fold_5.pth



Saved training history plot to: new_results_Multimodal_ResNet50_Concat_KFold\training_history_fold_5.png

=====

OVERALL PERFORMANCE (FOLD 5)

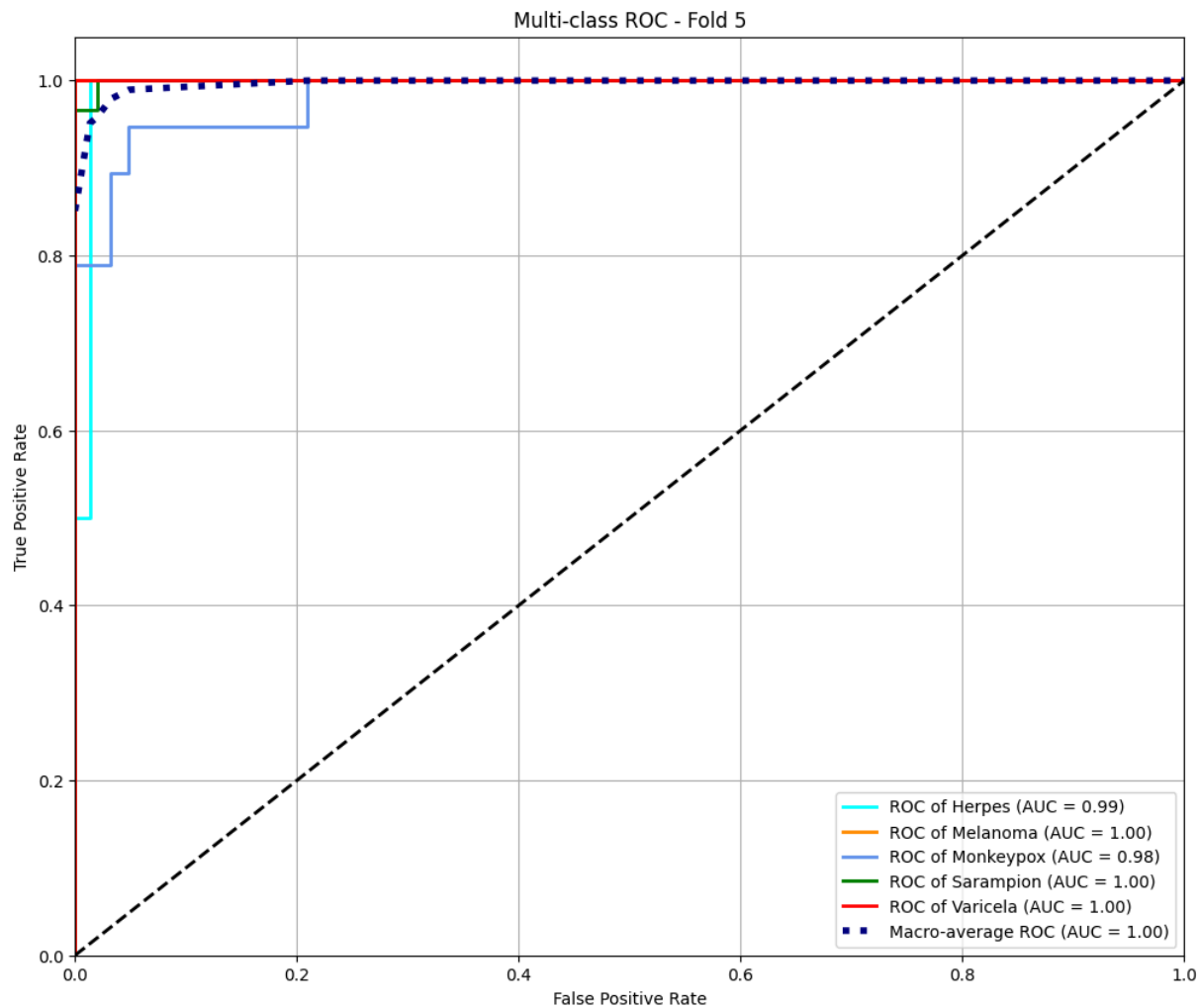
=====

	precision	recall	f1-score	support
Herpes	0.75	1.00	0.86	6
Melanoma	1.00	1.00	1.00	2
Monkeypox	0.89	0.89	0.89	19
Sarampion	1.00	0.90	0.95	30
Varicela	0.96	1.00	0.98	24
accuracy			0.94	81
macro avg	0.92	0.96	0.94	81
weighted avg	0.94	0.94	0.94	81

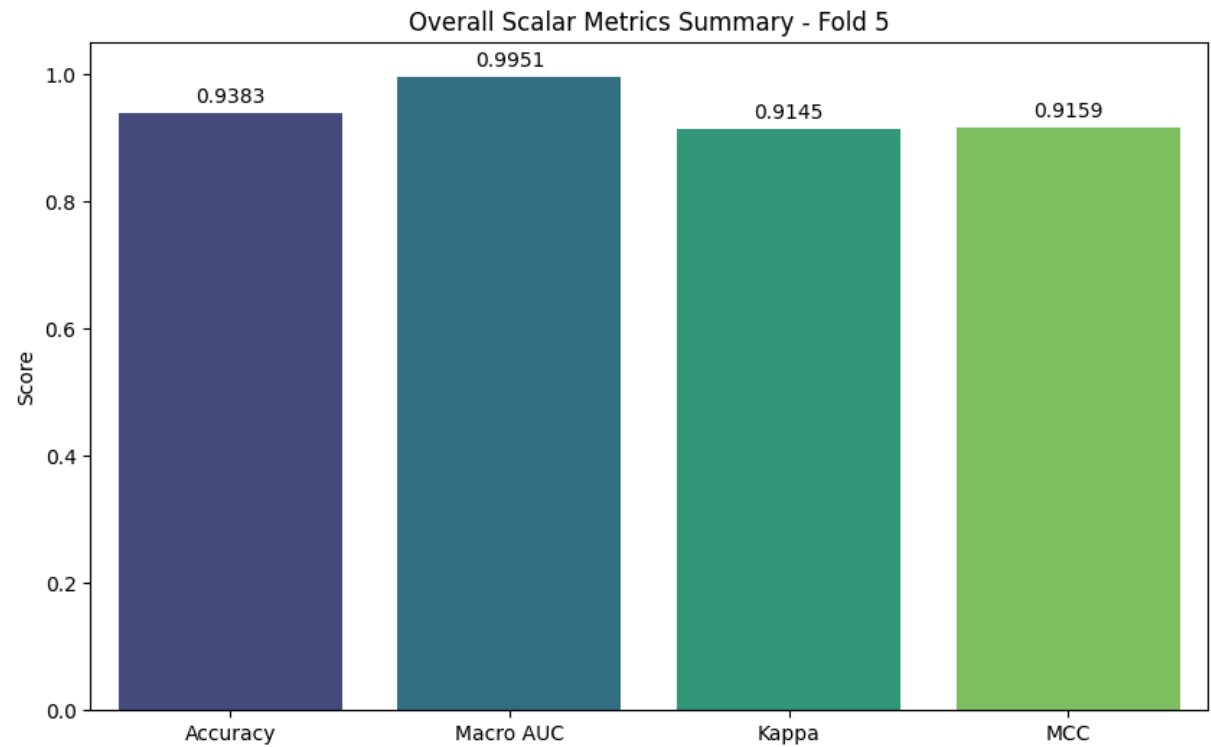
--- Overall Scalar Metrics ---
Accuracy: 0.9383
Macro ROC AUC: 0.9951
Cohen's Kappa: 0.9145
Matthews CorrCoef: 0.9159



Saved normalized confusion matrix to: new_results_Multimodal_ResNet50_Concat_KFold\confusion_matrix_fold_5.png



Saved ROC curves plot to: new_results_Multimodal_ResNet50_Concat_KFold\roc_curves_fold_5.png



Saved scalar metrics plot to: new_results_Multimodal_ResNet50_Concat_KFold\scalar_metrics_fold_5.png

=====

PERFORMANCE BY GENDER

=====

--- Subgroup: female ---

	precision	recall	f1-score	support
Herpes	0.67	1.00	0.80	2
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	0.75	0.86	4
Sarampion	1.00	0.93	0.97	15
Varicela	0.88	1.00	0.93	7
accuracy			0.93	28
macro avg	0.71	0.74	0.71	28
weighted avg	0.94	0.93	0.93	28

--- Subgroup: male ---

	precision	recall	f1-score	support
Herpes	0.80	1.00	0.89	4
Melanoma	1.00	1.00	1.00	2
Monkeypox	0.88	0.93	0.90	15
Sarampion	1.00	0.87	0.93	15
Varicela	1.00	1.00	1.00	17
accuracy			0.94	53
macro avg	0.93	0.96	0.94	53
weighted avg	0.95	0.94	0.94	53

=====

PERFORMANCE BY AGE

=====

--- Subgroup: 0-17 ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.00	0.00	0.00	0
Sarampion	0.00	0.00	0.00	1
Varicela	1.00	1.00	1.00	5
accuracy			0.83	6
macro avg	0.20	0.20	0.20	6
weighted avg	0.83	0.83	0.83	6

--- Subgroup: 18-59 ---

	precision	recall	f1-score	support
Herpes	0.75	1.00	0.86	6
Melanoma	1.00	1.00	1.00	2
Monkeypox	0.94	0.89	0.92	19
Sarampion	1.00	0.93	0.96	29
Varicela	0.95	1.00	0.97	19
accuracy			0.95	75
macro avg	0.93	0.97	0.94	75
weighted avg	0.95	0.95	0.95	75

=====

PERFORMANCE BY SKIN TONE

=====

--- Subgroup: asian ---

	precision	recall	f1-score	support
Herpes	1.00	1.00	1.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	1
Sarampion	1.00	0.75	0.86	4
Varicela	0.75	1.00	0.86	3
accuracy			0.89	9
macro avg	0.75	0.75	0.74	9
weighted avg	0.92	0.89	0.89	9

--- Subgroup: black ---

	precision	recall	f1-score	support
Herpes	1.00	1.00	1.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.80	1.00	0.89	4
Sarampion	0.00	0.00	0.00	1
Varicela	0.00	0.00	0.00	0
accuracy			0.83	6
macro avg	0.36	0.40	0.38	6
weighted avg	0.70	0.83	0.76	6

--- Subgroup: indian ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.67	0.67	0.67	3
Sarampion	1.00	0.75	0.86	4
Varicela	1.00	1.00	1.00	3
accuracy			0.80	10

macro avg	0.53	0.48	0.50	10
weighted avg	0.90	0.80	0.84	10

--- Subgroup: latino hispanic ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	2
Sarampion	1.00	1.00	1.00	7
Varicela	0.00	0.00	0.00	0
accuracy			1.00	9
macro avg	0.40	0.40	0.40	9
weighted avg	1.00	1.00	1.00	9

--- Subgroup: middle eastern ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	1
Sarampion	1.00	1.00	1.00	1
Varicela	0.00	0.00	0.00	0
accuracy			1.00	2
macro avg	0.40	0.40	0.40	2
weighted avg	1.00	1.00	1.00	2

--- Subgroup: white ---

	precision	recall	f1-score	support
Herpes	0.80	1.00	0.89	4
Melanoma	1.00	1.00	1.00	2
Monkeypox	1.00	0.88	0.93	8
Sarampion	1.00	1.00	1.00	13
Varicela	1.00	1.00	1.00	18
accuracy			0.98	45
macro avg	0.96	0.97	0.96	45
weighted avg	0.98	0.98	0.98	45

```
=====
FAIRNESS SUMMARY (ALL GROUPS) | WGA: 0.0000 EOD: 0.9714
Worst pair: 0-17|male|black vs 18-59|male|white
=====
```

```
=====
FAIRNESS SUMMARY (FILTERED, n>=15) | WGA: 0.9615 EOD: 0.0000
Worst pair: None vs None
Groups: ['18-59|male|white']
=====
```

```
#####  
####  
### FOLD 6 ###  
#####  
####
```

Unfreezing ResNet-50 layer4 for fine-tuning.

Total params: 26.43M | Trainable params: 17.89M (67.68%)

Using weighted CrossEntropyLoss for training.

START TRAINING (Resnet50 + Concat) for 200 epochs...

Epoch 1/200 | Train Loss: 1.8982 Acc: 0.1400 | Val Loss: 1.7870 Acc: 0.1882 | LR: 2.00e-05

Epoch 2/200 | Train Loss: 1.9305 Acc: 0.1400 | Val Loss: 1.7857 Acc: 0.2941 | LR: 3.00e-05

Epoch 3/200 | Train Loss: 1.9055 Acc: 0.1550 | Val Loss: 1.7763 Acc: 0.3882 | LR: 4.00e-05

Epoch 4/200 | Train Loss: 1.7903 Acc: 0.1850 | Val Loss: 1.7526 Acc: 0.3765 | LR: 5.00e-05

Epoch 5/200 | Train Loss: 1.7284 Acc: 0.2500 | Val Loss: 1.7128 Acc: 0.4353 | LR: 6.00e-05

Epoch 6/200 | Train Loss: 1.5752 Acc: 0.3000 | Val Loss: 1.6547 Acc: 0.5412 | LR: 7.00e-05

Epoch 7/200 | Train Loss: 1.5195 Acc: 0.3775 | Val Loss: 1.5755 Acc: 0.6235 | LR: 8.00e-05

Epoch 8/200 | Train Loss: 1.4589 Acc: 0.4200 | Val Loss: 1.4806 Acc: 0.6941 | LR: 9.00e-05

Epoch 9/200 | Train Loss: 1.3945 Acc: 0.4975 | Val Loss: 1.3722 Acc: 0.6824 | LR: 1.00e-04

Epoch 10/200 | Train Loss: 1.2763 Acc: 0.5850 | Val Loss: 1.2859 Acc: 0.7176 | LR: 1.00e-04

Epoch 11/200 | Train Loss: 1.2488 Acc: 0.6050 | Val Loss: 1.2123 Acc: 0.7294 | LR: 1.00e-04

Epoch 12/200 | Train Loss: 1.2047 Acc: 0.6450 | Val Loss: 1.1600 Acc: 0.7294 | LR: 1.00e-04

Epoch 13/200 | Train Loss: 1.2092 Acc: 0.6450 | Val Loss: 1.1205 Acc: 0.7412 | LR: 9.99e-05

Epoch 14/200 | Train Loss: 1.1060 Acc: 0.7025 | Val Loss: 1.0932 Acc: 0.7176 | LR: 9.99e-05

Epoch 15/200 | Train Loss: 1.0775 Acc: 0.7350 | Val Loss: 1.0699 Acc: 0.7294 | LR: 9.98e-05

Epoch 16/200 | Train Loss: 1.0511 Acc: 0.7425 | Val Loss: 1.0586 Acc: 0.7529 | LR: 9.98e-05

Epoch 17/200 | Train Loss: 1.0397 Acc: 0.7475 | Val Loss: 1.0406 Acc: 0.7294 | LR: 9.97e-05

Epoch 18/200 | Train Loss: 1.0144 Acc: 0.7825 | Val Loss: 1.0307 Acc: 0.7529 | LR: 9.96e-05

Epoch 19/200 | Train Loss: 1.0317 Acc: 0.7725 | Val Loss: 1.0153 Acc: 0.7529 | LR: 9.94e-05

Epoch 20/200 | Train Loss: 0.9763 Acc: 0.7900 | Val Loss: 1.0097 Acc: 0.7765 | LR: 9.93e-05

Epoch 21/200 | Train Loss: 0.9558 Acc: 0.8475 | Val Loss: 1.0002 Acc: 0.7765 | LR: 9.92e-05

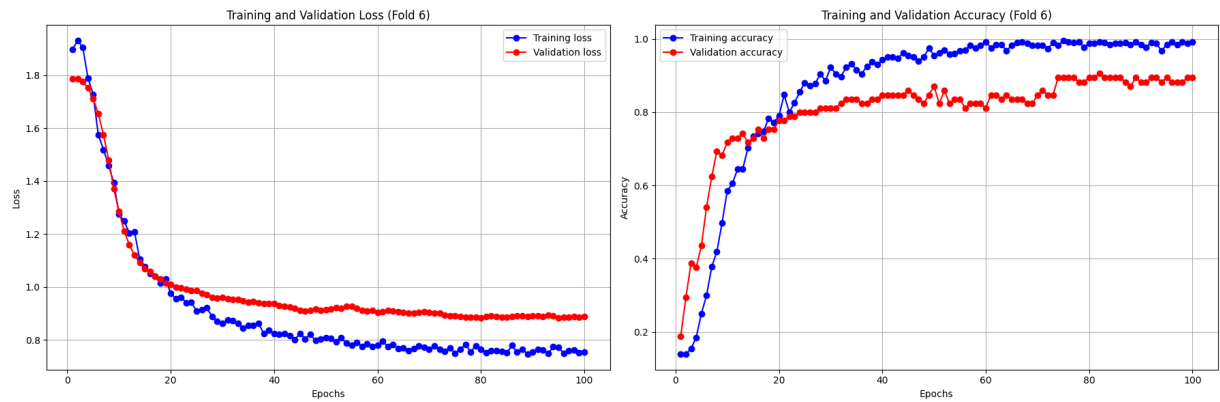
Epoch 22/200 | Train Loss: 0.9622 Acc: 0.8000 | Val Loss: 0.9969 Acc: 0.7882 | LR: 9.90e-05

Epoch 23/200 | Train Loss: 0.9415 Acc: 0.8250 | Val Loss: 0.9930 Acc: 0.7882 | LR:

9.88e-05
Epoch 24/200 | Train Loss: 0.9434 Acc: 0.8550 | Val Loss: 0.9872 Acc: 0.8000 | LR: 9.87e-05
Epoch 25/200 | Train Loss: 0.9089 Acc: 0.8800 | Val Loss: 0.9864 Acc: 0.8000 | LR: 9.85e-05
Epoch 26/200 | Train Loss: 0.9134 Acc: 0.8725 | Val Loss: 0.9763 Acc: 0.8000 | LR: 9.83e-05
Epoch 27/200 | Train Loss: 0.9212 Acc: 0.8775 | Val Loss: 0.9715 Acc: 0.8000 | LR: 9.80e-05
Epoch 28/200 | Train Loss: 0.8884 Acc: 0.9050 | Val Loss: 0.9620 Acc: 0.8118 | LR: 9.78e-05
Epoch 29/200 | Train Loss: 0.8714 Acc: 0.8850 | Val Loss: 0.9587 Acc: 0.8118 | LR: 9.76e-05
Epoch 30/200 | Train Loss: 0.8623 Acc: 0.9225 | Val Loss: 0.9599 Acc: 0.8118 | LR: 9.73e-05
Epoch 31/200 | Train Loss: 0.8763 Acc: 0.9050 | Val Loss: 0.9556 Acc: 0.8118 | LR: 9.70e-05
Epoch 32/200 | Train Loss: 0.8722 Acc: 0.8975 | Val Loss: 0.9542 Acc: 0.8235 | LR: 9.67e-05
Epoch 33/200 | Train Loss: 0.8638 Acc: 0.9225 | Val Loss: 0.9529 Acc: 0.8353 | LR: 9.64e-05
Epoch 34/200 | Train Loss: 0.8438 Acc: 0.9325 | Val Loss: 0.9474 Acc: 0.8353 | LR: 9.61e-05
Epoch 35/200 | Train Loss: 0.8550 Acc: 0.9150 | Val Loss: 0.9435 Acc: 0.8353 | LR: 9.58e-05
Epoch 36/200 | Train Loss: 0.8540 Acc: 0.9050 | Val Loss: 0.9453 Acc: 0.8235 | LR: 9.55e-05
Epoch 37/200 | Train Loss: 0.8632 Acc: 0.9250 | Val Loss: 0.9391 Acc: 0.8235 | LR: 9.51e-05
Epoch 38/200 | Train Loss: 0.8240 Acc: 0.9375 | Val Loss: 0.9379 Acc: 0.8353 | LR: 9.47e-05
Epoch 39/200 | Train Loss: 0.8367 Acc: 0.9300 | Val Loss: 0.9380 Acc: 0.8353 | LR: 9.44e-05
Epoch 40/200 | Train Loss: 0.8242 Acc: 0.9425 | Val Loss: 0.9376 Acc: 0.8471 | LR: 9.40e-05
Epoch 41/200 | Train Loss: 0.8223 Acc: 0.9500 | Val Loss: 0.9313 Acc: 0.8471 | LR: 9.36e-05
Epoch 42/200 | Train Loss: 0.8231 Acc: 0.9500 | Val Loss: 0.9274 Acc: 0.8471 | LR: 9.32e-05
Epoch 43/200 | Train Loss: 0.8176 Acc: 0.9475 | Val Loss: 0.9240 Acc: 0.8471 | LR: 9.27e-05
Epoch 44/200 | Train Loss: 0.8017 Acc: 0.9625 | Val Loss: 0.9190 Acc: 0.8471 | LR: 9.23e-05
Epoch 45/200 | Train Loss: 0.8242 Acc: 0.9550 | Val Loss: 0.9127 Acc: 0.8588 | LR: 9.19e-05
Epoch 46/200 | Train Loss: 0.8047 Acc: 0.9500 | Val Loss: 0.9089 Acc: 0.8471 | LR: 9.14e-05
Epoch 47/200 | Train Loss: 0.8218 Acc: 0.9400 | Val Loss: 0.9123 Acc: 0.8353 | LR: 9.09e-05
Epoch 48/200 | Train Loss: 0.7971 Acc: 0.9500 | Val Loss: 0.9183 Acc: 0.8235 | LR: 9.05e-05
Epoch 49/200 | Train Loss: 0.8037 Acc: 0.9750 | Val Loss: 0.9125 Acc: 0.8471 | LR: 9.00e-05
Epoch 50/200 | Train Loss: 0.8075 Acc: 0.9550 | Val Loss: 0.9157 Acc: 0.8706 | LR: 8.95e-05
Epoch 51/200 | Train Loss: 0.8067 Acc: 0.9625 | Val Loss: 0.9179 Acc: 0.8235 | LR:

8.89e-05
Epoch 52/200 | Train Loss: 0.7929 Acc: 0.9700 | Val Loss: 0.9219 Acc: 0.8588 | LR: 8.84e-05
Epoch 53/200 | Train Loss: 0.8076 Acc: 0.9575 | Val Loss: 0.9209 Acc: 0.8235 | LR: 8.79e-05
Epoch 54/200 | Train Loss: 0.7873 Acc: 0.9600 | Val Loss: 0.9263 Acc: 0.8353 | LR: 8.73e-05
Epoch 55/200 | Train Loss: 0.7794 Acc: 0.9675 | Val Loss: 0.9266 Acc: 0.8353 | LR: 8.68e-05
Epoch 56/200 | Train Loss: 0.7915 Acc: 0.9700 | Val Loss: 0.9190 Acc: 0.8118 | LR: 8.62e-05
Epoch 57/200 | Train Loss: 0.7763 Acc: 0.9825 | Val Loss: 0.9110 Acc: 0.8235 | LR: 8.56e-05
Epoch 58/200 | Train Loss: 0.7844 Acc: 0.9750 | Val Loss: 0.9101 Acc: 0.8235 | LR: 8.51e-05
Epoch 59/200 | Train Loss: 0.7763 Acc: 0.9825 | Val Loss: 0.9108 Acc: 0.8235 | LR: 8.45e-05
Epoch 60/200 | Train Loss: 0.7794 Acc: 0.9925 | Val Loss: 0.9053 Acc: 0.8118 | LR: 8.39e-05
Epoch 61/200 | Train Loss: 0.7949 Acc: 0.9750 | Val Loss: 0.9074 Acc: 0.8471 | LR: 8.33e-05
Epoch 62/200 | Train Loss: 0.7752 Acc: 0.9850 | Val Loss: 0.9123 Acc: 0.8471 | LR: 8.26e-05
Epoch 63/200 | Train Loss: 0.7841 Acc: 0.9850 | Val Loss: 0.9087 Acc: 0.8353 | LR: 8.20e-05
Epoch 64/200 | Train Loss: 0.7672 Acc: 0.9675 | Val Loss: 0.9079 Acc: 0.8471 | LR: 8.14e-05
Epoch 65/200 | Train Loss: 0.7693 Acc: 0.9825 | Val Loss: 0.9042 Acc: 0.8353 | LR: 8.07e-05
Epoch 66/200 | Train Loss: 0.7602 Acc: 0.9900 | Val Loss: 0.9027 Acc: 0.8353 | LR: 8.01e-05
Epoch 67/200 | Train Loss: 0.7680 Acc: 0.9925 | Val Loss: 0.9020 Acc: 0.8353 | LR: 7.94e-05
Epoch 68/200 | Train Loss: 0.7770 Acc: 0.9875 | Val Loss: 0.9029 Acc: 0.8235 | LR: 7.87e-05
Epoch 69/200 | Train Loss: 0.7727 Acc: 0.9825 | Val Loss: 0.9074 Acc: 0.8235 | LR: 7.80e-05
Epoch 70/200 | Train Loss: 0.7637 Acc: 0.9825 | Val Loss: 0.9043 Acc: 0.8471 | LR: 7.73e-05
Epoch 71/200 | Train Loss: 0.7770 Acc: 0.9825 | Val Loss: 0.9022 Acc: 0.8588 | LR: 7.67e-05
Epoch 72/200 | Train Loss: 0.7659 Acc: 0.9725 | Val Loss: 0.9007 Acc: 0.8471 | LR: 7.59e-05
Epoch 73/200 | Train Loss: 0.7582 Acc: 0.9900 | Val Loss: 0.8939 Acc: 0.8471 | LR: 7.52e-05
Epoch 74/200 | Train Loss: 0.7693 Acc: 0.9825 | Val Loss: 0.8910 Acc: 0.8941 | LR: 7.45e-05
Epoch 75/200 | Train Loss: 0.7502 Acc: 0.9950 | Val Loss: 0.8916 Acc: 0.8941 | LR: 7.38e-05
Epoch 76/200 | Train Loss: 0.7637 Acc: 0.9925 | Val Loss: 0.8884 Acc: 0.8941 | LR: 7.31e-05
Epoch 77/200 | Train Loss: 0.7840 Acc: 0.9900 | Val Loss: 0.8857 Acc: 0.8941 | LR: 7.23e-05
Epoch 78/200 | Train Loss: 0.7556 Acc: 0.9925 | Val Loss: 0.8856 Acc: 0.8824 | LR: 7.16e-05
Epoch 79/200 | Train Loss: 0.7787 Acc: 0.9775 | Val Loss: 0.8861 Acc: 0.8824 | LR:

7.08e-05
Epoch 80/200 | Train Loss: 0.7647 Acc: 0.9875 | Val Loss: 0.8844 Acc: 0.8941 | LR: 7.01e-05
Epoch 81/200 | Train Loss: 0.7529 Acc: 0.9875 | Val Loss: 0.8890 Acc: 0.8941 | LR: 6.93e-05
Epoch 82/200 | Train Loss: 0.7599 Acc: 0.9925 | Val Loss: 0.8918 Acc: 0.9059 | LR: 6.86e-05
Epoch 83/200 | Train Loss: 0.7599 Acc: 0.9900 | Val Loss: 0.8888 Acc: 0.8941 | LR: 6.78e-05
Epoch 84/200 | Train Loss: 0.7575 Acc: 0.9850 | Val Loss: 0.8874 Acc: 0.8941 | LR: 6.70e-05
Epoch 85/200 | Train Loss: 0.7520 Acc: 0.9875 | Val Loss: 0.8871 Acc: 0.8941 | LR: 6.62e-05
Epoch 86/200 | Train Loss: 0.7792 Acc: 0.9875 | Val Loss: 0.8892 Acc: 0.8941 | LR: 6.55e-05
Epoch 87/200 | Train Loss: 0.7534 Acc: 0.9900 | Val Loss: 0.8913 Acc: 0.8824 | LR: 6.47e-05
Epoch 88/200 | Train Loss: 0.7656 Acc: 0.9850 | Val Loss: 0.8905 Acc: 0.8706 | LR: 6.39e-05
Epoch 89/200 | Train Loss: 0.7473 Acc: 0.9925 | Val Loss: 0.8884 Acc: 0.8941 | LR: 6.31e-05
Epoch 90/200 | Train Loss: 0.7556 Acc: 0.9850 | Val Loss: 0.8904 Acc: 0.8824 | LR: 6.23e-05
Epoch 91/200 | Train Loss: 0.7648 Acc: 0.9775 | Val Loss: 0.8904 Acc: 0.8824 | LR: 6.15e-05
Epoch 92/200 | Train Loss: 0.7616 Acc: 0.9900 | Val Loss: 0.8894 Acc: 0.8941 | LR: 6.07e-05
Epoch 93/200 | Train Loss: 0.7493 Acc: 0.9875 | Val Loss: 0.8933 Acc: 0.8941 | LR: 5.99e-05
Epoch 94/200 | Train Loss: 0.7740 Acc: 0.9675 | Val Loss: 0.8909 Acc: 0.8824 | LR: 5.90e-05
Epoch 95/200 | Train Loss: 0.7730 Acc: 0.9850 | Val Loss: 0.8844 Acc: 0.8941 | LR: 5.82e-05
Epoch 96/200 | Train Loss: 0.7492 Acc: 0.9925 | Val Loss: 0.8862 Acc: 0.8824 | LR: 5.74e-05
Epoch 97/200 | Train Loss: 0.7585 Acc: 0.9850 | Val Loss: 0.8861 Acc: 0.8824 | LR: 5.66e-05
Epoch 98/200 | Train Loss: 0.7621 Acc: 0.9925 | Val Loss: 0.8882 Acc: 0.8824 | LR: 5.58e-05
Epoch 99/200 | Train Loss: 0.7512 Acc: 0.9875 | Val Loss: 0.8872 Acc: 0.8941 | LR: 5.50e-05
Epoch 100/200 | Train Loss: 0.7550 Acc: 0.9925 | Val Loss: 0.8889 Acc: 0.8941 | LR: 5.41e-05
Early stopping triggered.
Training complete in 7.5 minutes. Best Val Loss: 0.8844
Best model for fold saved to: new_results_Multimodal_ResNet50_Concat_KFold\best_model_fold_6.pth



Saved training history plot to: new_results_Multimodal_ResNet50_Concat_KFold\training_history_fold_6.png

=====

OVERALL PERFORMANCE (FOLD 6)

=====

	precision	recall	f1-score	support
Herpes	1.00	0.83	0.91	6
Melanoma	0.67	1.00	0.80	2
Monkeypox	0.94	0.94	0.94	18
Sarampion	0.96	0.90	0.93	29
Varicela	0.85	0.92	0.88	25
accuracy			0.91	80
macro avg	0.89	0.92	0.89	80
weighted avg	0.92	0.91	0.91	80

--- Overall Scalar Metrics ---

Accuracy: 0.9125

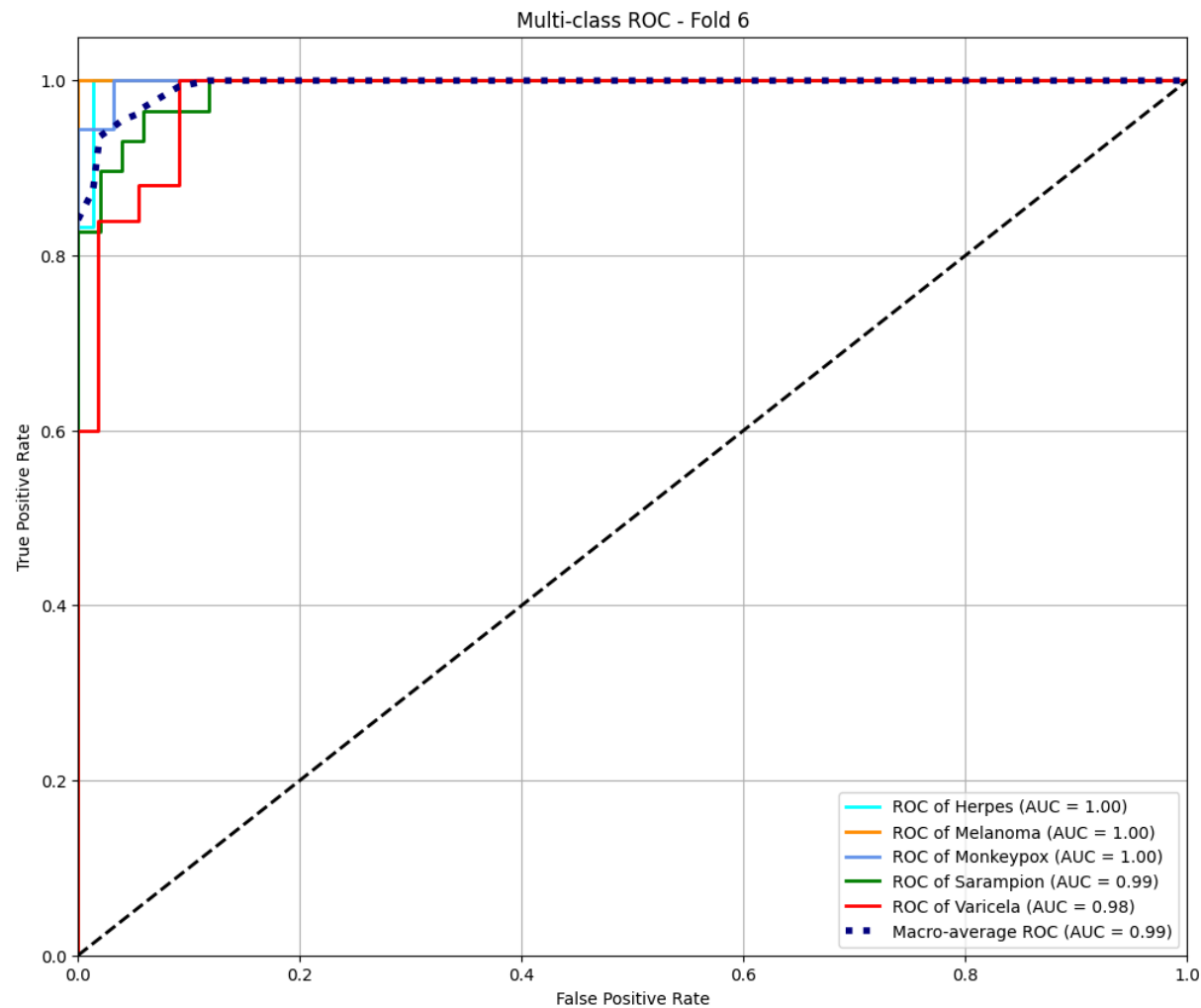
Macro ROC AUC: 0.9939

Cohen's Kappa: 0.8778

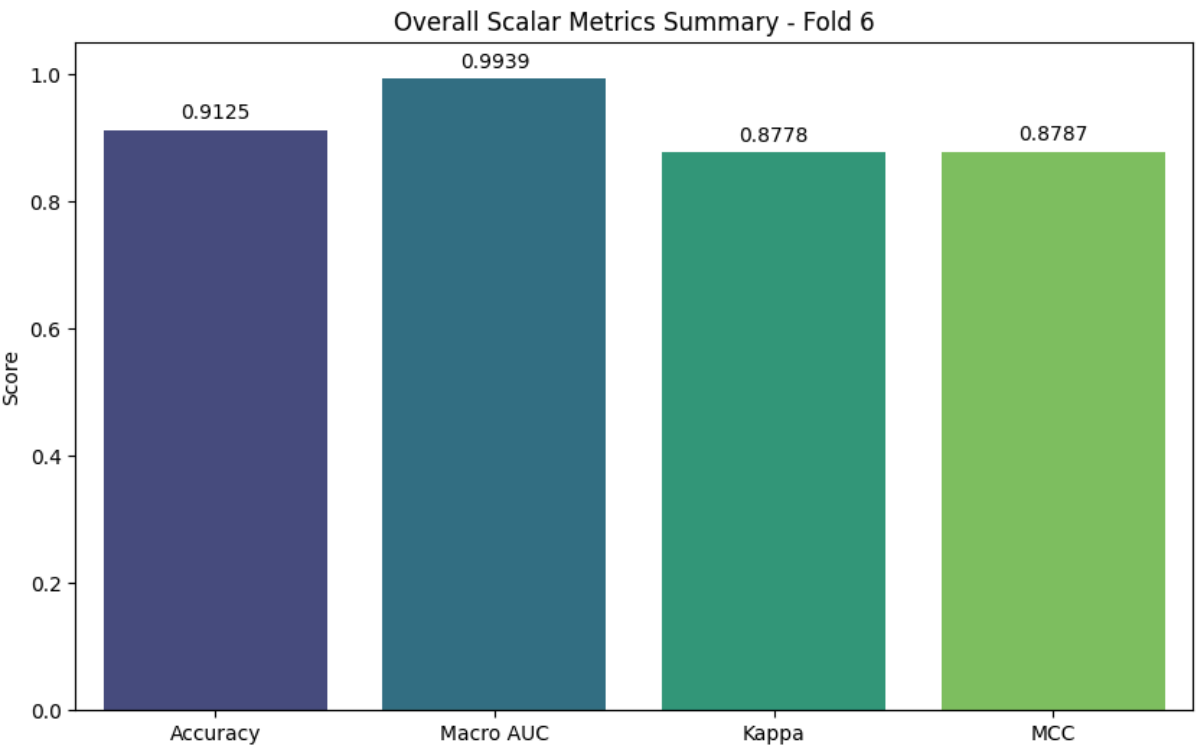
Matthews CorrCoef: 0.8787



Saved normalized confusion matrix to: new_results_Multimodal_ResNet50_Concat_KFold\confusion_matrix_fold_6.png



Saved ROC curves plot to: new_results_Multimodal_ResNet50_Concat_KFold\roc_curves_fold_6.png



Saved scalar metrics plot to: new_results_Multimodal_ResNet50_Concat_KFold\scalar_metrics_fold_6.png

=====

PERFORMANCE BY GENDER

=====

--- Subgroup: female ---

	precision	recall	f1-score	support
Herpes	1.00	1.00	1.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	0.50	0.67	2
Sarampion	0.91	1.00	0.95	10
Varicela	0.92	0.86	0.89	14
accuracy			0.89	27
macro avg	0.77	0.67	0.70	27
weighted avg	0.93	0.89	0.90	27

--- Subgroup: male ---

	precision	recall	f1-score	support
Herpes	1.00	0.80	0.89	5
Melanoma	1.00	1.00	1.00	2
Monkeypox	0.94	1.00	0.97	16
Sarampion	1.00	0.84	0.91	19
Varicela	0.79	1.00	0.88	11
accuracy			0.92	53
macro avg	0.95	0.93	0.93	53
weighted avg	0.94	0.92	0.92	53

=====

PERFORMANCE BY AGE

=====

--- Subgroup: 0-17 ---

	precision	recall	f1-score	support
Herpes	1.00	1.00	1.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	2
Sarampion	0.75	0.75	0.75	4
Varicela	0.75	0.60	0.67	5
accuracy			0.75	12
macro avg	0.70	0.67	0.68	12
weighted avg	0.81	0.75	0.78	12

--- Subgroup: 18-59 ---

	precision	recall	f1-score	support
Herpes	1.00	0.75	0.86	4
Melanoma	1.00	1.00	1.00	2
Monkeypox	0.94	0.94	0.94	16
Sarampion	1.00	0.92	0.96	25
Varicela	0.87	1.00	0.93	20
accuracy			0.94	67
macro avg	0.96	0.92	0.94	67
weighted avg	0.95	0.94	0.94	67

--- Subgroup: 60+ ---

	precision	recall	f1-score	support
Herpes	1.00	1.00	1.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.00	0.00	0.00	0
Sarampion	0.00	0.00	0.00	0
Varicela	0.00	0.00	0.00	0
accuracy			1.00	1
macro avg	0.20	0.20	0.20	1
weighted avg	1.00	1.00	1.00	1

=====

PERFORMANCE BY SKIN TONE

=====

--- Subgroup: asian ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	1
Sarampion	0.00	0.00	0.00	1
Varicela	0.83	1.00	0.91	5
accuracy			0.86	7
macro avg	0.37	0.40	0.38	7
weighted avg	0.74	0.86	0.79	7

--- Subgroup: black ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	3
Sarampion	1.00	1.00	1.00	1
Varicela	0.00	0.00	0.00	0
accuracy			1.00	4

macro avg	0.40	0.40	0.40	4
weighted avg	1.00	1.00	1.00	4

--- Subgroup: indian ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	0.50	0.67	2
Sarampion	1.00	1.00	1.00	7
Varicela	0.50	1.00	0.67	1
accuracy			0.90	10
macro avg	0.50	0.50	0.47	10
weighted avg	0.95	0.90	0.90	10

--- Subgroup: latino hispanic ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.00	0.00	0.00	0
Sarampion	1.00	1.00	1.00	3
Varicela	0.00	0.00	0.00	0
accuracy			1.00	3
macro avg	0.20	0.20	0.20	3
weighted avg	1.00	1.00	1.00	3

--- Subgroup: middle eastern ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	1
Sarampion	0.00	0.00	0.00	0
Varicela	0.00	0.00	0.00	0
accuracy			1.00	1
macro avg	0.20	0.20	0.20	1
weighted avg	1.00	1.00	1.00	1

--- Subgroup: white ---

	precision	recall	f1-score	support
Herpes	1.00	0.83	0.91	6
Melanoma	0.67	1.00	0.80	2
Monkeypox	0.92	1.00	0.96	11
Sarampion	0.94	0.88	0.91	17
Varicela	0.89	0.89	0.89	19
accuracy			0.91	55

macro avg	0.88	0.92	0.89	55
weighted avg	0.92	0.91	0.91	55

```
=====
FAIRNESS SUMMARY (ALL GROUPS) | WGA: 0.3333 EOD: 0.8267
Worst pair: 0-17|female|white vs 18-59|male|white
=====
```

```
=====
FAIRNESS SUMMARY (FILTERED, n>=15) | WGA: 0.9310 EOD: 0.5067
Worst pair: 18-59|female|white vs 18-59|male|white
Groups: ['18-59|female|white', '18-59|male|white']
=====
```

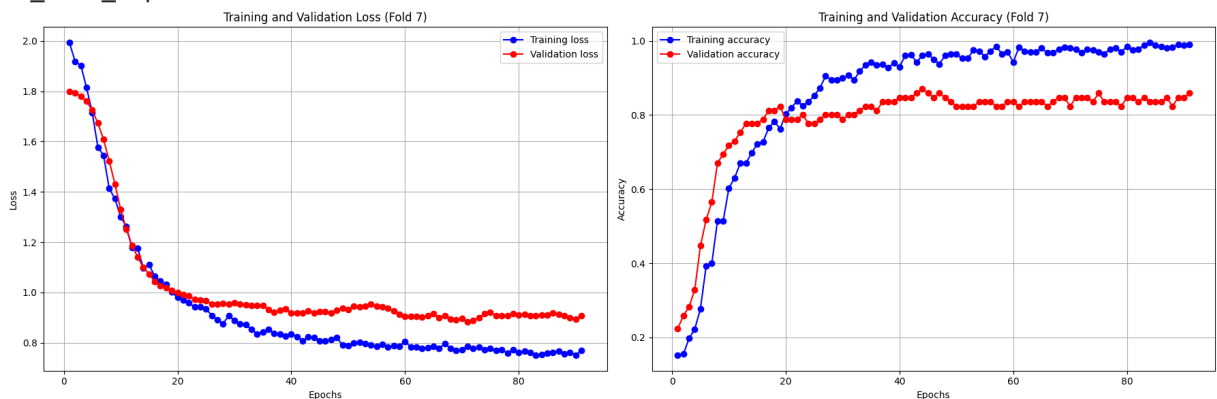
```
#####
####
### FOLD 7 ###
#####
####
Unfreezing ResNet-50 layer4 for fine-tuning.
Total params: 26.43M | Trainable params: 17.89M (67.68%)
Using weighted CrossEntropyLoss for training.
```

```
START TRAINING (Resnet50 + Concat) for 200 epochs...
Epoch 1/200 | Train Loss: 1.9929 Acc: 0.1525 | Val Loss: 1.7986 Acc: 0.2235 | LR: 2.00e-05
Epoch 2/200 | Train Loss: 1.9178 Acc: 0.1550 | Val Loss: 1.7919 Acc: 0.2588 | LR: 3.00e-05
Epoch 3/200 | Train Loss: 1.9026 Acc: 0.1975 | Val Loss: 1.7802 Acc: 0.2824 | LR: 4.00e-05
Epoch 4/200 | Train Loss: 1.8143 Acc: 0.2225 | Val Loss: 1.7602 Acc: 0.3294 | LR: 5.00e-05
Epoch 5/200 | Train Loss: 1.7157 Acc: 0.2775 | Val Loss: 1.7242 Acc: 0.4471 | LR: 6.00e-05
Epoch 6/200 | Train Loss: 1.5779 Acc: 0.3925 | Val Loss: 1.6732 Acc: 0.5176 | LR: 7.00e-05
Epoch 7/200 | Train Loss: 1.5443 Acc: 0.4000 | Val Loss: 1.6080 Acc: 0.5647 | LR: 8.00e-05
Epoch 8/200 | Train Loss: 1.4129 Acc: 0.5150 | Val Loss: 1.5222 Acc: 0.6706 | LR: 9.00e-05
Epoch 9/200 | Train Loss: 1.3728 Acc: 0.5150 | Val Loss: 1.4312 Acc: 0.6941 | LR: 1.00e-04
Epoch 10/200 | Train Loss: 1.3008 Acc: 0.6025 | Val Loss: 1.3305 Acc: 0.7176 | LR: 1.00e-04
Epoch 11/200 | Train Loss: 1.2621 Acc: 0.6300 | Val Loss: 1.2517 Acc: 0.7294 | LR: 1.00e-04
Epoch 12/200 | Train Loss: 1.1776 Acc: 0.6700 | Val Loss: 1.1866 Acc: 0.7529 | LR: 1.00e-04
Epoch 13/200 | Train Loss: 1.1764 Acc: 0.6700 | Val Loss: 1.1403 Acc: 0.7765 | LR: 9.99e-05
Epoch 14/200 | Train Loss: 1.0962 Acc: 0.6975 | Val Loss: 1.0989 Acc: 0.7765 | LR: 9.99e-05
Epoch 15/200 | Train Loss: 1.1104 Acc: 0.7225 | Val Loss: 1.0735 Acc: 0.7765 | LR: 9.98e-05
Epoch 16/200 | Train Loss: 1.0642 Acc: 0.7275 | Val Loss: 1.0423 Acc: 0.7882 | LR:
```

9.98e-05
Epoch 17/200 | Train Loss: 1.0450 Acc: 0.7650 | Val Loss: 1.0258 Acc: 0.8118 | LR: 9.97e-05
Epoch 18/200 | Train Loss: 1.0314 Acc: 0.7825 | Val Loss: 1.0176 Acc: 0.8118 | LR: 9.96e-05
Epoch 19/200 | Train Loss: 1.0013 Acc: 0.7625 | Val Loss: 1.0081 Acc: 0.8235 | LR: 9.94e-05
Epoch 20/200 | Train Loss: 0.9800 Acc: 0.8025 | Val Loss: 0.9988 Acc: 0.7882 | LR: 9.93e-05
Epoch 21/200 | Train Loss: 0.9707 Acc: 0.8200 | Val Loss: 0.9912 Acc: 0.7882 | LR: 9.92e-05
Epoch 22/200 | Train Loss: 0.9592 Acc: 0.8375 | Val Loss: 0.9850 Acc: 0.7882 | LR: 9.90e-05
Epoch 23/200 | Train Loss: 0.9425 Acc: 0.8250 | Val Loss: 0.9721 Acc: 0.8000 | LR: 9.88e-05
Epoch 24/200 | Train Loss: 0.9431 Acc: 0.8350 | Val Loss: 0.9696 Acc: 0.7765 | LR: 9.87e-05
Epoch 25/200 | Train Loss: 0.9352 Acc: 0.8525 | Val Loss: 0.9660 Acc: 0.7765 | LR: 9.85e-05
Epoch 26/200 | Train Loss: 0.9090 Acc: 0.8725 | Val Loss: 0.9548 Acc: 0.7882 | LR: 9.83e-05
Epoch 27/200 | Train Loss: 0.8907 Acc: 0.9050 | Val Loss: 0.9545 Acc: 0.8000 | LR: 9.80e-05
Epoch 28/200 | Train Loss: 0.8760 Acc: 0.8950 | Val Loss: 0.9568 Acc: 0.8000 | LR: 9.78e-05
Epoch 29/200 | Train Loss: 0.9079 Acc: 0.8950 | Val Loss: 0.9535 Acc: 0.8000 | LR: 9.76e-05
Epoch 30/200 | Train Loss: 0.8888 Acc: 0.9000 | Val Loss: 0.9584 Acc: 0.7882 | LR: 9.73e-05
Epoch 31/200 | Train Loss: 0.8765 Acc: 0.9075 | Val Loss: 0.9551 Acc: 0.8000 | LR: 9.70e-05
Epoch 32/200 | Train Loss: 0.8722 Acc: 0.8950 | Val Loss: 0.9516 Acc: 0.8000 | LR: 9.67e-05
Epoch 33/200 | Train Loss: 0.8531 Acc: 0.9175 | Val Loss: 0.9486 Acc: 0.8118 | LR: 9.64e-05
Epoch 34/200 | Train Loss: 0.8359 Acc: 0.9350 | Val Loss: 0.9495 Acc: 0.8235 | LR: 9.61e-05
Epoch 35/200 | Train Loss: 0.8427 Acc: 0.9425 | Val Loss: 0.9488 Acc: 0.8235 | LR: 9.58e-05
Epoch 36/200 | Train Loss: 0.8538 Acc: 0.9350 | Val Loss: 0.9323 Acc: 0.8118 | LR: 9.55e-05
Epoch 37/200 | Train Loss: 0.8369 Acc: 0.9375 | Val Loss: 0.9215 Acc: 0.8353 | LR: 9.51e-05
Epoch 38/200 | Train Loss: 0.8351 Acc: 0.9275 | Val Loss: 0.9287 Acc: 0.8353 | LR: 9.47e-05
Epoch 39/200 | Train Loss: 0.8276 Acc: 0.9400 | Val Loss: 0.9337 Acc: 0.8353 | LR: 9.44e-05
Epoch 40/200 | Train Loss: 0.8350 Acc: 0.9300 | Val Loss: 0.9184 Acc: 0.8471 | LR: 9.40e-05
Epoch 41/200 | Train Loss: 0.8250 Acc: 0.9600 | Val Loss: 0.9199 Acc: 0.8471 | LR: 9.36e-05
Epoch 42/200 | Train Loss: 0.8074 Acc: 0.9625 | Val Loss: 0.9192 Acc: 0.8471 | LR: 9.32e-05
Epoch 43/200 | Train Loss: 0.8225 Acc: 0.9425 | Val Loss: 0.9272 Acc: 0.8588 | LR: 9.27e-05
Epoch 44/200 | Train Loss: 0.8201 Acc: 0.9600 | Val Loss: 0.9192 Acc: 0.8706 | LR:

9.23e-05
Epoch 45/200 | Train Loss: 0.8088 Acc: 0.9650 | Val Loss: 0.9248 Acc: 0.8588 | LR: 9.19e-05
Epoch 46/200 | Train Loss: 0.8078 Acc: 0.9500 | Val Loss: 0.9240 Acc: 0.8471 | LR: 9.14e-05
Epoch 47/200 | Train Loss: 0.8141 Acc: 0.9375 | Val Loss: 0.9196 Acc: 0.8588 | LR: 9.09e-05
Epoch 48/200 | Train Loss: 0.8200 Acc: 0.9600 | Val Loss: 0.9307 Acc: 0.8471 | LR: 9.05e-05
Epoch 49/200 | Train Loss: 0.7922 Acc: 0.9650 | Val Loss: 0.9366 Acc: 0.8353 | LR: 9.00e-05
Epoch 50/200 | Train Loss: 0.7883 Acc: 0.9650 | Val Loss: 0.9329 Acc: 0.8235 | LR: 8.95e-05
Epoch 51/200 | Train Loss: 0.7988 Acc: 0.9525 | Val Loss: 0.9456 Acc: 0.8235 | LR: 8.89e-05
Epoch 52/200 | Train Loss: 0.8023 Acc: 0.9525 | Val Loss: 0.9427 Acc: 0.8235 | LR: 8.84e-05
Epoch 53/200 | Train Loss: 0.7974 Acc: 0.9750 | Val Loss: 0.9448 Acc: 0.8235 | LR: 8.79e-05
Epoch 54/200 | Train Loss: 0.7905 Acc: 0.9725 | Val Loss: 0.9529 Acc: 0.8353 | LR: 8.73e-05
Epoch 55/200 | Train Loss: 0.7861 Acc: 0.9575 | Val Loss: 0.9446 Acc: 0.8353 | LR: 8.68e-05
Epoch 56/200 | Train Loss: 0.7935 Acc: 0.9725 | Val Loss: 0.9417 Acc: 0.8353 | LR: 8.62e-05
Epoch 57/200 | Train Loss: 0.7830 Acc: 0.9850 | Val Loss: 0.9382 Acc: 0.8235 | LR: 8.56e-05
Epoch 58/200 | Train Loss: 0.7877 Acc: 0.9650 | Val Loss: 0.9264 Acc: 0.8235 | LR: 8.51e-05
Epoch 59/200 | Train Loss: 0.7858 Acc: 0.9700 | Val Loss: 0.9129 Acc: 0.8353 | LR: 8.45e-05
Epoch 60/200 | Train Loss: 0.8062 Acc: 0.9425 | Val Loss: 0.9047 Acc: 0.8353 | LR: 8.39e-05
Epoch 61/200 | Train Loss: 0.7828 Acc: 0.9825 | Val Loss: 0.9062 Acc: 0.8235 | LR: 8.33e-05
Epoch 62/200 | Train Loss: 0.7837 Acc: 0.9725 | Val Loss: 0.9050 Acc: 0.8353 | LR: 8.26e-05
Epoch 63/200 | Train Loss: 0.7786 Acc: 0.9700 | Val Loss: 0.9027 Acc: 0.8353 | LR: 8.20e-05
Epoch 64/200 | Train Loss: 0.7802 Acc: 0.9700 | Val Loss: 0.9070 Acc: 0.8353 | LR: 8.14e-05
Epoch 65/200 | Train Loss: 0.7870 Acc: 0.9800 | Val Loss: 0.9153 Acc: 0.8353 | LR: 8.07e-05
Epoch 66/200 | Train Loss: 0.7781 Acc: 0.9675 | Val Loss: 0.9002 Acc: 0.8235 | LR: 8.01e-05
Epoch 67/200 | Train Loss: 0.7968 Acc: 0.9675 | Val Loss: 0.9089 Acc: 0.8353 | LR: 7.94e-05
Epoch 68/200 | Train Loss: 0.7774 Acc: 0.9775 | Val Loss: 0.8949 Acc: 0.8471 | LR: 7.87e-05
Epoch 69/200 | Train Loss: 0.7687 Acc: 0.9825 | Val Loss: 0.8923 Acc: 0.8471 | LR: 7.80e-05
Epoch 70/200 | Train Loss: 0.7723 Acc: 0.9800 | Val Loss: 0.8970 Acc: 0.8235 | LR: 7.73e-05
Epoch 71/200 | Train Loss: 0.7856 Acc: 0.9775 | Val Loss: 0.8847 Acc: 0.8471 | LR: 7.67e-05
Epoch 72/200 | Train Loss: 0.7786 Acc: 0.9675 | Val Loss: 0.8876 Acc: 0.8471 | LR:

7.59e-05
Epoch 73/200 | Train Loss: 0.7826 Acc: 0.9775 | Val Loss: 0.9002 Acc: 0.8471 | LR: 7.52e-05
Epoch 74/200 | Train Loss: 0.7733 Acc: 0.9750 | Val Loss: 0.9151 Acc: 0.8353 | LR: 7.45e-05
Epoch 75/200 | Train Loss: 0.7765 Acc: 0.9700 | Val Loss: 0.9225 Acc: 0.8588 | LR: 7.38e-05
Epoch 76/200 | Train Loss: 0.7704 Acc: 0.9650 | Val Loss: 0.9075 Acc: 0.8353 | LR: 7.31e-05
Epoch 77/200 | Train Loss: 0.7724 Acc: 0.9775 | Val Loss: 0.9069 Acc: 0.8353 | LR: 7.23e-05
Epoch 78/200 | Train Loss: 0.7597 Acc: 0.9800 | Val Loss: 0.9077 Acc: 0.8353 | LR: 7.16e-05
Epoch 79/200 | Train Loss: 0.7728 Acc: 0.9700 | Val Loss: 0.9161 Acc: 0.8235 | LR: 7.08e-05
Epoch 80/200 | Train Loss: 0.7611 Acc: 0.9850 | Val Loss: 0.9098 Acc: 0.8471 | LR: 7.01e-05
Epoch 81/200 | Train Loss: 0.7676 Acc: 0.9750 | Val Loss: 0.9135 Acc: 0.8471 | LR: 6.93e-05
Epoch 82/200 | Train Loss: 0.7623 Acc: 0.9775 | Val Loss: 0.9074 Acc: 0.8353 | LR: 6.86e-05
Epoch 83/200 | Train Loss: 0.7512 Acc: 0.9875 | Val Loss: 0.9073 Acc: 0.8471 | LR: 6.78e-05
Epoch 84/200 | Train Loss: 0.7540 Acc: 0.9950 | Val Loss: 0.9114 Acc: 0.8353 | LR: 6.70e-05
Epoch 85/200 | Train Loss: 0.7597 Acc: 0.9875 | Val Loss: 0.9101 Acc: 0.8353 | LR: 6.62e-05
Epoch 86/200 | Train Loss: 0.7607 Acc: 0.9850 | Val Loss: 0.9181 Acc: 0.8353 | LR: 6.55e-05
Epoch 87/200 | Train Loss: 0.7675 Acc: 0.9800 | Val Loss: 0.9122 Acc: 0.8471 | LR: 6.47e-05
Epoch 88/200 | Train Loss: 0.7570 Acc: 0.9825 | Val Loss: 0.9071 Acc: 0.8235 | LR: 6.39e-05
Epoch 89/200 | Train Loss: 0.7622 Acc: 0.9900 | Val Loss: 0.8984 Acc: 0.8471 | LR: 6.31e-05
Epoch 90/200 | Train Loss: 0.7518 Acc: 0.9875 | Val Loss: 0.8947 Acc: 0.8471 | LR: 6.23e-05
Epoch 91/200 | Train Loss: 0.7685 Acc: 0.9900 | Val Loss: 0.9076 Acc: 0.8588 | LR: 6.15e-05
Early stopping triggered.
Training complete in 6.6 minutes. Best Val Loss: 0.8847
Best model for fold saved to: new_results_Multimodal_ResNet50_Concat_KFold\best_model_fold_7.pth



Saved training history plot to: new_results_Multimodal_ResNet50_Concat_KFold\training_history_fold_7.png

=====

OVERALL PERFORMANCE (FOLD 7)

=====

	precision	recall	f1-score	support
Herpes	1.00	0.67	0.80	6
Melanoma	1.00	1.00	1.00	2
Monkeypox	0.89	0.94	0.92	18
Sarampion	0.97	0.97	0.97	29
Varicela	0.92	0.96	0.94	25
accuracy			0.94	80
macro avg	0.96	0.91	0.93	80
weighted avg	0.94	0.94	0.94	80

--- Overall Scalar Metrics ---

Accuracy: 0.9375

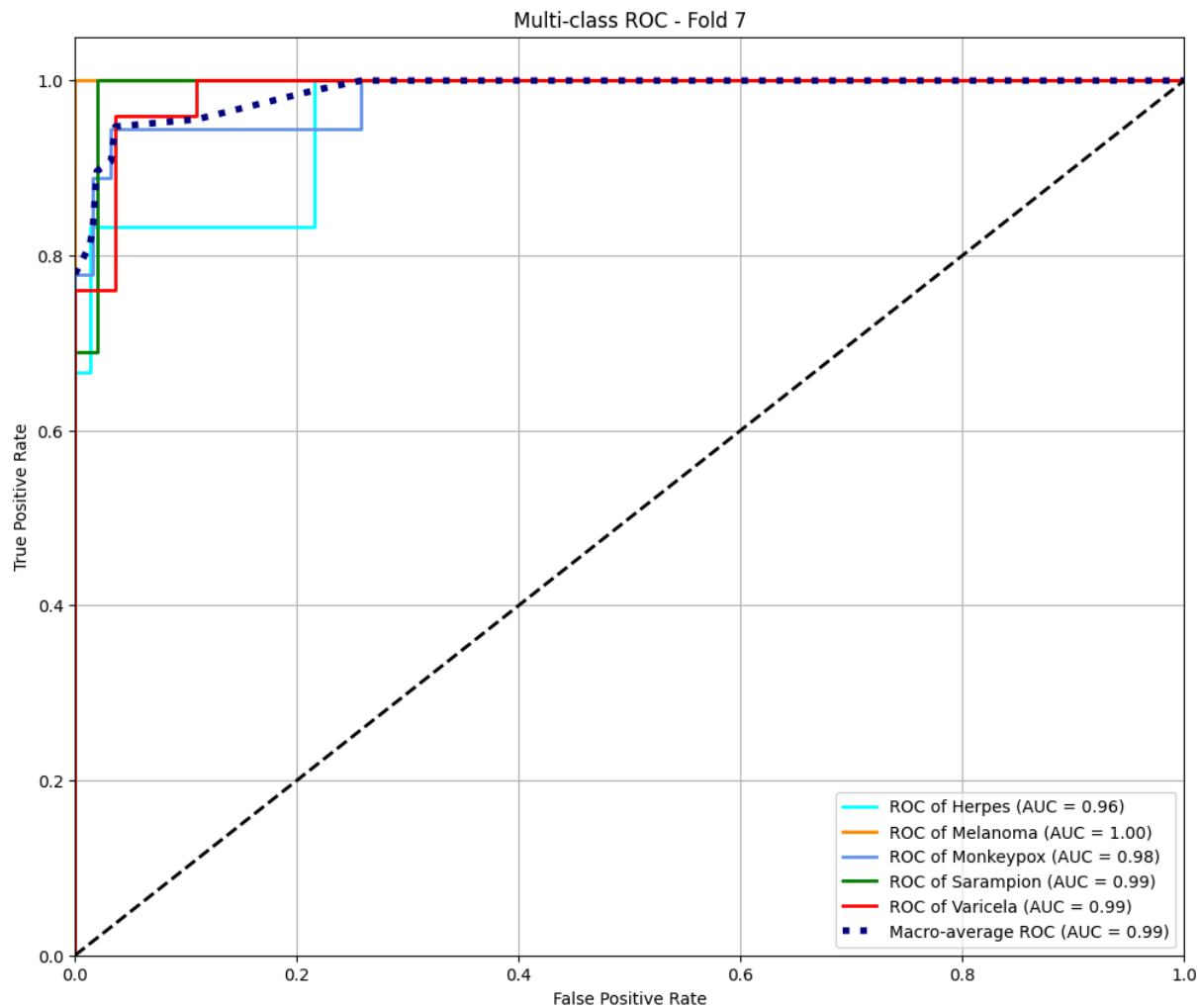
Macro ROC AUC: 0.9852

Cohen's Kappa: 0.9119

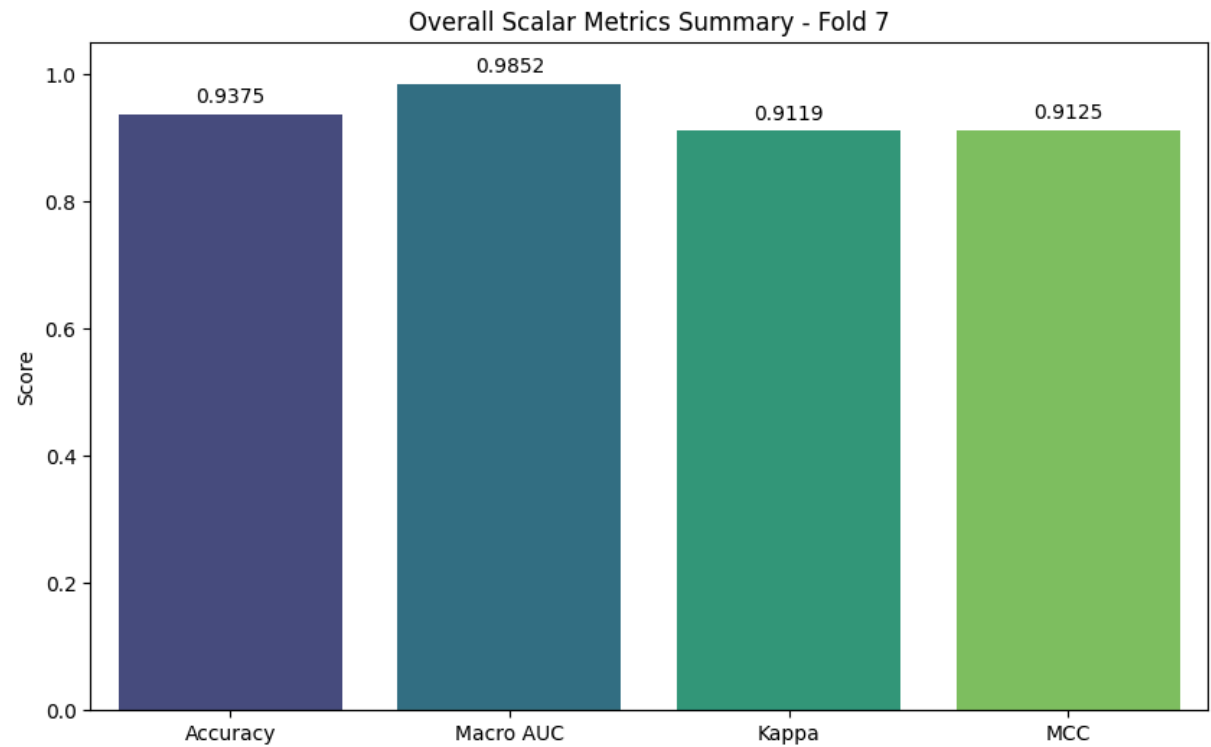
Matthews CorrCoef: 0.9125



Saved normalized confusion matrix to: new_results_Multimodal_ResNet50_Concat_KFold\confusion_matrix_fold_7.png



Saved ROC curves plot to: new_results_Multimodal_ResNet50_Concat_KFold\roc_curves_fold_7.png



Saved scalar metrics plot to: new_results_Multimodal_ResNet50_Concat_KFold\scalar_metrics_fold_7.png

=====

PERFORMANCE BY GENDER

=====

--- Subgroup: female ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	3
Sarampion	1.00	1.00	1.00	15
Varicela	1.00	1.00	1.00	12
accuracy			1.00	30
macro avg	0.60	0.60	0.60	30
weighted avg	1.00	1.00	1.00	30

--- Subgroup: male ---

	precision	recall	f1-score	support
Herpes	1.00	0.67	0.80	6
Melanoma	1.00	1.00	1.00	2
Monkeypox	0.88	0.93	0.90	15
Sarampion	0.93	0.93	0.93	14
Varicela	0.86	0.92	0.89	13
accuracy			0.90	50
macro avg	0.93	0.89	0.90	50
weighted avg	0.91	0.90	0.90	50

=====

PERFORMANCE BY AGE

=====

--- Subgroup: 0-17 ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.67	1.00	0.80	2
Sarampion	1.00	1.00	1.00	3
Varicela	0.80	0.80	0.80	5
accuracy			0.82	11
macro avg	0.49	0.56	0.52	11
weighted avg	0.76	0.82	0.78	11

--- Subgroup: 18-59 ---

	precision	recall	f1-score	support
Herpes	1.00	0.80	0.89	5
Melanoma	1.00	1.00	1.00	2
Monkeypox	0.93	0.93	0.93	15
Sarampion	0.96	0.96	0.96	26
Varicela	0.95	1.00	0.98	20
accuracy			0.96	68
macro avg	0.97	0.94	0.95	68
weighted avg	0.96	0.96	0.96	68

--- Subgroup: 60+ ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	1
Sarampion	0.00	0.00	0.00	0
Varicela	0.00	0.00	0.00	0
accuracy			1.00	1
macro avg	0.20	0.20	0.20	1
weighted avg	1.00	1.00	1.00	1

=====

PERFORMANCE BY SKIN TONE

=====

--- Subgroup: asian ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.80	1.00	0.89	4
Sarampion	1.00	1.00	1.00	5
Varicela	1.00	0.75	0.86	4
accuracy			0.92	13
macro avg	0.56	0.55	0.55	13
weighted avg	0.94	0.92	0.92	13

--- Subgroup: black ---

	precision	recall	f1-score	support
Herpes	1.00	1.00	1.00	1
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.83	1.00	0.91	5
Sarampion	1.00	0.50	0.67	2
Varicela	1.00	1.00	1.00	1
accuracy			0.89	9

macro avg	0.77	0.70	0.72	9
weighted avg	0.91	0.89	0.88	9

--- Subgroup: indian ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	4
Sarampion	1.00	1.00	1.00	1
Varicela	0.00	0.00	0.00	0
accuracy			1.00	5
macro avg	0.40	0.40	0.40	5
weighted avg	1.00	1.00	1.00	5

--- Subgroup: latino hispanic ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	1.00	1.00	1.00	2
Sarampion	1.00	1.00	1.00	3
Varicela	0.00	0.00	0.00	0
accuracy			1.00	5
macro avg	0.40	0.40	0.40	5
weighted avg	1.00	1.00	1.00	5

--- Subgroup: middle eastern ---

	precision	recall	f1-score	support
Herpes	0.00	0.00	0.00	0
Melanoma	0.00	0.00	0.00	0
Monkeypox	0.00	0.00	0.00	0
Sarampion	1.00	1.00	1.00	2
Varicela	0.00	0.00	0.00	0
accuracy			1.00	2
macro avg	0.20	0.20	0.20	2
weighted avg	1.00	1.00	1.00	2

--- Subgroup: white ---

	precision	recall	f1-score	support
Herpes	1.00	0.60	0.75	5
Melanoma	1.00	1.00	1.00	2
Monkeypox	1.00	0.67	0.80	3
Sarampion	0.94	1.00	0.97	16
Varicela	0.91	1.00	0.95	20
accuracy			0.93	46

macro avg	0.97	0.85	0.89	46
weighted avg	0.94	0.93	0.93	46

```
=====
FAIRNESS SUMMARY (ALL GROUPS) | WGA: 0.0000 EOD: 0.8833
Worst pair: 0-17|male|asian vs 18-59|male|white
=====
```

```
=====
FAIRNESS SUMMARY (FILTERED, n>=15) | WGA: 0.8889 EOD: 0.4833
Worst pair: 18-59|female|white vs 18-59|male|white
Groups: ['18-59|female|white', '18-59|male|white']
=====
```

```
=====
====
Cross-validated over 7 folds
=====
```

```
====
Accuracy          : 0.901 ± 0.034
Roc Auc Macro     : 0.984 ± 0.010
Kappa             : 0.863 ± 0.047
Mcc               : 0.864 ± 0.046
Wga All           : 0.205 ± 0.267
Eod All           : 0.871 ± 0.068
Wga Filtered      : 0.867 ± 0.082
Eod Filtered      : 0.428 ± 0.194
```

```
--- POOLED OVERALL ACCURACY BY GENDER (n>=15) ---
```

Group	Acc	Count
male	0.896	355
female	0.910	210

```
--- POOLED OVERALL ACCURACY BY AGE (n>=15) ---
```

Group	Acc	Count
18-59	0.908	501
0-17	0.836	61

```
--- POOLED OVERALL ACCURACY BY SKIN TONE (n>=15) ---
```

Group	Acc	Count
white	0.899	338
asian	0.900	70
indian	0.880	50
latino hispanic	0.940	50
black	0.921	38
middle eastern	0.842	19

```
--- POOLED OVERALL ACCURACY BY COMPOSITE (n>=15) ---
```

Group	Acc	Count
18-59 male white	0.873	166
18-59 female white	0.952	126
18-59 male asian	0.897	39
18-59 male indian	0.971	34
18-59 male black	0.933	30
0-17 male white	0.900	30

18-59 female asian	0.923	26
18-59 female latino hispanic	0.920	25
18-59 male latino hispanic	0.957	23
18-59 male middle eastern	0.833	18
0-17 female white	0.733	15

Results, models, and plots saved to 'new_results_Multimodal_ResNet50_Concat_KFold'

In []: